



B. Tech. ELECTRICAL ENGINEERING

Syllabus AY: 2019-20



CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Authored By: Charusat



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY



ACADEMIC REGULATIONS & SYLLABUS 2019-20 (Choice Based Credit System)

Faculty of Technology & Engineering
M & V Patel Department of Electrical Engineering
Bachelor of Technology Programme
(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	Preparation: To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	Core Competence: To build a broad engineering core competence and to enable the students to solve the problems systematically by providing them exposure to different dimensions of specializations.
PEO 3	Learning Environment: To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	Professionalism: To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	Engineering knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO 4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	The engineer and society: Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Electrical Engineering) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-265011, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2019-2020

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING ACADEMIC REGULATIONS Bachelor of Technology Programmes

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

Undergraduate programme	(B. Tech.)
Minimum	8 semesters (4 academic years)
Maximum	12 semesters (6 academic years)

3. Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – I attached

6. Attendance

6.1 All activities prescribed under these regulations and enlisted by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student regarding attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Principal.

6.2 Student's attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, the continuous assessment will be conducted by the respective department /institute.
- 7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these.
- 7.1.3 The weightages of continuous assessment and End-semester university examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.
- 7.1.4 The performance of candidate in continuous assessment and in end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.
- 7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examination

- 7.2.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per course	Minimum marks Overall per course
40%	45%

- 7.2.2 If a candidate obtains minimum required marks in each course but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

- 8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	<80 ≥73	<73 ≥66	<66 ≥60	<60 ≥55	<55 ≥50	<50 ≥45	<45
Corresponding Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his/her performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9 Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
Distinction	$CGPA \geq 7.5$ & ≤ 10.0
First Class	$CGPA \geq 6.0$ & < 7.5
Second Class	$CGPA \geq 5.0$ & < 6.0
Pass class	$CGPA < 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1 Maximum duration allowed for Completion of a programme

- Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10 Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

II Transcript:

- The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical/Tutorial	Total		Internal	External	Internal	External	
FY Sem-I	HS101.01A- HS110.01A	A Course on Liberal Arts	0	2	2	2			30	70	100
	MA143	Engineering Mathematics I	4	1	5	4	30	70	0	0	100
	ME146	Engineering Graphics	2	2	4	3	30	70	25	25	150
	PY141.01	Engineering Physics	3	2	5	4	30	70	25	25	150
	CL143	Engineering Mechanics	3	2	5	4	30	70	25	25	150
	ME142	Workshop Practices	0	2	2	1			25	25	50
			12	11	23	18	120	280	130	170	700
FY Sem-II	HS126.01A	Communication Skills -1	2		2	2	30	70			100
	MA144	Engineering Mathematics II	4	1	5	4	30	70			100
	CL142.01	Environmental Science	0	2	2	2			30	70	100
	IT143	Fundamentals of Computer Programming	2	2	4	3	30	70	25	25	150
	ME147	Basics of Civil & Mechanical Engineering	3	2	5	4	30	70	25	25	150
	EE145	Basics of Electronics & Electrical Engineering	3	2	5	4	30	70	25	25	150
			14	9	23	19	150	350	105	145	750

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – I)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS101.01 A/B/C/D/E/F/G/H: PAINTING

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				Total
				Contact Hours/Week	Theory		Practical		
					Internal	External	Internal	External	
I	HS101.01 A/B/C/D/E/F/G/H	Painting	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Painting <i>An Introduction to Painting</i> <i>Principles of Composition</i> <i>Medium and Techniques of Painting</i> <i>History of Painting: Folk Indian Painting / Western Painting</i> <i>2D and 3D Painting</i>	2
2	Drawing from Nature and Object <i>Objects of Drawing: Nature and Manmade / Artificial Objects</i> <i>Drawing Still / Live Objects</i> <i>Drawing from Memory</i> <i>Drawing from Life</i>	4
3	Colour Design and Colour Value Color Theory: <i>Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)</i> Color Contrast & Attributes: <i>Interaction, harmony, psychology/mood, culture & expression</i> Media Characteristics & Surfaces: <i>Acrylic, oil, paper, wood & canvas (primed/unprimed)</i>	6

4	Composition and Perspective • Composition: <i>Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting</i> • <i>Perspective: An approximate reproduction</i> • Types of Perspectives: <i>Linear Perspective, One-point Perspective, Twopoint Perspective, Three-point Perspective, Four-Point Perspective</i>	6
5	Figure Drawing and Proportion • <i>Proportions of the Human Body</i> • <i>Three views – Anterior (front), Lateral (side) and Posterior (back)</i> • <i>Fundamental Proportion – The Big Three</i>	4
6	Sketching • <i>Sketching and Freehand</i> • <i>Sketching Techniques</i> • <i>Sketch and Drawing Medium</i>	4
7	Contemporary Issues in Painting • <i>Contemporary Indian Art</i> • <i>Pioneers of Contemporary Indian Art</i> • <i>Contemporary Issues in Painting</i>	4
Total Hours		30

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05

3	Project	-	15	15
4	Attendance	-	05	05
	Total			30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
	Total			70

V. Course Outcome (COs):

At the end of the course, a student will

After completion of the course, the student would:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS102.01 A/B/C/D/E/F/G/H: PHOTOGRAPHY

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact	Theory		Practical		Total
				Hours/Week	Internal	External	Internal	External	
I	HS102.01 A/B/C/D/E/F/G/H	Photography	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Photography • <i>Art, Design and Visualization</i> • <i>Basics of Photography and Various Types of Photography</i> • <i>Basics of Post Production</i> • <i>A Brief History of Photography: Early Experiments and Later Developments</i>	03
2	Camera and Operating System • <i>Role of Camera in the Photography</i> • <i>Types of Camera</i> <i>Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera</i> • <i>Principal Parts of Photographic Camera</i> <i>Lens, Aperture, Shutters, various types and their functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer</i> • <i>Types of Lenses</i> <i>Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses</i> • <i>Different Models of Camera, their Features and Operating Systems</i>	05

	• <i>Camera and Size of the Image, Speed and Power of Lens</i>	
3	Light and Shade • <i>Reflection and refraction of light</i> • <i>Dispersion of light through a glass prism, lenses</i> • <i>Colour Filters:</i> <i>Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter</i> • <i>Photographic Light Sources:</i> <i>Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions</i> • <i>Artificial light sources:</i> <i>Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands</i> • <i>Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table</i>	10
4	Composition • <i>Different kinds of image formations</i> • <i>Principal focus and focal length of the lens</i> • <i>Depth of field, angle of view and perspective</i> • <i>Perspective and composition</i> • <i>Rules of composition</i>	09
5	Contemporary Issues in Photography • <i>Present Day Photography</i> • <i>Contemporary Photographers and their Contributions</i> • <i>Major Issues in Contemporary Photography</i>	03
Total Hours		30

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

V. Course Outcomes (COs):

- After completion of the course, the student would:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the

	mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	2	-	2	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS105.01 A/B/C/D/E/F/G/H : MEDIA AND GRAPHIC DESIGN**

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
I	HS105.01 A/B/C/D/E/F/G/H	Media and Graphic Design	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
------------	-------------	----------------------------

1	An Introduction to Media and Graphic Design • <i>Creating Art, Art in Context and Art as Inquiry</i> • <i>History of Graphic Design</i> • <i>Constructional, Representational, and Simplification Drawing</i>	03
2	Layout and Design • <i>Layout, Design and Aesthetics □ Elements of Design □ Principles of Design:</i> <i>Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion</i> • <i>Impact/function of Design</i> • <i>Indigenous design practices</i> • <i>Role of design in the changing social scenario</i>	07
3	Form and Space • <i>Types of Forms: Man-made, Nature</i> • <i>Types of Space: Negative and Positive</i> • <i>Composition of Form and Space to create Layout □ Exploring Creativity</i>	06
4	Computer Graphics • <i>An Introduction to Graphic Software</i> • <i>Flash, Coreldraw, Illustrator and Photoshop □ Pre-press Process</i>	04
5	Fonts • <i>Construction of Type</i> • <i>Anatomy of Type</i> • <i>Visual Language</i> • <i>Creating Logo and Symbol</i>	04
6	Basic Print Media • <i>An Introduction to Press and its Development Phases</i> • <i>Types of Press</i> • <i>Types of Printing Technologies</i> • <i>Post-press Processes</i>	03
7	Contemporary Issues in Graphic Design • <i>Present Day Graphic Designs</i> • <i>Contemporary Designers and their Contribution</i> • <i>Major Contemporary Issues in Graphic Design</i>	03
Total Hours		30

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

V. Course Outcomes (COs):

After completion of the course, the student would:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), No correlation -

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS109.01 A/B/C/D/E/F/G/H || DRAMATICS

I. Credits and Schemes

Sem.	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours / Week	Internal	External	Internal	External	
I	HS109.01 A/B/C/D /E/F/G/ H	Dramatics	02	02	--	--	30	70	100

II. Course Outline

Module	TITLE / TOPIC	Classroom Contact Sessions
1	Introduction to Drama • Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama	06
2	History of Drama and Contemporary Theatre • Important world dramatists & drama—from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama	06
3	Theatre Design and Techniques • Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance	06

4	Technicalities of Stage Performance • Selection of plot and character • Improvisation • Movement • Voice, Speech, Imagination • Character Development • Scene Enactment	08
5	Contemporary Trends in Drama • New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education	04

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation:

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

Course Outcome (COs):

After completion of the course, the student would:

CO1	be aware about the concept of performing art and its nuances.
-----	---

CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	2	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS110.01 A/B/C/D/E/F/G/H || CONTEMPORARY DANCE

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS110.01 A/B/C/D/E/F /G/H	Contemporary Dance	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Sessions
1.	Introduction to dance - Dance as a Performing Art - Dance as a Medium of Expression - History and Development of Dance	4
2.	Types of Dance - Western dance and classical dance - Salsa, rumba, hip hop, tap dance, belly dance, etc. - Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc. - Other Regional dance forms in India	6
3.	Basic Elements of Dance - Movements of different parts of a body for Expression - Concepts of: Nritya, Laya and Taal	4
4.	Technical Skills in Professional Contemporary Dance - Dance technique: alignment, balance, co-ordination, flexibility and control - Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. - Skills and processes of rehearsal and production: physical energy, stamina and athleticism - Musicality: clarity of timing and phrasing	6

5.	Contemporary Trends in Dance : • Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance	10
Total Hours		30

III. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

IV. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
			Total	30

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sr. No	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

V. Course Outcome (COs):

After completion of the course, the student would:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	-	-	1	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

**FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES**

MA143: ENGINEERING MATHEMATICS – I

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	1	5	4
Marks	100	0	-	100	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	Higher order derivatives and applications	16
2.	Complex numbers and Roots of polynomial Equations	14
3.	Matrix Algebra –I	12
4.	Partial differentiations	08
5.	Applications of Partial differentiations	10
	Total Hrs	60

C. Detailed Syllabus:

		Hours
1	Higher order derivatives and applications:	16
1.1	Set theory and Function	
1.2	Limit, Continuity, Differentiability for function of single variable and its uses. Mean Value Theorem, Local Maxima and Minima	
1.3	Successive differentiation: n^{th} derivative of elementary functions: rational, logarithmic, trigonometric, exponential and hyperbolic etc.	
1.4	Leibnitz rule for the n^{th} order derivatives of product of two functions	
		Hours
1.5	Tests of convergence of series viz., comparison test, ratio test, root test, Leibnitz test. Power series expansion of a function: Maclaurin's and Taylor's series expansion (one and two variables),	

1.6	L'Hospital's rule and related applications, Indeterminate forms	
2	Complex numbers and Roots of polynomial Equations:	14
2.1	Complex numbers and their geometric representation	
2.2	Complex numbers in polar and exponential forms	
2.3	De Moivre's theorem and its applications	
2.4	Exponential, Logarithmic, Trigonometric and hyperbolic functions.	
2.5	Statement of fundamental theorem of Algebra, Analytical solution of cubic equation by Cardan's method	
2.6	Analytic solution of Biquadratic equations by Ferrari's method with their applications.	
3.	Matrix Algebra- I:	12
3.1	Definition of Matrix, types of matrices and their properties	
3.2	Determinant and their properties	
3.3	Rank and nullity of a matrix	
3.4	Determination of rank	
3.5	The inverse of a matrix by Gauss Jordan method.	
3.6	Solution of a system of linear equations by Gauss elimination and Gauss Jordan Methods.	
4.	Partial differentiations:	08
4.1	Partial derivative and geometrical interpretation	
4.2	Euler's theorem with corollaries and their applications	
4.3	Chain rule	
4.4	Implicit functions	
4.5	Total differentials	
5.	Applications of Partial differentiations:	10
5.1	Tangent plane and normal line to a surface	
5.2	Maxima and Minima	
5.3	Lagrange's method of multiplier	
5.4	Jacobian	
5.5	Errors and approximations	

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quiz (surprise test) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	develop skill of successive differentiation, utilize appropriate theory and computational techniques to construct Taylor's series with its interval of convergence for using in a variety of applications such as approximating values, creating series representation and behaviour of a functions, use L'Hospital's rule to compute limits of the indeterminate forms.
CO2	perform basic mathematical operations with complex numbers in Cartesian and polar forms, know methods of finding the n^{th} roots of a complex number and solutions of simple polynomial equations, work with functions of complex variable.
CO3	find determinant and inverse of a square matrix, evaluate rank and nullity of a matrix, solve system of linear equations by using concept of matrices which are useful in various fields of engineering.
CO4	evaluate partial derivatives including higher order derivatives, solve problems using the chain rules, Euler's theorem with corollaries, implicit function and total differentials.
CO5	expand any function of two variables in ascending power of variables, solve problems using the techniques of multivariable calculus in various branches of engineering.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	--	-	-	-	2	-
CO2	3	1	-	-	-	-	-	-	--	-	-	-	2	-
CO3	3	2	1	-	2	-	-	-	--	-	-	-	3	1
CO4	3	-	-	-	1	-	-	-	--	-	-	-	2	-
CO5	3	2	1	1	1	-	-	-	--	-	-	-	3	1

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Recommended Study Material:

a. Text Books:

- Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
- H. K. Dass and Rajnish Verma, Higher Engineering Mathematics, S Chand & Co Pvt Ltd. 2012.

b. Reference Books:

- i. M. D. Weir et al: Thomas' Calculus, 11th Ed., Pearson Education, 2008.
- ii. Stewart James: Calculus Early Transcendental, 5th Ed., Thomson India, 2007
- iii. Wylie, C. R., and L. C. Barrett. Advanced Engineering Mathematics. 1982., McGrawHill Book Company.
- iv. Greenberg, Michael D. Advanced Engineering Mathematics. Prentice-Hall, 1988.

c. URL Links:

- <https://ocw.mit.edu/ans7870/resources/Strang/Edited/Calculus/Calculus.pdf>
- <http://nptel.ac.in/courses/111107108/>
- <http://nptel.ac.in/courses/122101003/> □ <http://nptel.ac.in/courses/111104085/>

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME146: ENGINEERING GRAPHICS

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	1	5	3
Marks	100	50	00	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of Engineering Graphics	05
2.	Projections of Points and Lines	03
3.	Projections of Planes	03
4.	Projections & Section of Solid	04
5.	Orthographic Projection*	05
6.	Isometric Projections*	04
7.	Computer Graphics*	03
8.	Development of Lateral Surfaces	03

* indicates the topics are discussed in laboratory

Total hours (Theory): 30

Total hours (Practical): 30

Total hours: 60

C. Detailed Syllabus:

1. **Fundamentals of Engineering Drawing 05 Hours 17%**
 - 1.1 Importance of engineering drawing, drawing instruments and materials, BIS and ISO
 - 1.2 Different types of lines used in engineering practice, methods of projections as per SP 46-1988.
 - 1.3 Engineering scale.
 - 1.4 Engineering curve.

2. Projections of Points and Lines	03 Hours	10%
2.1 Introduction to methods of projections		
2.2 Projections of lines inclined to both the planes		
3. Projections of Planes	03 Hours	10%
3.1 Projection of plane		
3.2 Auxiliary projection method		
4. Projections & Section of Solid	04 Hours	13%
4.1 Projection of solids		
4.2 Sectional view		
4.3 True shape of Sections		
4.4 Auxiliary Inclined Plane (AIP), Auxiliary Vertical Plane (AVP)		
5. Orthographic Projection	05 Hours	17%
5.1 Principle projection		
5.2 Methods of first and third angle projection with examples / problems		
6. Isometric Projections	04 Hours	13%
6.1 Terminology, Isometric scale		
6.2 Isometric view and Isometric projection with examples / problems		
7. Computer Graphics	03 Hours	10%
7.1 Introduce computer graphics		
7.2 Importance of computer graphics		
7.3 Demonstration of CAD Modeling software		
7.4 Training of Fusion 360 software		
8. Development of Lateral Surfaces	03 Hours	10%
8.1 Method of Development		
8.2 Developments of cylinder, cone, prism, pyramid		

D. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Drawing sheets/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory, discipline and behavior will be observed strictly.

E. Course Outcomes (COs):

At the end of the course, the students will be able to:

CO1	Realize and interpret various engineering drawings.
CO2	Understand the concept and application of engineering scale and engineering curve.
CO3	Visualize three-dimensional drawing of engineering components through orthographic, sectional orthographic and isometric drawing.
CO4	Explore the basic principle of projections and development of lateral surface of solid.
CO5	Use the computer for geometric modeling.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Text Books:

1. N. D. Bhatt & V. M. Panchal, “Engineering Drawing”, Charotar Publishing House Pvt. Ltd.
2. P. J. Shah, “Engineering Graphics”, S. Chand Publishing & Co.

Reference Books:

1. P.B. Patel & P.D. Patel, "Engineering Graphics", Mahajan Publishing House.
2. Arunoday Kumar, "Engineering Graphics", Tech-Max Publication.
3. Gopal Krishna K.L., "Engineering Drawing", Subhas Publications
4. Venugopal, K., "Engineering Drawing made Easy", Wiley Eastern Ltd.
5. M.L. Agrawal & R.K. Garg, "Engineering Drawing", Vol. I, Dhanpatrai & Co.
6. T.E. French, C.J. Vierck & R. J. Foster, "Graphic Science and Design", McGraw Hill.
7. W. J. Luzadder & J. M. Duff, "Fundamentals of Engineering Drawing", Prentice Hall.
8. K. Venugopal, "Engineering Drawing and Graphics", New Age international Pry. Ltd.

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in/courses/112103019/>
2. <http://nptel.ac.in/downloads/112105125/>
3. <http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101>
4. <http://nptel.ac.in/courses/105107122/>
5. <https://law.resource.org/pub/in/bis/S01/is.sp.46.2003.pdf>

**FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICS**

PY141.01: ENGINEERING PHYSICS

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Error Analysis	05
2.	Introduction to Quantum Mechanics	08
3.	Acoustics and Ultrasonics	08
4.	Physics of Laser and its applications	09
5.	Condensed Matter and Material Physics	10
6.	Nanoscience	05

Total hours (Theory): 45

Total hours (Practical): 30

Total hours: 75

C. Detailed Syllabus:

1. Error Analysis: 05 hours 11%

1.1 Uncertainties in Measurements: Sources and estimation of errors, accuracy and precision, systematic error, random error, Significant figure and round off

1.2 Uncertainties, Parent and Sample Distributions, Mean and Standard Deviation of Distributions

1.3 Average error, r.m.s error, probable error and error propagation		
1.4 Introduction to regression analysis and its uses		
1.5 Applications with numericals		
2. Introduction to Quantum Mechanics:	08 hours	18%
2.1 Origin of quantum mechanics		
2.2 Dual nature of matter, concept of wave group, Davisson and Germer experiment		
2.3 Heisenberg's uncertainty principle and its applications, Schrodinger's wave equation		
2.4 Interpretation of wave equation and wave function		
2.5 Applications of Schrodinger's wave equation		
3. Acoustics and Ultrasonics:	08 hours	18%
3.1 Introduction to waves		
3.2 Characteristics of Sound waves		
3.3 Architectural Acoustics, Absorption Coefficient, Reverberation, Sabine's formula, Factors affecting acoustics of buildings and their remedies		
3.4 Ultrasonics: Introduction and properties		
3.5 Production: piezoelectric and magnetostriction method		
3.6 Detection of Ultrasonic waves		
3.7 Applications with numericals		
4. Physics of Laser and its applications:	09 hours	20%
4.1 Lasers and its properties, Spontaneous and stimulated emission		
4.2 Einstein coefficients		
4.3 Gas laser (He-Ne Laser), Semiconductor Laser, Applications of Laser		
4.4 Basic Principle of Holography, Construction and reconstruction of hologram, Applications of Holography		
4.5 Introduction to optical fibre, Numerical Aperture of optical fibre		
4.6 Types of optical fibre, applications of optical fibre		
5. Condensed Matter and Material Physics:	10 hours	22%
5.1 Basics of Crystal structure		
5.2 X – Ray: properties, production, applications of X – Rays		
5.3 Conductors, Insulators and Semiconductors: Band theory of Solids		
5.4 Energy gap, Fermi energy, Electrical conductivity and mobility		
5.5 Hall effect		
5.6 Introduction to magnetic materials		

5.7 Basics of superconductivity and its applications

6. Nanoscience: 05 hours 11%

6.1 Introduction to nanomaterials

6.2 0D, 1D, 2D nanostructures

6.3 Bottom-up and Top-down approach to synthesis of nanomaterials

6.4 Structural characterization by XRD and SEM

6.5 Applications of nanomaterials

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 10 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	The student would be able to apply the concepts of physics in various branches of engineering.
CO2	An ability to identify, formulate and solve engineering problems.
CO3	An ability to use the techniques, skills and modern tools of physics necessary for engineering applications.
CO4	An ability to conduct experiments, analyse and interpret data.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2	-	-	-	-	-	-	-	-
CO2	3	3	3	1	2	-	-	-	-	-	-	-

CO3	3	3	-	3	3	-	-	-	-	-	-	-
CO4	3	3	2	3	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

a. Text Books:

- i. Vijayakumari, G., Engg. Physics, Vikas Publishing house Pvt. Ltd.
- ii. Rajagopal, K., Engg. Physics, Prentice Hall of India Pvt. Ltd.
- iii. Avadhalula, M. N. & Kshirsagar, P. G., A text book of Engg. Physics, S. Chand Pub.

b. Reference Books:

- i. Nayak Abhijit, Engg. Physics, S. K. Kataria and Sons Pub.
- ii. Topping, J., Errors of Observations and their Treatment, 3rd Ed. Chapman and Hall Ltd. London
- iii. Kittle, C., Solid State Physics
- iv. Resnick and Haliday, Physics Part-I & II, Wiley Eastern publication
- v. Beiser Arthur, Concept of Modern Physics
- vi. Ghatak, Optics, Tata McGraw Hill, 3rd Edition
- vii. Pillai, S.O., Solid State Physics, Wiley Eastern Ltd.

c. Web Materials:

1. http://www.nptel.iitm.ac.in/courses/Webcourse-contents/IIT%20Guwahati/engg_physics/index_cont.htm
2. http://ncert.nic.in/html/learning_basket.htm
3. <http://science.howstuffworks.com/laser1.htm>
4. <http://physics-animations.com/Physics/English/optics.htm>
5. <http://physics-animations.com/Physics/English/waves.htm>
6. <http://www.epsrc.ac.uk>
7. <http://www.pitt.edu/~poole/physics.html#light>
8. <http://de.physnet.net/PhysNet/optics.html>

**FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI SHIVABHAI PATEL DEPARTMENT OF CIVIL
ENGINEERING**

CL143 ENGINEERING MECHANICS

A. Credits and Hours:

Teaching Scheme	Theory	Tutorial	Practical	Total	Credit
Hours/week	3	1	2	6	4
Marks	100	-	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	04
2	Fundamental of Statics	20
3	Friction	05
4	Truss	05
5	Centroid and Centre of Gravity	06
6	Fundamentals of Kinematics and Kinetics of Particles	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

- 1 Introduction** **04 hrs 08%**
Introduction of Mechanics, Fundamental concepts and idealization of mechanics, Fundamental principles & Laws of mechanics, Scalar and Vector Quantities,
Components, unit vectors and position vector, Composition and resolution of vector, System of Units

- 2 Fundamental of Statics** **20 Hrs 45%**
- 2.1 Coplanar Concurrent Force system**
Introduction of Force, Effect of force and Characteristics of force, Types of force, Type of force systems, Principle of Transmissibility, Resultant of force systems, Resolution of a single force, Composition and Resolution of force system, Resolution method for coplanar concurrent force system
- 2.2 Moments and Couples**
Moment of a force, Principle of moments, Coplanar applications, Parallel force system, Couples, Equivalent couples, Operations with couples, Equivalent system of forces
- 2.3 Coplanar Non-Concurrent Force system**
Introduction, Resultant of coplanar non-concurrent force system, Concentrated and distributed loads
- 2.4 Equilibrium of Rigid bodies**
Equilibrium, Resultant & Equilibrant, Principle of action and reaction, Free body diagram & Lami's theorem, Tensions of strings, condition of equilibrium for Coplanar concurrent forces & Coplanar non-concurrent forces, Equilibrium of Coplanar concurrent forces.
Equilibrium of Coplanar non-concurrent forces
- 3 Friction** **5 Hours 11%**
Friction and its applications, Types of friction and Laws of dry friction, Angle of friction, Angle of repose, Coefficient of friction, Block Friction, Ladder friction, Wedge friction
- 4 Truss** **5 Hours 11%**
Classification of Truss, Perfect & Imperfect truss, Analysis of pin jointed perfect truss using method of joints and Method of section.
- 5 Centroid and Centre of Gravity** **6 Hours 14%**
Introduction, basic definitions and their understanding, Concept of centre of gravity, Centroids of Linear elements & Planar elements, Determination of centroids by integrations, Centroids of Composite sections (1D & 2D).

**6 Fundamentals of Kinematics and Kinetics of Particles 5 Hours
11%**

Rectilinear motion, Curvilinear motion, Motion of rigid bodies, Velocity and acceleration, Newton's law of motion, Energy and momentum.

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Two internal exams will be conducted as a part of continuous theory evaluation.
- Assignments/ Tutorial based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar/ presentation/ model making will be carried out as a part of continuous theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments related to course content will be carried out in the laboratory.

E. Course Outcomes (COs):

On the completion of the course one should be able to:

CO1	Understand laws of mechanics and their application to engineering problems.
CO2	Understand the fundamentals of statics and apply them to simple structural problems
CO3	Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple practical problems.
CO4	Compute centroid of composite sections.
CO5	Recognize different types of friction and its effects on an object.
CO6	Classify different types of truss and analyze the pin jointed perfect truss.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	1	-	-	-	-	-	-	-	1	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	1	-
CO4	2	1	1	-	-	-	-	-	-	-	-	-	1	-
CO5	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO6	1	1	2	-	1	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

- Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I & II, Charotar Publishing House
- Junnarkar, S. B. & Shah, H. J., Applied Mechanics, Charotar Publishing House
- Beer and Johnston, Engineering Mechanics (Statics & Dynamics)

Reference Books:

- Beer and Johnston, Mechanics of Materials
- Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
- Hibbler, R.C., Engineering Mechanics, Pearson Education
- Popov, E.P., Engineering Mechanics of Solids, Prentice Hall of India, New Delhi
- Meriam, J. L. & Kraige, L. G., Engineering Mechanics Statics, John Wiley & Son, Singapore
- A K Tayal, Engineering Mechanics (Statics & Dynamics), Umesh Publications

Web Materials:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Mechanics%20Of%20Solids/index.htm>
- <http://nptel.iitm.ac.in/video.php?subjectId=105106116>

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME142: WORKSHOP PRACTICES

A. Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	2	-	2	1
Marks	-	50	-	50	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to workshop facility.	02
2.	Carpentry shop	06
3.	Fitting shop.	08
4.	Different Metal Joining Processes.	04
5.	Smithy shop.	02
6.	Sheet metal working.	02
7.	Plumbing shop.	02
8.	Introduction to machine tools.	02
9.	Injection molding process.	02

Total hours (Lab): 30

Total hours: 30

C. Detailed Syllabus:

1. Introduction to workshop facility.	02 hours	07%
1.1 Familiarization with workshop facility.		
1.2 Introduction to different shops of the workshop.		
2. Carpentry Shop.	06 hours	20%
2.1 Introduction to different tools of carpentry shop.		

2.2 Making of drawing of the job to be made.

2.3 Making of finished job as per drawing out of the given raw material of wood.

Identification on the job for traceability.

3. Fitting Shop. **08 hours** **27%**

3.1 Introduction to different tools of fitting shop.

3.2 Making of drawing of the job to be made.

3.3 Making of finished job as per drawing out of the given raw material. Identification on the job for traceability.

4. Different Metal Joining Processes. **04 hours** **14%**

4.1 Introduction to different tools of welding shop.

4.2 Making of drawing of the job to be made.

4.3 Making or demonstration of finished job as per drawing.

4.4 Introduction to Soldering and brazing of metal joining process.

4.5 Joining of two metal sheet or plate by Riveting.

4.6 Making of drawing of the job to be made by riveting.

4.7 Making or demonstration of finished job as per drawing.

5. Smithy Shop. **02 hours** **07%**

5.1 Introduction to different tools of smithy shop.

5.2 Making of drawing of the job to be made for Cold smithy.

5.3 Making or demonstration of finished job as per drawing.

5.4 Making of drawing of the job to be made for Hot smithy.

5.5 Making or demonstration of finished job as per drawing.

6. Sheet Metal Working. **02 hours** **06%**

6.1 Introduction to different tools of sheet metal working shop.

6.2 Making of drawing of the job to be made from sheet metal.

6.3 Making or demonstration of finished job as per drawing.

7. Plumbing Shop. **02 hours** **06%**

7.1 Introduction to all plumbing tools.

7.2 Demonstration of plumbing on the piping model.

8. Introduction to Machine Tools. **02 hours** **07%**

8.1 Detailed introduction to Lathe machine, Shaping machine, Drilling machine, Grinding machine, Milling machine, Bending machine, Mechanical press.

9. Injection molding process.

02 hours 06%

9.1 Introduction and demonstration to Injection Molding Process for making job out of plastic material.

D. Instructional Method and Pedagogy:

- Attendance is compulsory in laboratory session.
- Journal writing based on above course content and practical work in form of performance practical's by preparing job at the workshop floor.
- In the laboratory discipline and behavior will be observed strictly.
- All the students must follow code of conduct during working at the shop floor.
- Journal should be submitted to the respective course teacher within the given time limit.

E. Course Outcome (COs):

On the completion of the course one should be able to:

1. Recognize essential tools and process of carpentry.
2. Understand the joining process like welding, soldering and brazing.
3. Recognize essential tools and process of fitting.
4. Recognize various forging and forming processes with the aid of smithy process.
5. Understand different types of sheet metal joints which are useful in working.
6. Recognize different types of fittings and plumbing tools.
7. Identify different machines in workshops along with use.
8. Recognize plastic molding processes.

Course Articulation Matrix:

	P O1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	-	-	-	-	-	-	-	-	-	-

CO6	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO7	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO8	1	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

1. Anderson, James, and Earl E. Tatro. "Shop Theory. 5th ed." (1968).
2. Bawa, H. S. Workshop Technology. India: Tata McGraw-Hill, (1995).
3. Choudhury, Hajra. "Elements of Workshop Technology, Vol. I & II." Media Promoters Pvt Ltd (2009).
4. Raghuwanshi, B. S. Course in Workshop Technology. Dhanpat Rai and Company (P) Limited, (2009).

Reference Books:

1. Chapman, W. A. J. "Workshop Technology Part 1-3." (1998).
2. Tejwani V.K., “Basic Machine Shop Practical Vol. I, II”, Tata McGraw Hill Pub. Co., New Delhi, (1989).
3. Arora B.D., “Workshop Technology Vol. I, II”, Satya Prakashan, New Delhi, (1981).

Other materials:

Workshop book and manual.

Web sites: <http://nptel.ac.in/courses/112107145/>

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – II)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS126.01 (A/B/C/D/E/F/G/H): Communication Skills – I

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS126.01	Communication Skills – I	02	02	30	70	--	--	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	Communicative English • Time and Tenses • Active and Passive Voice • Direct-Indirect Speeches • Prepositions and Conditionals	04
2	Functional English • Introduction to Functional English • Describing Actions and Processes, Offering, Requests, • Routines/Timetable, Making Comparisons, Sharing Interests and Experience	04
3	Conversational Skills • An Introduction to Conversations for various purposes • Importance of acquiring Conversational Skills • Models, Techniques and Types of Conversations	04
4	Introduction to Communication and Key Concepts in Communication An Introduction to Communication <i>Basic Terms, Concepts, and Contexts of Communication</i> <i>Factors influencing message encoding, the nature of message,</i>	06

	<i>and message uses and effects.</i> <i>Importance, Types and Principles of Communication</i>	
5	Effective Listening and Reading Skills An Introduction to Listening and Reading <i>Purposes, Types and Techniques of Listening and Reading</i> <i>Barriers to effective Listening & Reading and overcoming the Barriers</i> <i>Note-taking and Note-making</i>	06
6	Writing Skills An Introduction to Writing Importance of effective Writing <i>Paragraph Development: Coherence –Topic Sentence, Supporting Sentence & Data etc.</i> Business Letter Writing	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on pragmatic learning. Classroom Teaching will be facilitated by Reading Material, Classroom Discussions, Task-based learning, projects, assignments and various interpersonal activities like case-studies, critical reading, group-work/pair-work, and presentations.

IV. Evaluation:

Students will be evaluated continuously in the form of internal as well as external examinations. Evaluation (Theory) is schemed as 30 marks for internal evaluation and 70marks for external evaluation in the form of University examination.

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks Per incidence	Total Marks
1	Assignment / Project Work	2	25	25
2	Attendance and Class Participation			05
Total				30

External Evaluation

The University Theory examination will be of 70 marks and will test the reasoning, logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications. There will be at least one question on case analysis relevant to the components of the course.

V. Course Outcome (COs):

After completion of the course, the student would :

CO1	have polished their grammar and developed the ability to communicate effectively in business situations,
CO2	be able to communicate message accurately, handle situation that require thoughtful communication, to use appropriate words and tones and so on.
CO3	be able to use the language effectively for various functions
CO4	develop familiarity with varied styles of communication and gain insights into how to deal with people with different communication styles
CO5	hone basic linguistic and communication skills (of students) required in a business organization, namely: Listening, Speaking, Reading and Writing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Reference Books:-

Sanjay Kumar and PushpLata (FirstEdition, 2011), *Communication Skills*, Oxford University Press, New Delhi
 Krishna Mohan and MeenaBanerji (2010), *Developing Communication Skills*, Macmillan Publications India Ltd., New Delhi
 M V Rodriques (2013), *Effective Business Communication*, Concept Publishing Company (P) Ltd., New Delhi

Mohanand Meenakshi Raman(2006), *Effective English Communication Krishna*, McGraw-Hill Publishing Company Limited, New Delhi
Geoffrey Leech & Jan Swartvik (1994), *A Communicative Grammar of English*, Longman Publications, New York
Jones Leo (1979), *Functions of English*, Cambridge University Press, UK

Reference Reading:-

- <http://www.communicationskills.co.in/index.html>
- <http://www.hodu.com/default.htm>
- <http://www.bbc.co.uk/worldservice/learningenglish>
- <http://www.englishlearner.com/tests/test.html>
- <http://www.englishclub.com/vocabulary/idioms-body.htm>
- <http://dictionary.cambridge.org>

FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA144: ENGINEERING MATHEMATICS –II

A. Credits and Hours:

Teaching Scheme	Theory	Tutorial	Total	Credit
Hours/week	4	1	5	4
Marks	100	-	100	

B. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	First order and First degree Ordinary Differential Equations	08
2	Higher Order Ordinary Linear Differential Equations	12
3	Partial Differential Equations and Applications	10
4	Matrix Algebra –II	10
5	Improper and Multiple Integrals	10
6	Probability and Statistics	10
	Total hours	60

C. Detailed Syllabus:

		Hours
1.	First order and First degree Ordinary Differential Equations:	08
1.1	Modeling of real world problems in terms of first order ODE	
1.2	Concept of general and particular solutions	
1.3	Initial and boundary value problems	
1.4	Existence and Uniqueness of solutions by illustrations	
1.5	Solutions of first order and first degree differential equations	
1.6	Linear, Bernoulli, Exact and non-exact differential equations	
2.	Higher Order Ordinary Linear Differential Equations:	12
2.1	Model of real world problems of higher order LDE	

2.2	General Solution of Higher Order Ordinary Linear Differential Equations with Constant coefficients	
2.3	Methods for finding particular integrals viz. variation of parameters and undetermined coefficients	
2.4	LDE of higher order with variable coefficients: Legendre's Equations (Special case: Cauchy-Euler equation)	
2.5	System of simultaneous first order linear differential equations	
3.	Partial Differential Equations and Applications:	10
3.1	Boundary valued conditions	
3.2	Methods of solutions of first order PDE	
3.3	Lagrange's Linear Partial Differential Equations.	
3.4	Special types of Nonlinear PDE of the first order	
3.5	Solutions of Heat, Wave and Laplace equations using separation of variables.	
3.6	Modeling of real world problem in terms of PDE	
4.	Matrix Algebra –II:	10
4.1	Revision of matrices, determinant.	
4.2	Eigen values and Eigen vectors of matrices	
4.3	Eigen values and Eigen vectors of special matrices	
4.4	Cayley-Hamilton's Theorem and its applications.	
4.5	LU decomposition	
5.	Improper and Multiple Integrals:	10
5.1	Improper integrals and their convergence	
5.2	Definitions, properties and examples of Beta, Gamma and error functions	
5.3	Evaluation of double and triple integrals	
5.4	Change of order of double integration	
5.5	Transformation to polar and cylindrical coordinates	
5.6	Applications of double and triple integrals	
6.	Probability and Statistics:	10
6.1	Mean, median, mode and standard deviation	
6.2	Combinatorial probability	
6.3	Joint and Conditional probability and Bayes theorem	

6.4	Random variables, probability distribution functions - Binomial, Poisson, exponential and normal.	
-----	---	--

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/tutorials which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quiz (surprise test) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

E. Student Learning Outcomes (Course Outcome):

At the end of the course, the students will be able to

CO1	formulate models of natural phenomena using differential equations and find its solution using standard methods.
CO2	identify, analyze and subsequently solve physical problems analytically whose behaviour can be described by linear and nonlinear differential equations.
CO3	find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix, construct LU decomposition of a square matrix.
CO4	use advanced techniques to evaluate improper integrals, apply multiple integrals to find area, volume and mass in engineering field.
CO5	recognize the difference between different measure of central tendency, summarize and interpret data.
CO6	understand and solve the problems using probability axioms, rules and Bayes theorem, use distributions such as Binomial, Poisson, Exponential and Normal to solve real world problems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	-	3	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	3	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO5	2	2	-	1	1	-	-	-	-	-	-	-	2	1

CO6	2	2	1	2	1	1	1	-	-	-	-	-	2	1
-----	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. **Recommended Study Material:**

❖ **Text Books:**

1. Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
2. H. K. Dass and Rajnish Verma, Higher Engineering Mathematics, S Chand & Co Pvt. Ltd.
3. Ross, Sheldon. A first course in probability. Pearson, 2014.

❖ **Reference Books:**

1. M. D. Weir *et al*: Thomas' Calculus, 11th Ed., Pearson Education, 2008.
2. Stewart James: Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. Wylie, C. R., and L. C. Barrett. Advanced Engineering Mathematics. 1982, McGraw-Hill Book Company.
4. Greenberg, Michael D. Advanced engineering mathematics. Prentice-Hall, 1988.
5. Hogg, R V, E A Tanis and D L Zimmerman, Probability and Statistical Inference, 9th edition, Prentice Hall, 2015.

❖ **Web Materials:**

1. <http://nptel.ac.in/courses/122107037/>
2. <http://nptel.ac.in/courses/111107108/>
3. <http://nptel.ac.in/courses/122103012/>
4. <http://nptel.ac.in/courses/122104018/>
5. <http://nptel.ac.in/courses/111106100/>
6. <http://nptel.ac.in/courses/122101003/>
7. <https://ocw.mit.edu/courses/mathematics/18-02-multivariable-calculus-fall2007/lecture-notes/>
8. <https://nptel.ac.in/courses/111105041/>

FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI SHIVABHAI PATEL DEPARTMENT OF CIVIL ENGINEERING

CL142.01: ENVIRONMENTAL SCIENCES

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	2
Marks	0	100	100	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction	05
2	Environmental Pollution	12
3	Ecology & Ecosystems	10
4	Natural Resources	03

Total Hours (Theory): 00

Total Hours (Lab): 30

Total Hours: 30

C. Detailed Syllabus:

1 Introduction 05 Hours 24%

- 1.1 Basic definitions
- 1.2 Objectives and guiding principles of environmental studies
- 1.3 Components of environment
- 1.4 Structures of atmosphere
- 1.5 Man-Environment relationship
- 1.6 Impact of technology on the environment

2 Environmental Pollution 12 Hours 33%

- 2.1 Environmental degradation

- 2.2 Pollution, sources of pollution, types of environmental pollution
- 2.3 Air pollution: Definition, sources of air pollution, pollutants, classifications of air pollutants (common like SO_x & NO_x), sources & effects of common air pollutants
- 2.4 Water pollution: Definition, sources water pollution, pollutants & classification of water pollutants, effects of water pollution, eutrophication
- 2.5 Noise pollution: Sources of noise pollution, effects of noise pollution
- 2.6 Ill Effects of Fireworks: Severity of toxicity, environmental effects and health hazards
- 2.7 Current environmental global issues, global warming & green houses, effects, acid rain, depletion of Ozone layer

3 Ecology & Ecosystems **10 Hours 33%**

- 3.1 Ecology: Objectives and classification
- 3.2 Concept of an ecosystem: Structure & function
- 3.3 Components of ecosystem: Producers, consumers, decomposers
- 3.4 Bio-Geo-Chemical cycles & its environmental significance
- 3.5 Energy flow in ecosystem
- 3.6 Food Chains: Types & food webs
- 3.7 Ecological pyramids
- 3.8 Major ecosystems

4 Natural Resources **03 Hours 10%**

- 4.1 Natural resources: Renewable resources, non-renewable resources, destruction versus conservation
- 4.2 Energy resources: Conventional energy sources & its problems, Non-conventional energy sources advantages & its limitations, problems due to overexploitation of energy resources

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Sessions will be conducted with the aid of multi-media projector, black board, OHP, etc.
- Attendance is compulsory in sessions which carry weightage.
- Assignments/ Projects/ Presentation/ Seminars/ Quiz, etc. will be conducted which will be the part of internal evaluation.
- Practical Examination/ Presentation with report/ Case Studies/ MOOCs, etc. will be conducted and which will be the part of the external evaluation.

E. Course Outcomes (COs):

At the end of the course, the students will be able:

CO1	To perceive the elementary knowledge about natural environment and its relation with science.
CO2	To identify and analyze human impacts on the environment.
CO3	To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.
CO4	To Initiate new and innovative environmental friendly practices.
CO5	To communicate on recent environmental problems thereby creating awareness among society.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	3
CO2	-	-	-	-	-	-	3	-	-	-	-	-	-	3
CO3	-	-	-	-	-	2	3	-	-	-	-	-	-	3
CO4	-	-	2	-	-	-	3	-	-	-	-	-	2	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material: Text Books:

1. Varandani, N.S., Basics of Environmental Studies

2. Sharma, J. P., Basics of Environmental Studies

Reference Books:

1. Shah Shefali & Goyal Rupali, Basics of Environmental Studies
2. Agrawal, K.C., Environmental Pollution : Causes, Effects & Control
3. Dameja, S. K., Environmental Engineering & Management
4. Rajagopalan, R., Environmental Studies, Oxford University Press
5. Wright Richard T. & Nebel Bernard J., Environmental Science
6. Botkin Daniel B. & Edward A. Keller, Environmental Science
7. Shah, S.G., Shah, S.G. & Shah, G. N., Basics of Environmental Studies, Superior Publications, Vadodara

Web Materials:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Environmental%20Air%20Pollution/index.htm>
- <http://nptel.iitm.ac.in/video.php?subjectId=105104099>
- http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html
- <http://www.epa.gov>
- <http://www.globalwarming.org.in>
- <http://nopr.niscair.res.in>
- <http://www.indiaenvironmentportal.org.in>

FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY

IT143: FUNDAMENTALS OF COMPUTER PROGRAMMING

A. Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	02	02	04	03
Marks	100	50	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Computation and Algorithm	03
2	Introduction to Programming and 'C' language	02
3	Constants, Variables & Data Types in 'C'	03
4	Operators and Expression in 'C'	03
5	Managing Input & Output Operations	02
6	Conditional Statements ,Branching & Looping	04
7	Arrays	04
8	Character Arrays	03
9	User-Defined Functions in 'C'	03
10	Structures and Basics of pointer	03

Total Hours (Theory): 30

Total Hours (Lab): 30

Total Hours: 60

C. Detailed syllabus

Following contents will be delivered to the students during laboratory sessions.

- | | | | |
|-----|--|-----------------|------------|
| 1 | Introduction to Computation and Algorithm | 03 Hours | 9% |
| 1.1 | Various number systems: Decimal, Binary, Octal, Hexadecimal, conversion from one number system to another | | |
| 1.2 | What is computer, Algorithms, Flow-charts | | |
| 1.3 | Solve Various types of algorithms like Exchanging values of two variables, (using 3 variables & 2 variables), Arrange numbers in ascending order, Evaluate various series e.g.: $\sin x$, $1^2-2^2+3^2-\dots$, $1+2/2!+3/3!+\dots$, | | |
| 2 | Introduction to Programming and 'C' language | 02 Hours | 7% |
| 2.1 | What is program & programming, programming languages and it's types, compiler, and interpreter, History of C, Characteristics of C, Basic structure of a program, Compiling process of C a Program | | |
| 3 | Constants, Variables & Data Types in 'C' | 03 Hours | 9% |
| 3.1 | Character set, C tokens, Keyword, Constants, and Variables | | |
| 3.2 | Data types – declaration & initialization, User-defined type declaration - typedef, enum, Symbolic constant (#define) | | |
| 4 | Operators and Expression in 'C' | 03 Hours | 13% |
| 4.1 | Classification of operators: arithmetic, relational, logical, assignment, increment / decrement, bitwise, special operators. Unary, binary and ternary operators | | |
| 4.2 | Arithmetic expression, evaluation, type conversion: implicit & explicit, precedence and associativity, use of math.h file | | |
| 5 | Managing Input & Output Operations | 02 Hours | 6% |
| 5.1 | Basic input and output operations, Input a character, introduction to ASCII code, various library functions from ctype.h | | |
| 5.2 | Formatted input using scanf(), formatted output of integer and real data using printf () | | |
| 6 | Conditional Statements, Branching & Looping | 04 Hours | 14% |
| 6.1 | Decision making using if, if...else statement, nesting of if...else, else...if Ladder | | |

- 6.2 *switch*, use of if...else instead of conditional operator, *goto* statement
- 6.3 Need of looping, entry-controlled loop: while, for, exit-controlled loop: do...while, difference
- 6.4 Nesting of looping statements, use of break and continue, use of if, if...else in loop
- 7 Arrays 04 Hours 10%**
- 7.1 Need of array, declaration & initialization 1D array, various programs of 1D
- 7.2 2D array and their memory allocation, 2D array basic programs and matrix operations
- 8 Character Arrays 03 Hours 8%**
- 8.1 Difference of character array with numeric array and importance of NULL character
- 8.2 Declaration, initialization and various input and output methods of string.
- 9 User-Defined Functions in 'C' 03 Hours 12%**
- 9.1 Need of modularization, advantages, introduction to user-defined function, form of C functions, function prototype, function call, function body
- 9.2 Call by value, actual & formal arguments, use of return, nesting of functions, recursion, Array as function arguments
- 9.3 storage classes: scope, life of a variable in C
- 10 Structures and Basics of pointer 03 Hour 12%**
- 10.1 Need of user-defined data type, structure definition, declaration and initialization of variables
- 10.2 Background of memory, variable, value, address of variable, introduction to pointer, declaration & initialization, access value using pointer, indirection (*) operator

D. Instructional method and pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed. Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory and tutorial session respectively.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify situations where computational methods would be apply and develop various problem solving skills.
CO2	Define, select, and compare data types, operators, and functions to solve mathematical as well as scientific problems.
CO3	Describe, design and develop modular programs using control structures.
CO4	Illustrate the use of different data structures with practical aspect.
CO5	Validate the logic building and code formulation by designing code capable of passing various test cases.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	1	1	1	-	-	-	-	-	-	-	-	-	1	-
CO3	2	-	-	-	1	-	-	-	-	-	-	-	1	-
CO4	-	-	-	1	-	-	-	-	-	-	-	-	1	1
CO5	2	1	2	-	-	-	-	-	-	-	-	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended study material:

❖ Text book:

1. Programming in ANSI C, E.Balagurusamy, Tata McGraw Hill.

❖ Reference books:

1. C Programming Language (2nd Edition), Brian W. Kernighan, Dennis M. Ritchie, Prentice Hall (PHI)
2. C: The Complete Reference, Herbert Schildt, McGrawHill
3. Let us C: Yashwant Kanetkar, BPB publications new Delhi
4. Computer programming and utilization: M.T.Savaliya, Atul Prakashan
5. Computer concepts and Programming , Vikas Gupta, DreamTech
6. Computer fundamentals and Programming in C, Pradip dey and Manas Ghosh, Oxford.

❖ URL Links:

1. <http://www.tutorials4u.com/c/>
2. <http://www.cprogramming.com/tutorial.html>
3. <http://www.howstuffworks.com/c.htm>
4. <http://www.programmingtutorials.com/c.aspx>
5. http://www.physics.drexel.edu/courses/Comp_Phys/General/C_basics/

FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS MATRUSANSTHA DEPARTMENT OF MECHANICAL
ENGINEERING

ME147: BASICS OF CIVIL AND MECHANICAL ENGINEERING

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction of Mechanical Engineering	06
2.	Steam and Steam Generator	05
3.	Internal Combustion Engines	06
4.	Pumps and Compressors	03
5.	Refrigeration and Air Conditioning Systems	05
6.	Power and Motion Transmission	05
7.	Scope of Civil Engineering	02
8.	Introduction to Surveying	06
9.	Elements of building Construction	07

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. Detailed Syllabus:

PART A:

1. Introduction of Mechanical Engineering 06 Hours 13%

1.1 Prime movers and its types, Sources of energy

1.2 Basic terminology: Force and mass, Pressure, Work, Power, Energy, Heat, Temperature, Units of heat, Specific heat capacity, Interchange of heat, Change of state, Internal energy, Enthalpy, Entropy, Efficiency

- 1.3 Zeroth Law and First Law of Thermodynamic, Boyle's law, Charle's law and Combined gas law, Relation between C_p and C_v
 1.4 Constant volume process, Constant pressure process, Isothermal process, poly-tropic process, adiabatic process
 1.5 Numerical

2. Steam and Steam Generator **05 Hours** **11%**

- 2.1 Introduction to steam formation and its types
 2.2 Introduction to steam table
 2.3 Calorimeter and its types
 2.4 Boiler definition and its classification
 2.5 Cochran boiler, Babcock and Wilcox boiler and its mountings and accessories
 2.6 Efficiencies of boiler

3. Internal Combustion Engines **06 Hours** **13%**

- 3.1 Introduction
 3.2 Basic terminology of I.C. engine
 3.3 Types of I. C. engines
 3.4 Efficiencies of an engine
 3.5 Numerical Practice

4. Pumps and Compressors **03 Hours** **7%**

- 4.1 Introduction
 4.2 Classification and application of pumps and compressors,
 4.3 Working of Reciprocating and Centrifugal Pump

5. Refrigeration and Air Conditioning Systems **05 Hours** **11%**

- 5.1 Introduction to refrigeration and air conditioning
 5.2 Basic terminology, Principal and application of refrigeration
 5.3 Vapour compression refrigeration system
 5.4 Domestic refrigerator
 5.5 Window and split air conditioning systems

6. Power and Motion Transmission **05 Hours** **11%**

- 6.1 Introduction
 6.2 Types of couplings, brakes and clutches.
 6.3 Belt drive and its types
 6.4 Gear drives and its types, Gear trains, Chain drives

PART B:

7. Scope of Civil Engineering **02 Hours** **05%**

- 7.1 Scope of Civil Engineering, Branches of civil engineering, Role of civil engineer

8. Introduction to Surveying **06 Hours** **13%**

- 8.1 Definition of surveying, Objects of surveying, Uses of surveying,
 8.2 Primary divisions of surveying, Principles of surveying,
 8.3 List of classification of surveying, Definition: Plan and Map,
 8.4 Introduction to linear and angular measurement

9. Elements of building Construction **7Hours** **16%**

- 9.1 Types of building,
- 9.2 Design loads
- 9.3 Building components (super structure and substructure)
- 9.4 Principles of Planning
- 9.5 Basics Requirements of a building Planning
- 9.6 Types of Residential Building

D. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

E. Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand fundamental principles of mechanical engineering
CO2	Learn the formation of steam and different types of the boilers and its mounting and accessories
CO3	Explore the basics of internal combustion engine and pumps & compressors
CO4	Learn basics of refrigeration and air conditioning system and power & motion transmission systems
CO5	Learn importance and application of civil engineering
CO6	Understand the fundamentals of surveying and building planning and Identify different building components

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	1	-	-	-	-	-	-	-
CO5	2	-	-	-	-	1	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	1	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:**Text Books:**

1. S. M. Bhatt, H. G. Katariya, J. P. Hadiya, “Elements of Mechanical Engineering”, Books India Publication, Ahmedabad.
2. P. S. Desai, S. B. Soni, “Elements of Mechanical Engineering”, Atul Prakashan, Ahmedabad

Reference Book:

1. Dr. Sadhu Singh, “Elements of Mechanical Engineering”, S.CHAND Publication, New Delhi
2. V. K. Manglik, “Elements of Mechanical Engineering”, PHI Learning, Delhi.
3. Khasia R.B. and Shukla R.N., “Elements of Civil Engineering”, Mahajan Publication.
4. Kandya Anurag, “Elements of Civil Engineering”, Charotar Publishing House Pvt. Ltd.
5. Punamia B.C., “Surveying”, Vol. I & II.

Reading Materials, web materials with full citations:

1. <http://nptel.ac.in/downloads/112105125/>
2. http://www.slideshare.net/allsaintsscience/7th-grade-ch-2-sec-3-behavior-ofgases?qid=75b08741-fb53-4413-b434-5982afe602bf&v=&b=&from_search=12
3. http://www.slideshare.net/Arjun_Dedaniya/properties-of-steam-62226458?qid=fa8777fd-b5434128-813c-cf3af3b86579&v=&b=&from_search=2
4. http://www.slideshare.net/shanus1/i-c-engines-a-study?qid=69826356-b9ed-4618-9c77b2d5a3eac2e3&v=&b=&from_search=8
5. <http://nptel.ac.in/syllabus/syllabus.php?subjectId=105104101>
6. <http://nptel.ac.in/courses/105107122/>

FACULTY OF TECHNOLOGY & ENGINEERING
M & V PATEL DEPARTMENT OF ELECTRICAL ENGINEERING

**EE145: BASICS OF ELECTRONICS & ELECTRICAL
ENGINEERING**

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	4
Marks	100	50	0	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total
Internal	External	Internal	External	Marks
30	70	25	25	150

C. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basic Electrical Terms and Units	04
2	Electrical Circuit Analysis	07
3	Electrostatic	08
4	Electromagnetism	05
5	AC Fundamentals	05
6	Single Phase AC Series Circuits	05
7	Polyphase Circuits	04
8	Basics of Electronics	07

Total hours (Theory) : 45 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 75 Hrs

D. Detailed Syllabus

- | | | | |
|-----|---|-----------------|------------|
| 1 | Basic Electrical Terms and Units | 04 Hours | 08% |
| 1.1 | Ohm's law, resistor and its coding, properties, temperature co-efficient of resistance, resistance variation with temperature, examples | | |

2	Electrical Circuit Analysis	07 Hours	15%
2.1	Kirchoff's current and voltage law, mesh and nodal analysis, Examples		
2.2	Series parallel circuits, star-delta transformation		
3	Electrostatic	08 Hours	18%
3.1	Capacitors, charge and voltage, capacitance, electric fields, electric field strength and electric flux density, relative permittivity, dielectric strength, Examples		
3.2	Capacitors in parallel and series, Calculation of capacitance of parallel plate and multi plate capacitor, examples		
4	Electromagnetism	05 Hours	12%
4.1	Magnetic field, its direction and characteristics, magnetic flux and flux density, magneto motive force and magnetic field strength, examples		
4.2	Faraday's law of electromagnetic induction, Fleming's left hand and right hand rule, Lenz law, force on a current carrying conductor, examples		
4.3	Self and mutual inductance		
5	AC Fundamentals	05 Hours	12%
5.1	AC Waveform and definition of its terms, relation between speed and frequency		
5.2	Average and RMS value and its determination for sinusoidal wave shapes, examples		
6	Single Phase AC Series Circuits	05 Hours	12%
6.1	R-L and R-C series circuit, power in ac circuits, examples		
6.2	R-L-C series circuit, resonance in R-L-C series circuit, relevant examples		
7	Polyphase Circuits	04 Hours	08%
7.1	Phase sequence, voltage and current relations in star and delta connected system		
8	Basics of Electronics	07 Hours	15%
8.1	Electronic Systems: Basic amplifier, voltage, current and power gain, Basic attenuators, CRO		
8.2	Transmission and Signals: Analog and digital signals, bandwidth,		
8.3	Forward and reverse bias of PN junction diode, zener diode		
8.4	Rectifiers: Half Wave, Full Wave - Centre Tap, Bridge		
8.5	Transistor: Bipolar junction transistor, construction and biasing, configuration		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
25	25	10	20	10	10

F. Course Outcomes (COs):

On the completion of the course, one should be able to:

CO1:	Describe resistors, capacitors and inductors properties, readings and calculation
CO2:	State the basic electrical laws and apply these laws to solve electrical network.
CO3:	Identify the property of magnetic materials and understand the laws of emf generation
CO4:	Solve the series and parallel DC circuits and AC circuits for single and poly-phase networks
CO5:	Develop skill and design AC-DC rectification circuits, operate basic electrical and electronics instruments

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO2:	2	3	-	-	-	-	-	-	-	-	-	-	1	-
CO3:	1	3	-	-	-	-	-	-	-	-	-	-	1	-
CO4:	1	2	-	2	-	-	-	-	-	-	-	-	3	-
CO5:	1	2	3	-	3	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), No correlation “-”

G. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, pre-requisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, blackboard, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries 5 Marks weightage.
- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

H. Recommended Study Material:

- **Text Books:**

1. Elements of Electrical Engineering and Electronics by U. A. Patel and R. P. Ajwalia
2. A Text Book of Electrical Technology by B. L. Thareja, S. Chand
3. Principles of Electrical Engineering and Electronics by V. K. Mehta, S. Chand

- **Reference Books:**

1. Hughes, Electrical Technology, Pearson Education
2. Electrical Engineering by Del Toro

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY												
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING												
Sem	Course Code	Course Title	Teaching Scheme					Examination Scheme				
			Contact Hours				Credit	Theory		Practical		Total
			Theory	Practical	Tutorial	Total		Internal	External	Internal	External	
SY Sem-III	XXXXX	University Elective	0	2	0	2	2			30	70	100
	MA241	Engineering Mathematics-III	4	0	0	4	4	30	70	0	0	100
	EE241	Analog and Digital Electronics	4	2	0	6	5	30	70	25	25	150
	EE242	Circuit Theory	4	2	0	6	5	30	70	25	25	150
	EE243	Electrical Measurement and Industrial Instrumentation	4	2	0	6	5	30	70	25	25	150
	HS122A	Values and Ethics	2			2	2	30	70	0	0	100
			18	8	0	26	23	150	350	105	145	750
SY Sem-IV	XXXXX	University Elective	0	2	0	2	2			30	70	100
	EE244	Electrical Power Generation and Its Economics	4	0	0	4	4	30	70	0	0	100
	EE246	Microprocessor and Microcontrollers	4	2	0	6	5	30	70	25	25	150
	EE247	Transformers and Induction Machines	4	4	0	8	6	30	70	50	50	200
	EE245	Control Systems	4	2	0	6	5	30	70	25	25	150
	HS133A	Creativity Problem Solving and Innovation	2			2	2	30	70	0	0	100
			18	10	0	28	24	150	350	130	170	800

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 3rd Semester Undergraduate Programme
(Syllabus is attached in separate Annexure)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC281.01: Introduction to MATLAB Programming	EC / FTE
2	CE281.01: Art of Programming	CE / FTE
3	CL281.01: Environmental Sustainability and Climate Change	CL / FTE
4	EE283: Python for Electrical Engineers	EE/FTE
5	IT281.01: ICT Resources and Multimedia	IT/FTE
6	ME281.01: Engineering Drawing	ME/FTE
7	PH233.01: Fundamentals of Packaging	RPCP/FPH
8	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS
9	NR251.01: First Aid & Life Support	NURSING / FMD
10	PT191.01: Health Promotion and Fitness	ARIP / FMD
11	CA224: Introduction to Web Designing	CMPICA / FCA
12	BM231: Banking and Insurance	I2IM / FMS
13	PD261: Astrophysics, Space and Cosmos-I	PDPIAS/FAS

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses
for 4th Semester Undergraduate Programme
(Syllabus is attached in separate Annexure)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course
1	EC282.01: Prototyping Electronics with Arduino	EC / FTE
2	CE282.01: Web Designing	CE / FTE
3	CL282.01: Basics of Environmental Impact Assessment	CL / FTE
4	EE286: Computer Programming for Electrical Engineering	EE/FTE
5	IT282.01: Internet Technology and Web Design	IT/FTE
6	ME282.01: Material Science	ME/FTE
7	PH238.01: Cosmetics in daily life	RPCP/FPH
8	NR261.01: Life Style Diseases & Management	NURSING / FMD
9	PT192.01: Occupational Health & Ergonomics	ARIP / FMD
10	CA225: Programming the Internet	CMPICA / FCA
11	BM241: Health Care Management	I2IM / FMS
12	PD262: Astrophysics, Space and Cosmos-II	PDPIAS/FAS

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – III)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MA241: ENGINEERING MATHEMATICS-III
B TECH 3rd SEMESTER (CL/EE/ME)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	-	4	4
Marks	100	-	100	

A. Outline of the course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fourier series	10
2.	Laplace Transforms	12
3.	Applications of Differential Equations	10
4.	Matrix Algebra- II	10
5.	Vector Differential Calculus	08
6.	Vector Integral Calculus	10

Total hours: 60

B. Detailed Syllabus:

1.	Fourier series	10 Hours	16%
1.1	Periodic functions, Dirichlet's conditions, Trigonometric series		
1.2	Euler formulae, Fourier series of periodic function of period 2π		
1.3	Discontinuous functions, Even and odd functions, Half range series		
1.4	Fourier series of functions of arbitrary period		
2.	Laplace Transforms	12 Hours	22%
2.1	Laplace transforms as an improper integral and its existence. Laplace transforms of elementary functions.		
2.2	Inverse Laplace transforms, linearity property.		
2.3	First and second shifting theorems, Laplace transforms of derivatives and integrals.		
2.4	Convolution theorem and its application to obtain inverse Laplace transform		
2.5	Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function)		

2.6	Solving Differential equations using Laplace transforms	
3	Applications of Differential Equations	10 Hours
3.1	Applications of ODE: Mechanical vibration system, Electrical circuit system, deflection of beams.	
3.2	Application of PDE: Heat, wave, Laplace equations and their solution by method of separation of variables and Fourier series.	
4	Matrix Algebra -II	10 Hours
4.1	Revision of Determinant and Matrix	
4.2	Eigen values and Eigen vectors of Matrices	
4.3	Eigen values and Eigen vectors of Special Matrices	
4.4	Applications of Cayley - Hamilton Theorem	
5	Vector Differential Calculus	08 Hours
5.1	Revision of concepts of Vector algebra, Scalar and Vector fields.	
5.2	Gradient of a scalar functions, Directional derivatives.	
5.3	Divergence and Curl of a vector field and their properties.	
5.4	Physical interpretations of gradient, divergence and curl. Irrotational, solenoidal and conservative vector fields.	
6	Vector Integral Calculus	10 Hours
6.1	Line integrals, Surface integrals, Volume integrals	
6.2	Statement and examples of Green's theorem, Stokes' and Divergence theorem	
6.3	Applications of vector calculus in engineering systems.	

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures/laboratory which carries a 5% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5%.
- Two Quizzes (surprise tests) will be conducted which carries 5% component of the overall evaluation.

E. Student Learning Outcomes:

- At the end of the course, the students will be able to know the various applications of Engineering Mathematics in their respective field e.g. Vector calculus mainly useful for Electromagnetism. Fourier series, Laplace transforms and applications of differential equations widely applicable in Control theory, Structure engineering and Fluid mechanics.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Represent continuous time periodic signals using concepts of Fourier series and determine the solution of complex engineering problems heat, wave and Laplace's equations using it.
CO2	Classify and apply the necessary Laplace transform and inverse Laplace transform techniques to solve ordinary differential equations of real world problems.
CO3	Differentiate the dot product, cross product, length of vectors, partial derivatives, derivatives of vector-valued functions, gradient functions and these tools are employed successfully in different branches of engineering and physics.
CO4	Analyze parameterize curves and calculate line integrals using them and Use vector operators and implement into surface integrals, Classify and examine the Green, Stokes and Divergence theorem
CO5	find and explain significant of Eigenvalues and Eigenvectors of a square matrix, use Cayley-Hamilton's theorem to find inverse and power of a square matrix.

- Course Articulation Matrix (ME/CL/EE):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO3	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO4	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO5	3	1	-	-	-	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), no correlation : “-”

F. Recommended Study Material:

❖ Text Books:

- Erwin Kreyszig: Advanced Engineering Mathematics, 8th Ed., Jhon Wiley & Sons, India, 1999.
- Thomas, G. B., and R. L. Finney. "Calculus with Analytic Geometry (9th Edition), 1996.", Addison Wesley Publishing.

❖ **Reference Books:**

1. Ahsaan, Zafar. Differential equations and their applications. PHI Learning Pvt. Ltd., 2004.
2. Stewart James: Calculus Early Transcendental, 5th Ed., Thomson India, 2007
3. Wylie & Barrett: Advanced Engineering Mathematics, Mcgraw Hill pub.
4. Greenberg M D: Advanced Engineering Mathematics, 2nd ed., Pearson.
5. Anton, Howard. Elementary linear algebra. John Wiley & Sons, 2010.
6. Grewal, B. S. Higher engineering mathematics. Khanna Publisher, New Delhi, 1996.
7. Dass, H. K. Advanced engineering mathematics. S. Chand, 2008.
8. Debnath, Lokenath, and DambaruBhatta. Integral transforms and their applications. CRC press, 2014.
9. Stroud, Kenneth Arthur, and Dexter J. Booth. Advanced engineering mathematics. Palgrave Macmillan, 2011.

URL link: Web site: <http://mathworld.wolfram.com> , <http://en.wikipedia.org/wiki/Math>

EE241: ANALOG AND DIGITAL ELECTRONICS

3rd Semester and 2nd Year (Level II)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	BJT and FET	06
2	Operational Amplifiers	06
3	Applications of op-amp	12
4	Specialized ICs	06
5	Introduction to Digital Electronics and Numbering Systems	05
6	Digital Logic families and Design	09
7	Combinational Logic	08
8	Flip-flops, Timers and Counters	08

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

D. Detailed Syllabus:

1	BJT and FET	10%	06 Hrs
	BJT: V-I Characteristics, Transistor DC load line and Operating point, Transistor as a		

	switch and an amplifier, Faithful Amplification, Transistor biasing and its Methods, Transistor Stability and Stability Factor, Testing and Transistor data-sheet. FET: Types of FET, Construction and working principle of JFET, Difference between BJT and JFET, V-I characteristics, Pinch-off Voltage, Shorted-gate Drain Current, Parameters of JFET, Biasing of JFET, Working of MOSFET and V-I Characteristics, Difference between JFET and MOSFET.		
2	Operational Amplifiers	10%	06 Hrs
	Introduction to Integrated Circuits, IC classifications, packages and symbols, Pin diagram of 741 Op-amp, Internal block diagram of a Typical op-amp, Ideal op-amp and its equivalent circuit, Specification and Datasheet parameters of op-amp (Differential Input Resistance (R_i), Output Resistance(R_o), Common Mode Rejection Ratio (CMRR), Slew Rate, Power Supply Rejection Ratio (PSRR), Input offset Voltage, Input offset Current, Input bias Current, Output offset Voltage), Feedback Configurations, An op-amp with and without Feedback, Closed-loop Voltage Gain.		
3	Applications of op-amp	20%	12 Hrs
	Linear applications: DC and AC Amplifier, Voltage follower, Peaking, Summing, Scaling, Averaging Amplifier, Differential Amplifiers and Instrumentation Amplifier, V to I converter, I to V converter, Integrator, Differentiator Active filters & Oscillators: First order Low-Pass and High-Pass Filters, Second order Low-Pass and High-Pass Filters, Band-Pass and Band-Reject filters, All-Pass filters, Phase Shift, Wien Bridge Oscillators; Square, Triangular, Sawtooth Wave Generators. Comparators & Converters: Basic Comparator, Zero crossing Detector, Schmitt Trigger, Basic Positive and Negative Clippers and Clampers, Absolute-Value output Circuit, Peak Detector, ADC and DAC using op-amp		
4	Specialized ICs	10%	06 Hrs
	555 Timer IC, 555 timer IC as Monostable, bistable and astable Multivibrator, Three terminal regulator ICs (Fixed Voltage Regulator ICs, Adjustable output voltage regulator ICs, their use and basic design considerations for designing regulated power supplies), Analog Multiplier MPY634.		
5	Introduction to Digital Electronics and Numbering Systems	8.34%	05 Hrs

	Digital circuits, Digital signals, Binary number system, Signed & unsigned binary number system, 1's complement and 2's complement, Binary arithmetic, octal, hexadecimal and decimal Number systems and their inter conversion, BCD numbers, ASCII and EBCDIC codes		
6	Digital Logic families and Design	15%	09 Hrs
	Boolean Algebra: Basic Logic gates and their truth tables, Universal Gates, Laws of Boolean algebra, De-Morgan's theorem, Min term, Max term, POS, SOP, Karnaugh Maps, Minimization of logical function, Don't care condition Logic families and Design: Introduction, Resistor transistor logic, Direct coupled transistor logic, Integrated injection logic, DTL logic, TTL logic, CMOS Logic		
7	Combinational Logic	13.33%	08 Hrs
	Half adder, Full adder, Binary subtraction, Arithmetic logic unit, Digital comparators, Priority encoder, decoders, Multiplexer & de-multiplexer		
8	Flip-flops, Timers and Counters	13.33%	08 Hrs
	S-R flip-flop, D- flip-flop, T-flip-flop, J-K flip-flop, Clocked flip-flop design, Registers, Asynchronous counter, Synchronous counter		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs)

Upon successful completion of this course, a student will be able to

CO1:	Develop & Enhance their skills in basics of analog and digital circuits
CO2:	Describe the performance and classification of ICs
CO3:	State linear and non-linear applications of operational amplifiers
CO4:	Design and troubleshoot hardware circuits like filters, comparators, PID controllers, oscillators, digital logic gates, timer and counters
CO5:	Analyze characteristics of popular MOS and bipolar logic families, with emphasis on CMOS and TTL technologies

Mapping of course outcomes with program outcomes
Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	2	-
CO3	1	3	-	-	-	-	-	-	-	-	-	-	2	-
CO4	1	2	2	3	3	-	-	-	-	-	-	-	3	1
CO5	1	1	3	3		-	-	-	-	-	-	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Principles of Electronics by V. K. Mehta S. Chand & Company Ltd.
2. Op-Amp and Linear integrated Circuit technology- Ramakant A Gayakwad, PHI Publication
3. Analog and Digital Electronics by J. S. Katre, Techmax Publications
4. Modern digital electronics , Second edition, R.P.Jain

5. Fundamental of digital circuits by A.ANANDKUMAR, PHI Publication

Reference Book:

1. Operational Amplifier and Linear integrated Circuits By Robert F.Coughlin, Frederick F. Driscoll
2. Operational Amplifier and Linear integrated Circuits By K.LAL kishore. Pearsons
3. Electronics Devices and Circuits An Introduction By A. Mottershed, PHI Publication.
4. Digital Fundamentals by Morris and Mano, PHI Publication

Web Material:

- <https://www.facstaff.bucknell.edu/mastascu/eLessonsHTML/EEIndex.html>
- <http://www.electronics-tutorials.ws/>
- <http://nptel.ac.in/courses/108106068/>
(Videos and Notes by Prof. K. Radhakrishna Rao, IIT Madras)
- <http://nptel.ac.in/courses/108106069/>
(Videos and Notes by Dr. Amitava Dasgupta, IIT Madras)

I. List of experiments:

1. (A) To study and perform Differential Amplifier using op-amp.
(B) To study and perform Summing, Scaling, Averaging Amplifier using op-amp.
2. To study and perform Integrator and Differentiator using one op-amp.
3. (A) To study and perform Comparator using op-amp.
(B) To study and perform Zero Crossing Detector using op-amp.
4. (A) To study and perform 1st order Low Pass filter.
(B) To study and perform all-Pass filters.
5. (A) To study and perform Square Wave and Triangular Wave Generators.
(B) To study and perform 555 timer as an astable Multivibrator.
6. To study and perform logic gates.
7. To study and perform half-adder and half subtractor circuit.
8. To Study and Perform Full Adder circuit diagram using two half adders circuit diagram.
9. To study and perform J-K Flip-Flop and D Flip-Flop.
10. To study and perform the operation of a universal Bi-directional shift Register.

EE242: CIRCUIT THEORY
3rd Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1.	Circuit elements and energy sources	3
2	Network Topology	7
3.	Network Equations	7
4	Network theorems and Impedance function	8
5	Two-Port Network Analysis	8
6	Laplace Transformation, its Application and Special Signal Waveforms	19
7	Initial conditions in networks	8

Total hours (Theory) :60 Hrs
Total hours (Lab) :30 Hrs
Total hours : 90 Hrs

D. Detailed Syllabus:

1	Circuit elements and energy sources	05%	03 Hrs
	Circuit Elements And Their Characteristics, Energy Sources, Source Transformation.		

2	Network Topology	11.67%	07 Hrs
	Concept Of Network Graph, Terminology Used In Network Graph, Number Of Trees In A Graph, Tie-Set Matrix, Fundamental Tie-Set Matrix, Fundamental Cut-Set, Cut-Set Matrix, The Dot Convention For Coupled Circuit, Principle Of Duality		
3	Network Equations	11.67%	07 Hrs
	Kirchhoff's laws, Current division, Voltage division in series circuits, Nodal and mesh analysis of electric circuits in DC and AC excitations.		
4	Network theorems and Impedance function	13.33%	08 Hrs
	Superposition theorem, Thevenin theorem, Norton Theorem, Reciprocity and Maximum power transfer theorem in Electrical Circuits with DC and AC excitations, Transform impedance and transform circuits		
5	Two-Port Network Analysis	13.33%	08 Hrs
	Network Elements, Classification of Network, Network Configuration, Recurrent Network, Z-parameters, Y-parameters, ABCD Parameters, Inverse ABCD Parameters, Interrelationship between parameters, Concept of Reciprocity & Symmetricity for network parameters		
6	Laplace Transformation, its Application and Special Signal Waveforms	31.67%	19 Hrs
	Basic Types Of Special Signals, Laplace Transformation Of Special Signal Waveforms, Step Response of R-L, R-C, R-L-C Series & Parallel Circuit, Transient Response of Series R-L, R-C, R-L-C With DC Excitation, Transient Response of Series R-L, R-C, R-L-C With AC Excitation.		
7	Initial conditions in networks	13.33%	08 Hrs
	Initial conditions in elements, Geometrical Interpretation of Derivatives, A Procedure of Evaluating Initial conditions, R-L, R-C & R-L-C Circuits analysis using Initial Conditions.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level

Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	20	20	15	5

F. Course Outcomes (Cos):

Upon successful completion of this course, a student will be able to

CO1:	To understand basics of network elements and their behaviour.
CO2:	To understand basics of network topologies, tie set and cutset schedules
CO3:	To apply concept of mesh, nodal analysis technique, various network theorems such as Superposition, Thevenin, Norton, Reciprocity, and Maximum Power Transfer.
CO4:	To calculate two port parameters such as Z, Y, ABCD, etc. for the given two port network.
CO5:	Develop the skills related to analyse circuit taking into account of initial conditions and solving the DC and AC circuits with differential equations with the help of Laplace approach.

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	3	-	2	-	3	1	3	3	-
CO2	3	3	-	-	-	-	-	-	-	1	-	-	3	-
CO3	2	1	1	3	2	1	-	-	-	1	-	2	2	3
CO4	3	1	1	2	2	-	2	2	3	-	2	2	2	2
CO5	3	1	1	3	3	-	2	2	3	-	2	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Circuit Theory by A. Chakrabarti , Dhanpat Rai Publications
2. Network Analysis by Van Valkenburg , Prentice Hall College
3. Network Analysis and Synthesis by Ravish R Singh, McGraw Hill Education Pvt. Ltd

Reference Book:

1. U. A. Patel, Network analysis and Synthesis, Mahajan Publisher
2. G. K. Mithal, Network analysis, Khanna publications
3. F. F. Kuo, Network Analysis, John Wiley
4. Electric Circuits and Networks :- By K. S. Suresh Kumar – Pearson Education
5. Linear Circuits Analysis 2nd edition :-By DeCarlo/ Lin – Oxford University Press(Indian edition)

Web Material:

- <http://nptel.ac.in/courses/108I02042/9>
- <http://freevideolectures.com/Course/2350/Networks-Signals-and-Systems>

I. List of experiments:

1. Application of Nodal analysis on DC Resistive Circuit.
2. Application of Mesh analysis on a DC resistive circuit.
3. Application of Thevenin's theorem on a DC resistive circuit.
4. Application of Norton's theorem on a DC resistive circuit.
5. Application of Superposition theorem on a DC resistive circuit.
6. To verify maximum power transfer theorem.
7. To verify Reciprocity theorem.
8. To find Y-parameters of a given network.
9. To find ABCD-parameters of a given network.

10. Introduction to MATLAB/SIMULINK and solve the experiments 1 to 9 in MATLAB/SIMULINK.
11. To analyse the transient response of RL-RC, RLC series & parallel circuit under step, ramp, impulse input.

EE243: Electrical Measurements & Industrial Instrumentation
3rd Semester and 2nd Year (Level II)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Instrumentations and Measurement	04
2	Temperature and Strain Measurement	08
3	Flow and Pressure Measurement	08
4	Measurement of torque, velocity and vibration	04
5	Miscellaneous Instruments in Industrial, Biomedical and Environmental Applications	08
6	Recorders	03
7	Instrument Transformer	09
8	Analog and Digital instruments	10
9	Recent Development in instrumentation and measurements	06

Total hours (Theory) :60Hrs

Total hours (Lab) :30 Hrs

Total hours : 90Hrs

D. Detailed Syllabus:

1	Introduction of Instrumentations and Measurement	06%	04 Hrs
	History to the Instrumentation and Measurements, Application of Instrument		

	Systems, Functional Elements of Measurement system, classification of Instruments, Microprocessor based Instrument, Static and Dynamic Performance Characteristics of Instruments, Standards and Calibration.		
2	Temperature and Strain Measurement	13%	08Hrs
	Temperature Measurement: Temperature Scales, International Practical Temperature Scale, Measurement of Temperature, Non-Electrical Methods, Electrical Methods and Radiation Methods (Pyrometer) Strain Measurement: Construction, Operation, Theory and Factors Affecting Strain Measurements, Applications of Strain Gauges, Load Cells, Advantages, Disadvantages.		
3	Flow and Pressure Measurement	13%	08Hrs
	Flow Measurement Devices: Venture Tube, Orifice Plate, Flow Nozzle, Rotameter, Pitot Tube, Ultrasonic Flow Detection, Turbine Flow Meter, Electromagnetic Flow Meter, Hot Wire Anemometer Pressure Measurement Devices: Construction, Operation, Advantages, Disadvantages and Applications of Elastic Transducers (Diaphragm, Bellow, Bourdon Tube), Variable Inductance and Capacitance Transducers, Vacuum Pressure Measuring Devices (Pirani Gauge, Ionization Gauge, McLeod Gauge, Piezo-Electric Transducers		
4	Measurement of torque, velocity and vibration	06%	04Hrs
	Torque measurement: Strain Gauge Torque Meters and Magneto-Strictive Transducers Vibration measurement: Nature of Vibrations, Quantities Involved in Vibration, Piezoelectric Transducer (Accelerometer) Velocity measurement: Moving Magnet Type, Moving Coil Type Velocity Transducer, Tachometers		
5	Miscellaneous Instruments in Industrial, Biomedical and Environmental Applications	16%	10Hrs
	Specific gravity measurement: Bubbler System, Hydrometer Method, Totally Immersed Float Method, Nuclear Absorption Method, Fixed Volume Method Liquid Level measurement: <u>Direct method:</u> Dip-Stick Method, Sight Glass Method, Hook Gauge, Float Gauge <u>Indirect method:</u> Hydrostatic Pressure Level Measurement Device, Bubbler or Purge Technique, Resistive, Inductive and Capacitive Transducers, Ultrasonic Level Gauge and Radiation Method. Viscosity measurement: Capillary Tube Viscometer, Efflux Type Viscometer, Rotation Concentric Cylinder Viscometer, Variable Area Viscometer. Humidity and Moisture measurement: Dew Point Meter, Hair Hygrometer, Psychrometer, Resistive and Capacitive Hygrometer. Measurement of PH value & Conductivity: Ph meter& Conductivity meter		

	Biomedical measurement: Blood Pressure Measurement, Measurement of Bioelectric Potentials Measurement of Environmental air pollution parameters: Orsat Apparatus For Exhaust Gas Analysis, Gas Chromatography, Non-Dispersive Infrared and Non-Dispersive Ultraviolet Gas Analyzers, and Smoke Density Measurement.		
6	Recorders	05%	03Hrs
	Strip Chart Recorders, X-Y Recorders, Circular Chart recorders, Digital Recorders, Data Acquisition System, Data Display and Storage.		
7	Instrument Transformer	15%	09Hrs
	Application of instrument transformers, CT, PT, testing of CT and PT Current Transformer: Relationship Between Various Quantities in CT, Errors in CT, Characteristics of CT, types of CT Potential Transformer: Relationship Between Various Quantities in PT, Errors on PT, Characteristics of PT, Types of PT		
8	Analog and Digital instruments	16%	10Hrs
	Analog instruments: Electrodynamic and Induction Type Instruments, Power Factor Meter, Analog Frequency Meter, Synchroscope, Phase Sequence Indicator, Extension of Instrument Ranges and Multipliers. Digital Instruments: Introduction, Digital Voltmeter, Digital Multimeter, Digital Frequency Meter and Analyzer, Signal Generator, Digital Signal Oscilloscope, Harmonic Analyzer		
9	Recent Development in instrumentation and measurements	10%	06Hrs
	Computer-Aided measurement: Introduction, Microcomputers for Data Acquisition Purposes, Data Transfer and Communication, Data Acquisition System Fiber optic transducer: Introduction, types of Configuration, Applications. Advantages and Disadvantages, Micro Sensors, Smart Sensors, Virtual Instrumentation		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (Cos):

Upon successful completion of this course, a student will be able to

CO1:	Recognize different sensors, transducers and various measuring instruments
CO2:	Understanding the constructional and operational aspects of various sensors, transducers and measuring instruments used for industrial, biomedical and environmental applications
CO3:	Aware with the performance characteristics of instrument transformer used for measurement of currents and voltages
CO4:	Aware about the recent developments in measuring instruments and sensors

Course Articulation Matrix

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	2	-	-	3	2	-
CO2:	3	-	2	2	-	-	-	-	2	-	-	-	2	-
CO3:	3	-	-	-	-	-	-	-	-	-	-	3	3	-
CO4:	2	2	-	3	3	-	-	-	-	-	-	2	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. B.C. Nakra & K.K. Chaudhry, Instrumentation, Measurement and Analysis, TATA McGraw Hill.
2. A. K. Shawhney, Electrical and Electronics Measurement and Instrumentation, Dhanpat Rai Publishers

Reference Book:

- i. Rangan, Sharma, Mani, Industrial Instrumentation, Tata McGraw Hill
- ii. Murthy, Transducers and Instrumentation, Tata McGraw Hill
- iii. Patranabis, 'Principles of Industrial Instrumentation', Tata McGraw Hill Publishing Company Ltd, Revised edition
- iv. R. K. Rajput, Electrical and Electronics Measurement and Instrumentation, S. Chand Limited.
- v. S. K. Singh, Industrial Instrumentation & Control, Tata McGraw Hill.R.S. Khandpur, Handbook of Analytical instruments, Tata McGraw Hill.

Web Material:

- www.isa.org
- <http://nptel.iitm.ac.in/video.php?courseId=1062>
- http://www.mywbut.com/syllabus.php?mode=SM&paper_id=57&dept_id=6

I. List of experiments:

1. To study various types of sensors and measurement devices used in industries.
2. Measurement of strain and pressure using Piezo-resistive transducer.
3. Measurement of temperature using (1) Resistance Temperature Detector (RTD) (2) Thermocouple and (3) Thermistor.
4. Measurement of flow using orifice plate.
5. Measurement of liquid level using Air bubbler method and capacitive type level sensor.
6. Measurement of relative humidity using humidity sensor.
7. Measurement of torque using torque transducer.
8. To study one phase induction type energy meter.
9. To perform the extension range of Ammeter and Voltmeter.
10. Determination of transformation ratio, nominal ratio, phase angle error and ratio error for Potential Transformer.
11. Determination of transformation ratio, nominal ratio, phase angle error and ratio error for Current Transformer.
12. Calibration of Ammeters, Voltmeters and Wattmeters.

HS122A: VALUES AND ETHICS

3rd Semester and 2nd Year

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/ Week	Theory		Practical		Total
					Internal	External	Internal	External	
III	HS122 A/B/C/D/E/F/G/H	VALUES AND ETHICS	02	02	30	70	-	-	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Introduction to Values and Ethics - Need, Relevance and Significance of Values and Ethics : General - Concept and Meaning of Values and Ethics	06
2	Elements and Principles of Values - Universal & Personal Values - Social, Civic & Democratic Values - Adaptation Models & Methods of Values	08
3	Applied Ethics - Universal Code of Ethics - Professional Ethics - Organizational Ethics - Ethical Leadership - Domain Specific Ethics	08
4	Value, Ethics & Global Issues - Cross-Cultural Issues - Role of Ethics & Values in Sustainability - Case Studies	08
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various

interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations etc.

IV. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
3	Assignment / Project Work	2	25	25
4	Attendance and Class Participation			05
Total				30

External Evaluation

The University Theory examination will be of 70 marks and will test the reasoning, logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications. There will be at least one question on case analysis relevant to the components of the course.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Theory Paper	01	70	70
Total				70

V. Course Outcome (COs):

- After completion of the course, the student would :

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C01	-	-	-	-	-	-	1	3	2	1	-	-	-	-
C02	-	-	-	-	-	-	-	-	1	-	-	2	-	-
C03	-	-	-	-	-	1	1	-	-	-	-	-	-	-
C04	-	-	-	-	-	-	1	1	-	-	-	-	-	-
C05	-	-	-	-	-	-	-	-	2	-	-	-	-	-
C06	-	-	-	-	-	-	-	3	-	2	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Reference Books / Reading

- Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
- Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>)
- Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)
- www.ethics.org

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – IV)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE244: ELECTRICAL POWER GENERATION AND ITS ECONOMICS
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	0	4	4
Marks	100		100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	0	0	100

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	01
2	Steam Power Plants	08
3	Hydro Electric Power Plant	04
4	Nuclear Power Plant	04
5	Combined cycle power plant	03
6	Wind Power Generation	04
7	Solar Power Generation	03
8	Other Renewable Energy Sources	04
9	Insulated Cables	05
10	Components of Overhead Lines	10
11	Mechanical Design of Transmission Lines (Sag-Tension Relationship)	03
12	Economics of Power System	07
13	Power Factor Improvement in Power System	04

Total hours (Theory) :60 Hrs
Total hours (Lab) :0 Hrs

Total hours : 60 Hrs

D. Detailed Syllabus:

- | | | | |
|----------|---|------------|---------------|
| 1 | Introduction | 01% | 01 Hrs |
| | Power and Energy, sources of energy such as coal, water, nuclear, wind, solar, tidal etc | | |
| 2 | Steam Power Plants | 14% | 08 Hrs |
| | Classification of steam power plants, layout of steam power plant, Component of steam power plants, Essential requirement for steam power plant, selection of site, Choice of steam condition, fuel handling, Combustion equipments for steam boilers, Ash handling, electrostatic precipitator and dust collection, Boilers, steam turbines, pressure and velocity compounding, Steam condenser, feed water treatment, advantages and disadvantages of steam power plant | | |
| 3 | Hydro Electric Power Plant | 07% | 04 Hrs |
| | Application, advantages and disadvantages of hydro power plant, Site selection for hydro power plant, layout and elements of hydro power plant, Classification of hydro power plant, Hydraulic turbines. | | |
| 4 | Nuclear Power Plant | 07% | 04 Hrs |
| | General aspect of nuclear engineering, layout of nuclear power plant, Nuclear reactors, Main components of nuclear power plants, advantages and disadvantages of nuclear power plant. | | |
| 5 | Combined Cycle Power Plant: | 04% | 03 Hrs |
| | Various combined cycle power plant, working, advantages and disadvantages, comparison of various power plant. | | |
| 6 | Wind Power Generation | 06% | 04 Hrs |
| | Applications, Merits and demerits of wind energy, nature and origin of wind, variables in wind energy conversion system, wind velocities and height from ground and site selection, Types of wind energy system, wind turbine generator unit with battery storage facilities, solar wind hybrid, Wind farm siting. | | |
| 7 | Solar Power Plant | 04% | 03 Hrs |
| | Solar energy routes and prospects, merits and limitations of solar energy conversion and utilization, Types of solar thermal collectors, comparison between conventional and solar thermal power plant, ratings of solar power plant heat transfer fluids, solar pond and binary cycle solar thermal power plant. | | |

8 Other Renewable Energy Sources 07% 04 Hrs

Fuel cell based power plant: Introduction, concept, types, Electrochemical Reactions, Hydrogen, Oxygen Fuel cells.

MHD Power Generation: Principles, MHD Systems, Advantages of MHD Systems, Electrical conditions.

Geothermal energy: Applications, utilization of geothermal energy, origin of geothermal resources, liquid dominated geothermal electric power plant.

Ocean and tidal energy: Ocean energy resources, advantages and limitations of ocean energy conversion, technologies, ocean thermal energy conversion, principle of OTEC. Tidal energy conversion : High and low tides, tidal energy conversion.

9 Insulated Cables 09% 05 Hrs

Requirement of insulated cables, properties of conductor and insulating material for cables, construction of cables, types of cables, types of insulating material for cable, Insulation resistance of a single core cable, capacitance of single and three core cable, Examples, Electrostatic stresses in a single core cable, most economical conductor size in cable, examples, Capacitance grading and inter sheath grading, examples, Calculation of losses in cable and factors affecting it, current rating of a cable.

10 Components of Overhead Lines 17% 10 Hrs

Basic power system structure, generation, transmission, substation, sub-transmission, distribution, Complex power, single phase transmission, three phase transmission. Conductors and line supports, Types of Insulators, potential distribution over a string of suspension insulators, String efficiency and methods to improve it, Examples on string efficiency.

11 Mechanical Design of Transmission Lines (Sag-Tension Relationship) 05% 03 Hrs

Sag-tension relation and its example with different configurations.

12 Economics of Power System 12% 07 Hrs

Loads in power system, effects of variable load on power system, load curves and load duration curves, Important terms and factors related to power system economics of power system, importance of those factors, Examples on load curves, Load curves and selection of generating units, Examples, Cost of electrical energy and its expression, methods of determining the depreciation, examples, Types of Tariffs, Examples

13 Power Factor Improvement in Power System:**07% 04 Hrs**

Causes of low power factor, disadvantages of low power factor, power factor improvement, Power factor improvement equipments, calculations of power factor correction, most economical power factor, examples.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Acquire the knowledge regarding the basic functioning of conventional and non-conventional power plants
CO2:	Describe the working of various power plants and various equipments of power plants
CO3:	Develop skill and Identify the key point to improve the performance of the power plants.
CO4:	Aware with the new era of energy generation technology like wind, solar and fuel cell etc.
CO5:	Analyze and select the proper cable for particular application.
CO6:	Determine the basic structure of the tariff and means to reduce the electricity charges by improving the power factor

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3		-	-	-	-	-	-	-	-	-	-	1	-
CO2:	2	3	1	-	-	-	-	-	-	-	-	-	2	-
CO3:	-	3	-	-	-	2	3	1	1	-	-	-	3	2
CO4:	-	2	-	-	3	-	-	-	-	-	-	-	2	2
CO5:	1	3	3	-	-	-	-	-	-	-	-	-	3	-
CO6:	1	3	3	-	2	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of power plant will clear the concept of power generation and student will realise the actual size of generating equipments.

H. Recommended Study Material:

Text Book:

1. Energy Technology by S. Rao & Dr. B.B.Parulekar, Khanna Publication, New Delhi.
2. Principles of Power system by V.K. Mehta, Publisher: S. Chand & Company Ltd.
3. Renewable energy sources and conversion technology by N.K. Bansal, Tata McGraw Hill, New Delhi.
4. A Course in electrical power by Soni & Bhatnagar, Dhanpat Rai & Sons.
5. Energy sources, by G.D. Rai, Khanna Publication, New Delhi

Reference Book:

1. Power Generation Technologies by Paul Breeze, Newnes
2. Renewable energy in Power system by Leon Freris, David Infield, John Wiley Inc.
3. Renewable Energy Resources by John Twidell and Tony Weir, Taylor & Francis

Web Material:

- <http://nptel.ac.in/courses/108105058/>
- <http://nptel.ac.in/courses/108108078/>
- <https://www.youtube.com/watch?v=u0jAVR-j60o>

EE245: CONTROL SYSTEMS
4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to control Systems	03
2	Mathematical Modeling of Control systems	14
3	Time response analysis	11
4	Stability of Control Systems	09
5	Frequency response analysis	08
6	Compensation Technique	07
7	State Variable Analysis of Control Systems	08

Total hours (Theory) :60 Hrs
Total hours (Lab) :30 Hrs
Total hours : 90 Hrs

D. Detailed Syllabus:

1	Introduction to control System	05%	03 Hrs
	Introduction, Classification of control systems: open loop, closed loop, continuous, discrete, linear & nonlinear control system, Examples of control system application, Effect of feedback on control system performance, Feed forward control system.		

2	Mathematical modeling of control systems	23.33%	14 Hrs
	Transfer function representation, Block diagram algebra, Block diagram reduction technique, Signal flow graph, Mason's gain formula, Modeling of DC motors, Servomechanism		
3	Time response analysis	18.34%	11 Hrs
	Introduction, Standard test signals, Time response of first order and second order system, Damping ratio, natural un damped frequency, Transient response specifications, Steady state error and error constant, Effect of integral and derivative control action on system performance.		
4	Stability of Control Systems	15%	09 Hrs
	Concept of stability, Necessary condition for stability, Absolute and relative stability, Routh Hurwitz stability criteria, Root locus concept, Construction of root loci.		
5	Frequency response analysis	13.33%	08 Hrs
	Bode plot, Procedure for drawing bode plot and determination of gain margin, phase margin and stability, Polar plot.		
6	Compensation Technique	11.67%	07 Hrs
	Phase lead compensation, Phase lag compensation, Phase lead lag compensation, Selection of Compensator, Design of Phase Lag, Phase Lead and Phase Lag-Lead Compensator.		
7	State Variable Analysis of Control Systems	13.33%	08 Hrs
	State space representation of dynamic system, Concept of state, state variable and state model, Concept of controllability and Observability, State –Transition Matrix ,Computation of State Transition Matrix, Eigen Values and Eigen Vectors.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (Cos):

Upon successful completion of this course, a student will be able to

CO1:	Demonstrate an understanding of the fundamentals of control systems
CO2:	Acquire the skill related to development and usage of models for physical which are suitable for use in the analysis and design of control systems
CO3:	Express and solve system equations in state-variable form (state variable models)
CO4:	Determine the (absolute) stability of a closed-loop control system.
CO5:	Determine the time and frequency-domain responses of first and second-order systems to step and sinusoidal (and to some extent, ramp) inputs with controller designs

Mapping of course outcomes with program outcomes

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	2	-	-	-	-	-	2	3	-
CO2	3	3	2	2	1	-	-	-	-	-	-	-	-	1
CO3	3	2	2	3	3	1	-	-	-	-	-	2	1	3
CO4	3	2	2	3	2	2	-	-	-	2	2	2	2	-
CO5	3	2	3	3	3	-	-	3	-	2	1	1	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.

- Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Automatic control system by B.C. Kuo
2. Modern control engineering by K. Ogata
3. Control System Engineering by J Nagrath & M Gopal New Age International Publishers

Reference Book:

1. Control Systems by A. Anand Kumar, PHI Learning Pvt. Ltd.
2. Automatic Control Systems by S. Hasan Saeed, Katson Books.
3. Control Systems by Ashfaq Hussain, Haroon Ashfaq, Dhanpat Rai & Co.

Web Material:

- http://nptel.iitg.ernet.in/courses/Elec_Engg/IIT%20Delhi/Control%20Systems.htm
- <http://nptel.ac.in/courses/108101037/>
- <http://nptel.ac.in/courses/108102043/>
- <http://nptel.ac.in/courses/108103007/>

I. List of experiments:

1. To study basic feedback control system and perform the feedback control test.
2. To study the various test signal generator and plot the graph for each signal.
3. To find out the transfer function of the system with Block Diagram Reduction Method using MATLAB programming.
4. To generate different LTI 2nd order system responses using script file & model file.
5. To analyze the effect of various controllers on 2nd order System with MATLAB-Simulink.

6. To study and perform various commands of State Space Representation and Controllability and Observability.
7. To plot Root Locus, Bode Plot and Nyquist Plot with the help of MATLAB command.
8. To study the operation of Synchro Transmitter & Receiver.
9. To study the characteristics of various controller actions using hardware kits.
10. Steady state and frequency response behavior of systems
 - (i) To study the steady state and frequency response behavior of type "0" control system, on standard test signals.
 - (ii) To study the steady state and frequency response behavior of type "1" control system, on standard test signals.
 - (iii) To study the steady state and frequency response behavior of type "2" control system, on standard test signals.
11. To study and plot SPEED TORQUE characteristics of D.C. Servomotor.
12. To study and plot SPEED TORQUE characteristics of A.C. Servomotor.

EE246: MICROPROCESSOR & MICROCONTROLLER
4th Semester and 2nd Year (Level II)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of 8085	10
2	Introduction to Microcontroller	8
3	8051 Programming	12
4	On chip Hardware Modules/Peripherals in 8051	15
5	Interfacing Microprocessor/Microcontroller & Applications	15

Total hours (Theory) :60 Hrs
Total hours (Lab) :30 Hrs
Total hours :90 Hrs

D. Detailed Syllabus:

- 1 Basic of 8085 17% 10 Hrs
Introduction to Microprocessor, Block Diagram of 8085: Various Registers, stack pointer, stack memory, Program Counter, Instruction decoder, Bus organization; Pin diagram and Pin functions, Multiplexing/Demultiplexing of address/data bus, Types

of Memory (i.e. various type of RAM and ROM memory), External Memory Interfacing and required control signal interfacing.

- 2 **Introduction to Microcontroller** 13% 08 Hrs
Comparison between Microprocessor and Microcontroller, Introduction to 8051 Family (MCS-51) microcontrollers and salient features, Architectural block Diagram of INTEL 8051, Pin diagram and Pin Functions, General Purpose and Special Function Registers, Memory organization, Crystal Frequency & Crystal Connection, Calculation of Speed of Execution, 6/12 clock mode operation, PORT Structure.
- 3 **8051 Programming.** 20% 12 Hrs
Software & Hardware Required for Programming: Concept of Assembler, Editor, Linker, Loader, Debugger in IDE, Hardware Loader Circuit for ISP featured 8051.
Programming Language:Assembly Instruction Sets: Data Transfer & Addressing modes, Logical, Branching, Arithmetic & Bit wise Operation, Comparison of Assembly and C Programming, Data type in C, Keywords in C, Various Operator in C, Control loops, Functions and Arrays, Special Keywords & Interrupt function in Keil Microvision IDE.
- 4 **On chip Hardware Modules/Peripherals in 8051** 25% 15 Hrs
Internal Architecture of Timer/Counter and Related C Programs with all modes, Basics of Interrupt: Mask-able/Nonmask-able Interrupt, Enabling the mask-able Interrupt, Interrupt vector table, interrupt latency time and interrupt priority, External Interrupt and Related C Programs, Basics of Serial communication Protocol, Difference Between Synchronous and Asynchronous Communication, Internal Architecture of Serial Communication module in 8051 (UART) and related C programs with all modes (using SBUF register and printf library function).
- 5 **Interfacing Microcontroller & Applications** 25% 15 Hrs

Interfacing: Hardware interfacing diagram for driving a relay by ULN 2803/Transistor ,solid state relay(162341), Controlling a Lamp using Opto-isolator(ILD74),External Memory Interfacing (RAM and ROM), Keyboard Interfacing (Key Debouncing Concept, Simple Key interfacing Hardware & Software program, Matrix key board), Displays Interfacing (LED, 7-Segment display and LCD display), ADC & DAC interfacing (Serial and parallel output ADC/DAC).

Applications: Temperature measurement(LM35) and display, Counting Cars passing through passage/bottles on belt, Unidirectional and Bidirectional DC Motor Control with PWM using L293 and ILD74, Stepper Motor Drive with ULN2003.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
15	30	30	10	05	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Understand the basic building blocks, architecture of a microcontroller/ microprocessor, its salient feature and deference between various microcontrollers of MCS-51 family with basic overview of other microcontrollers
CO2:	Work on Integrated development Environment(IDE)/Compiler to Analyze and develop programs using assembly code and embedded 'c'
CO3:	Gain hands-on experience in doing experiments on microcontroller by using hardware kit in the laboratory and present the report
CO4:	Interface memory and I/O devices with 8 bit microcontroller
CO5:	Develop small embedded system and applications

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	2	-	-	-	-	-	-	-	-	-	-	1	-

CO2:	1	2	3	-	3	-	-	-	-	-	-	-	2	-
CO3:	1	1	2	3	-	-	-	-	-	1	-	-	2	-
CO4:	1	1	1	3	-	-	-	-	-	-	-	-	2	-
CO5:	1	-	3	2	3	-	-	-	-	-	-	-	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Microprocessor Architecture, Programming, and Applications with the 8085, By Ramesh Gaonkar, Penram International Publishing (India) Ltd.
2. Exploring C for Microcontrollers by JivanParab, VinodShelake, RajanishKamat, GourishNaik, Springer
3. The 8051 Microcontroller Architecture by Kenneth Ayala
4. The 8051 Microcontroller and Embedded Systems Using Assembly And C, by M A Mazidi, Janice Mazidi, RolinKinlay, Pearson Publication
5. Microcontrollers and PLCs by Sanjay Attri, Dhanpatrai& Co.

Reference Book:

1. Programming and Customizing the 8051 Microcontroller, 1999, by MykePredko, Tata McGraw Hill
2. 8051 Microcontroller – Internals, Instructions, Programming and Interfacing, 2010, by Subrata Ghoshal, Pearson
3. 8051 Microcontrollers MCS 51 family and its variants, 2010, by S. Shah, Oxford Higher Education
4. User guide for keil (c51) and Related document from: www.keil.com
5. Datasheet Intel8051 Microcontroller, ADC 0809 (Parallel ADC), MAX 1112 (Serial ADC), DAC 0808 (Parallel DAC), ULN2003, ILD74, L293
6. Programming in ANSI C by E. Bala gurusamy, McGraw Hill
7. MCS@51 Microcontroller Family User's Manual by INTEL

Web Material:

- A web tutorial on 8051:
http://old.zntu.edu.ua/base/lection/rpf/lib/zhzh03/8051_tutorial.pdf

I. List of experiments:

1. To understand 8085 Dynalog Kit and to write Assembly programs for it,
2. To Study C Programming concepts required for Embedded Microcontroller programming using Keil Microvision IDE.
3. To develop a Program for 8051 microcontroller's Port & Port Pin and to use it for Digital input and Digital Output.
4. To develop a program that toggles the status of port pin P1.0 when a falling edge comes on INT0 pin.
5. For 8051 microcontroller, develop programs for Timer 0 covering various mode of operation. (mode 0, 1, 2).
6. Write Program for 8051 microcontroller, which configure Timer0 module in counter mode & count the no of pulses coming to T0 input pin with in a duration of Timer 1 overflow.
7. To develop a program to display a string and a variable floating point number on 16 x 2 LCD (JHD162A module with HD44780 LCD controller).

8. To use serial port of 8051 for serial communication between 8051 & personal computer.
9. To develop a program for 8051 that generates a PWM signal (of a duty cycle) using two timers Timer 0 & Timer 1 individually controlling the ON time & Total time T_s respectively.
10. Develop a program that starts a Timer at zero crossing of a voltage cycle & provides a ON pulse till desired instant in a half cycle. Discuss the role of such application in converter control.

EE247: TRANSFORMERS & INDUCTION MACHINES
4th Semester and 2nd Year (Level II)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	4	8	6
Marks	100	100	200	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	50	50	200

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electromechanical energy conversion	04
2	Fundamentals of single phase transformer	09
3	Testing and performance analysis of single phase transformers	06
4	Fundamentals of three phase transformers.	07
5	Operation and performance of three phase transformers.	05
6	Construction and operation of polyphase induction machines.	09
7	Testing and analysis of three phase induction machines.	08
8	Starting and speed control of three phase induction machines.	04
9	Single phase induction motor.	08

Total hours (Theory) :60 Hrs
Total hours (Lab) :60 Hrs
Total hours : 120 Hrs

D. Detailed Syllabus:

1 Electromechanical energy conversion 06% 04 Hrs

Basic principles of electro-mechanical energy conversion, Force and torque in magnetic field systems, energy balance, Singly, multiply excited magnetic field system, Energy flow in electro mechanical devices, Force and torque in permanent magnet system, Energy conversion via electrical field, Dynamical equations of Electromechanical energy systems.

- 2 **Fundamentals of single phase transformer** 15% 09 Hrs
 Introduction; Basics of magnetic circuits, Ampere's law, linear and nonlinear magnetic circuits; Faraday's law of electromagnetic induction; Concept of an ideal transformer, assumptions; ideal transformer at no load and on load, phasor diagram, voltage current and power relations; Transformer on DC; Emf equation, Voltage and current transformation ratio, Effects of voltage and frequency variations, No load current wave shape; Basic construction of a practical single phase transformer and its types, conceptual differences with an ideal transformer; Constructional details of a practical single phase transformer, core, winding, insulation and cooling methods; Circuit model of a practical transformer incorporating an ideal transformer; Analysis of a practical single phase transformer, exact and approximate equivalent circuits, phasor diagram, relation between referred variables and parameters, per unit system of representation; Examples.
- 3 **Testing and performance analysis of single phase transformers.** 10% 06 Hrs
 Open circuit and short circuit tests, per unit parameters; Experimental determination of single phase transformer equivalent circuit parameters, example; Percentage regulation of a single phase transformer, phasor diagram, derivation, per unit representation; example; Efficiency of a single phase transformer, maximum efficiency, all day efficiency; Sumpner's test for experimental determination of a single phase transformer efficiency; Auto transformer; Examples.
- 4 **Fundamentals of three phase transformers.** 12% 07 Hrs
 Bank of three single phase transformers, voltage, current and kVA rating with different connections, open Δ connection; example; Three phase transformer as a single unit; constructional features, three limb, five limb, core type and shell type constructions; comparison with bank of three single phase transformers; Three phase transformer connections, polarity and terminal convention, vector groups; Y/Y and Δ/Δ connections, connection and phasor diagram, per phase equivalent circuit based

analysis; Y/ Δ and Δ /Y connections. Connection and phasor diagrams, per phase equivalent circuit based analysis; examples. Y/Z and Δ /Z connections. Connection and phasor diagrams, voltage, current relations, Phase conversion: 3 to 6 phase and 3 to 2 phase conversions (Scott Connections).

- 5 **Operation and performance of three phase transformers.** 08% 05 Hrs
Tap changing transformer and voltage control; Parallel operation of single and three phase transformers, conditions and connection diagrams, per phase equivalent circuit based analysis with equal and unequal no load voltages, load sharing; Harmonics and switching transients in transformers, effect of transformer connections, inrush current; Examples
- 6 **Construction and operation of polyphase induction machines.** 15% 09 Hrs
Introduction, Classification of three phase induction motor, Advantages, Disadvantages and Application of Induction Motor, Production of three phase rotating magnetic field, Locking of stator and rotor field, Construction, Principle of operation, slip, Frequency of rotor current, Speed of rotor field, Rotor EMF, Rotor current, Rotor reactance under starting and running condition, Relation between torque and rotor power factor, Torque equation, Torque under running condition, Torque-Speed curves and effect of change in rotor resistance, Operating region, Starting torque, Full load torque, Breakdown torque, Condition for maximum torque, Effect of change in supply voltage and frequency on torque and slip, Braking of induction motor, examples. Losses in induction machines, Power flow diagram in a three phase induction machine, air gap power, slip power, mechanical power; Torque, Synchronous watt, Efficiency, examples.
- 7 **Testing and analysis of three phase induction machines.** 14% 08 Hrs
Induction motor as generalized transformer, Phasor diagram of induction motor, Exact and approximate per phase equivalent circuit; Induction motor no load test and blocked rotor test, determining equivalent circuit parameters; losses and efficiency, examples; Concept of circle diagram, Series circuit and current locus, Construction of circle diagram, Induction machine performance computation from circle diagram; examples. Effect of space harmonic fields, harmonics induction torques, Harmonic synchronous torques, Crawling and Cogging, Deep bar and double cage induction motor (High torque motors), Equivalent circuit, Characteristics, examples.

8 Starting and speed control of three phase induction machines. 06% 04 Hrs

Starting of squirrel cage and slip ring induction motor; Direct on line starting, stator resistance and reactance based starting, rotor resistance based starting; Y/ Δ and auto transformer based starting; examples; Speed control of induction motors by stator voltage variation and pole changing techniques, Variable frequency speed control of induction machines, v/f control method, slip power control method.

9 Single phase induction motor. 14% 08 Hrs

General description of the motor and constructional features; Rotating and pulsating field, development of equivalent circuit based on double revolving field theory; starting of single phase induction motor, Determination of parameters by test, performance analysis, Types of induction motor: split phase or resistance start, Capacitor-start, Capacitor start capacitor-run, Permanent capacitor, Shaded pole motor: Construction, Working, Starting and running performance, Characteristics and Applications; selection of capacitor value for starting and running conditions, Selection capacitance values for maximum torque, examples.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Get in-depth knowledge in fundamental concepts of electromechanical conversion equipment and transformer
CO2:	Acquire the knowledge about the construction details, operating principle, performance characteristics of induction machines
CO3:	Acquire the knowledge of testing and applications of induction machines
CO4:	Acquire the knowledge about the construction details, performance characteristics, testing and applications of transformers
CO5:	Familiarize with the recent developments in transformers and induction machines to get

the opportunity to work as a team member/leader in multi-disciplinary branches

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	2	2	1	1	-	-	-	-	-	-	-	2	2	-
CO2:	2	2	2	1	-	-	-	-	-	-	-	2	2	-
CO3:	1	2	1	1	-	-	-	-	-	2	-	1	2	-
CO4:	2	2	2	1	-	-	-	-	-	2	-	2	2	-
CO5:	2	1		2	-	-	-	-	3	-	-	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Theory and performance of electrical machines by J.B. Gupta, S.K. Kataria and sons
2. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
3. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill

4. The performance and design of alternating current machines by M.G. Say, CBS Publishers & Distributors
5. A textbook of electrical technology VOL II (AC& DC machines) by B.L.Theraja & A.K. Theraja S. Chand Publication.

Reference Book:

1. Electrical Machines by D.P. Kothari & I. J. Nagrath,
2. Electrical Machinery by Dr. P.S. Bimbhra, Khanna Publisher
3. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill
4. Fundamentals of Electrical Machines by B.R. Gupta, New Age International Publishers
5. Electrical machine Drives and Power System by Wildi, 6th Edition, Pearson Publications.
6. Charles .I. Hubert, Electrical Machines (Theory, operation, application, adjustment and control) Pearson Publication, 2nd edition.
7. Theory of Alternating of Alternating Current Machinery by A.S Langsdorf.

Web Material:

- <http://nptel.ac.in/courses/108105017/>
(video lectures Prof.D. Kastha, IIT Kharagpur)
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_Machines_I
- www.elk.itu.edu.tr/~ozdemir/DC-notes-1
- <http://www.ece.ualberta.ca/~knight/ee332/ee332.htm>

I. List of experiments:

1. Open circuit and short circuit tests on single-phase transformer.
2. Determination of equivalent circuit parameters of transformer.
3. Load test on single-phase transformer and three phase transformers.
4. Parallel operation of single phase transformer
5. Parallel operation of three phase transformer
6. Sumpner's test on single phase transformers.
7. Scott Connection
8. Separation of no-load losses in single-phase transformer.

9. No load and Blocked rotor test of single-phase induction motor.
10. Determination of equivalent circuit parameters and performance of single phase induction motor
11. No load and Blocked rotor test of poly phase induction motor.
12. Load test of poly phase induction motor
13. Speed control of wound rotor induction motor
14. Speed control of squirrel cage induction motor
15. Computation of circle diagram.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS133 A/B/C/D/E/F/G/H
Creativity, Problem Solving and Innovation (Sem-IV)

I. Credits and Schemes:

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
IV	HS133 A/B/C/E/F/G/H	Creativity, Problem Solving and Innovation	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Introduction to Creativity, Problem Solving and Innovation - Definitions of Creativity and Innovation - Need for Problem Solving and Innovation - Scope of Creativity in various Domains - Types and Styles of Thinking - Strategies to develop Creativity, Problem Solving and Innovation skills	06
2	Questioning, Learning and Visualization - Strategy and Methods of Questioning - Asking the Right Questions - Strategy of Learning and its Importance - Sources and Methods of Learning - Purpose and Value of Creativity Education in real life - Visualization strategies - Making thoughts Visible - Mind Mapping and Visualizing Thinking	06

3	Creative Thinking and Problem Solving Creative Thinking and its need Strategy of Thinking Fluency Generating all Possibilities SCAMPER Technique Divergent Vs Convergent Thinking Lateral Vs Vertical Thinking Fusion of Ideas for Problem Solving Applying strategies for Problem Solving	06
4	Logic, Language and Reasoning • <i>Basic Concepts of Logic</i> • <i>Statement Vs Sentence</i> • <i>Premises Vs Conclusion</i> • <i>Concept of an Argument</i> • <i>Functions of Language: Informative, Expressive and Directive</i> • <i>Inductive Vs Deductive Reasoning</i> • <i>Critical Thinking & Creativity</i> • <i>Moral Reasoning</i>	06
5	Contemporary Issues and Practices in Creativity and Problem Solving • <i>Cognitive Research Trust Thinking for Creatively Solving Problems</i> • <i>Case Study on Contemporary Issues and Practices in Creativity and Problem Solving</i>	06
Total		30

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

IV. Evaluation:

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
---------	-----------	--------	---------------------	-------------

1	Attendance	100 %	--	20
2	Individual Activity Participation	As stipulated by the Resource Person(s) in the Training		20
3	Group Activity Participation			20
4	Presentation			30
5	Feedback on Improvement			10
Total				100

V. Course Outcome (COs):

After completion of the course, the student would :

CO1	Demonstrate creativity in their day to day activities and academic output
CO2	Solve personal, social and professional problems with a positive and an objective mindset
CO3	Think creatively and work towards problem solving in a strategic way
CO4	Initiate new and innovative practices in their chosen field of profession

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	1	-	-	-	-	-	-	-	-	-
CO2	1	2	3	1	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	1	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	1	2	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Books / Reference Books / Reading

Text Books

1. R Keith Sawyer, Zig Zag, The Surprising Path to Greater Creativity, Jossey-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genius, Ten Speed Press 2001

Reference Books

3. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
4. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994

5. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
6. Tony Buzan, How to Mind Map, Thorsons Publication 2002
7. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
8. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
9. Ira Flatow, The all Laughed, Harper Publication 1992
10. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

11. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
12. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
13. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
14. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
15. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
16. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
17. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
18. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
19. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
20. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
21. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
22. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
23. Howard Gardner, Creating minds, Basic Books Publication 1993
24. Mihaly Csikszentmihalyi, Creativity–Flow and Psychology of Discovery and Invention, Harper Publication 1996
25. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
26. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994
27. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

□ Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Th.	Pr.	Total		Int.	Ext.	Int.	Ext.	
TY Sem– V	EE341	Electrical Power Transmission and Distribution	4		4	4	30	70			100
	EE342	Synchronous and DC Machines	4	4	8	6	30	70	50	50	200
	EE343	Power Electronics and Drives – I	4	2	6	5	30	70	25	25	150
	EE344	Minor Project I		4	4	2			50	50	100
	EE345	Electrical Product Survey		2	2	2			50	50	100
	EE350	Summer Internship I		3	3	3			75	75	150
	HS124.01A	Professional Communication		2	2	2			30	70	100
	EE371/EE372	Advanced Microcontrollers/ VLSI Technology and Design	4	2	6	5	30	70	25	25	150
	TOTAL:		16	19	35	29	120	280	305	345	1050
TY Sem - VI	EE346	Digital Signal Processing	3	2	5	4	30	70	25	25	150
	EE347	Programmable Logic Controllers and Industrial Automation	2	4	6	4	30	70	50	50	200
	EE348	Power Electronics and Drives II	3	2	5	4	30	70	25	25	150
	EE349	Fault Analysis and Switchgear	4	2	6	5	30	70	25	25	150
	EE360	Minor Project II		4	4	2			50	50	100
	HS134A	Contributor Personality Development	2		2	2	30	70	-	-	100
	EE376/EE377	Special Electrical Machines and Applications/ Embedded Systems	3	2	5	4	30	70	25	25	150
	TOTAL:		17	16	33	25	180	420	200	200	1000

B. Tech. (Electrical Engineering) Programme

SYLLABI

(Semester – V)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE341: ELECTRICAL POWER TRANSMISSION AND DISTRIBUTION
5th Semester and 3rd Year (Level III)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	00	4	4
Marks	100	00	100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70			100

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Transmission Line Parameters	10
2	Characteristic and Performance of Transmission Line	10
3	Design of Transmission system	10
4	Design of Power System and Substation	08
5	Design of Distribution system	10
6	Economics of Distribution System	05
7	Power System Planning	07

Total hours (Theory) : 60 Hrs
 Total hours (Lab) : 00 Hrs
 Total hours : 60 Hrs

D. Detailed Syllabus:

- 1 **Transmission Line Parameters** 17% 10 Hrs
 Basics of Electric field.
 Resistance of line, basic concepts of inductance and capacitance.

Flux linkages of an isolated current carrying conductor.

Inductance of a single phase two wire line, examples.

Flux linkages of one conductor in a group and inductance of composite conductor lines, examples.

Inductance of three phase line with unsymmetrical spacing and symmetrical spacing, importance of transposition of tower, examples. Inductance of double circuit line, examples.

Inductance of bundled conductors and importance of bundled conductors, examples.

Electric field of a long straight conductor, potential difference between two conductors of a group of parallel conductors.

Capacitance of two wire line. Capacitance of three phase line with equal and unequal spacing, examples.

Effect of earth on transmission line capacitance.

2 Characteristic and Performance of Transmission Line 17% 10 Hrs

Voltage regulation of short transmission line, example.

T and π representation of medium transmission line and their Phasor diagram, example.

Performance analysis of long transmission line and its ABCD parameter representation.

Surge impedance and surge impedance loading, Equivalent T and π representation of long line, Interpretation of long line equation, Ferranti effect. Examples.

Power flow through the single transmission line between two bus and important observations.

3 Design of Transmission system 17% 10 Hrs

Requirement of Transmission lines, selection of voltage for high-voltage transmission lines. choice of conductors , spacing of conductor, corona.

Surge impedance loading of transmission lines.

Electrical design of transmission lines, Earth Wires and towers.

Transmission of Electrical Power at extra-high voltage. Design consideration of EHV lines.

Selection & spacing of conductors for EHV line.

Radio & television interference.

Insulation co-ordination & towers.

4 Design of Power System and Substation 13% 08 Hrs

Introduction ,Selection of size and location of generation stations.

Selection of specification of transmission lines.

Size and locations of substations. Role of vector group in designing of sub-station.

Sub-station Grounding, Earthing, step and touch potential.

Interconnection.

Bus-bar arrangements in sub-station.

5 Design of Distribution system 17% 10 Hrs

Distribution system, requirements of distribution system, design consideration of distribution system, classification of distribution system.

Connection schemes of Distribution system.

D.C. distribution system, Types of distribution system.

A.C. distribution system, A.C. distribution system calculations.

6 Economics of Distribution System 08% 05 Hrs

Comparison of overhead- transmission line and distribution systems.

Effect of voltage

Selection of equipment of power systems.

Economic size of power factor improvement apparatus.

Economic selection of Distribution system.

7 Power System Planning 11% 07 Hrs

Introduction, Methods of Power system planning.

Forecasting load and energy requirements.

Generation Planning , Transmission System planning.

Distribution System Planning.

Reliability of electrical power system, Method of measuring power system reliability.

Trends in power system Planning in India.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
10	15	25	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Calculate the different parameters of transmission line with different configuration of transmission line & bundled conductor
CO2:	Analyze & calculate the performance of short, medium & long transmission line and power transferred through transmission line
CO3:	Design of EHV transmission line with tower and Demonstrate the knowledge regarding the surge impedance loading of transmission line.
CO4:	Chose the size and location of generating station and sub-station. Prepare the specification of transmission line, Construct the arrangement of different parts of sub-station and bus-bar arrangement
CO5:	Design and Connection schemes of A.C. & D.C. distribution system can create the best alternative based on economics
CO6:	Acquire the knowledge of Power system planning for Generation, Transmission & Distribution System

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	3	-	-	-	-	-	-	-	-	-	-	1	1
CO2:	3	3	-	-	-	-	-	-	-	-	-	-	2	1
CO3:	2	2	3	-	3	-	-	-	3	3	-	-	3	3
CO4:	-	-	3	3	3	3	-	2	3	2	-	-	3	3
CO5:	-	3	3	3	-	-	-	1	-	-	-	-	3	3
CO6:	-	-	-	2	3	3	2	-	3	3	3	3	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The visit of nearby substation will clear the concept of transmission line design and student will realise the actual size of transmission towers and spacing of conductors.
- In lab session students have hands on practical about the different configuration of transmission line.

H. Recommended Study Material:

Text Book:

1. Electrical Power System Design, by M. V. Deshpande, Tata Mcgraw Hill.
2. Principles of Power system by V.K.Mehta, Publisher: S.Chand & Company Ltd.
3. A Course in electrical power by Soni & Bhatnagar, Dhanpat Rai & Sons.
4. Sub-station Design and Equipment by P.V. Gupta and P.S.Satnam, Dhanpat Rai & Sons.

Reference Book:

1. Handbook of Electrical Power Distribution, by Ramamurthy University Press (I) Pvt. Ltd
2. Power System Analysis and Design by B.R. Gupta, S.Chand Publications
3. Power system analysis by Hadi Saadat, Tata McGraw Hill Publishing Company, New Delhi
4. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath, Tata Mcgraw Hill
5. Power System Analysis by Grainger & Stevenson, Tata Mcgraw Hill

Web Material:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/powersystem/ui/TOC.htm>
- http://en.wikipedia.org/wiki/Electric_power_transmission

EE342: SYNCHRONOUS AND DC MACHINES
5th Semester and 3rd Year (Level III)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	4	8	6
Marks	100	100	200	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	50	50	200

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Fundamentals of DC Machines	12
2	Operation and performance Characteristics of DC Machine	12
3	Construction and Operation of Synchronous Generator	12
4	Voltage Regulation, Power Equations and Synchronization of Alternators	14
5	Synchronous Motor	10

Total hours (Theory) :60 Hrs

Total hours (Lab) :60 Hrs

Total hours :120 Hrs

D. Detailed Syllabus:

1 Fundamentals of DC Machines 20% 12 Hrs

Classification of rotating electrical machines, Construction of DC machines, Comparison of AC and DC Generators, Principle of DC generator and motor, Comparison of motor and generator action, Elementary DC generator, essential parts,

Types of DC machines (methods of excitation), EMF equation, Voltage build up process, Causes of its failure, Critical resistance and speed, Armature reaction and its effect, Commutation in DC machines, Reactance voltage, methods of improving commutation, compensating winding.

- 2 **Operation and performance Characteristics of DC Machine.** 20% 12 Hrs
 Characteristics of separately excited DC machines (motor and generator), Characteristics of shunt, series and compound machines (generators & motors), Power flow diagram of DC machines, Losses in DC machines, Efficiency, condition for maximum efficiency, Voltage regulation in DC generator, Speed regulation in DC motor, Parallel operation of DC machines, Examples, Necessity of starter, types of starter, Speed control of DC shunt and series motors, Braking of DC motors, Examples, Applications of all types of DC generators and DC motors.
- 3 **Construction and Operation of Synchronous Generator** 20% 12 Hrs
 Introduction, classification of synchronous machine, details of construction, advantages of rotating field type of construction, difference between salient and cylindrical alternator, damper winding, operating principle, Speed and Frequency, Synchronous speed, pitch factor, distribution factor and emf equation, emf waveform for concentrated and distributed winding, Examples, armature leakage reactance, synchronous reactance and impedance, armature reaction, Equivalent circuit, phasor diagram, Synchronous generator operating alone, operation of alternator on no load and on load, effect of variation in load on synchronous generator, Laboratory methods of determination of synchronous reactance, oc/sc test, determination of effective resistance of armature, Short circuit ratio, Losses and efficiency, power flow diagram, Examples.
- 4 **Voltage Regulation, Power Equations and Synchronization of Alternators** 23% 14 Hrs
 Voltage Regulation, methods of calculating voltage regulation, Direct load test, EMF method, MMF method, ZPFC method, Comparison of methods, examples, Active, Reactive and Complex Input and Output Power Equations for cylindrical synchronous generator, Maximum input and output power and torque developed, Power Angle Characteristics, Two reactance theory for salient pole synchronous machine, Phasor Diagram, Power Equations and Torque Angle Characteristics, Determination of X_d and X_q by slip test, Capability Curves for Synchronous

Generators, Prime mover characteristics, House Diagram, Frequency Power and Voltage Reactive Power relations (F-P and Q-V), Power shared by two alternators, Necessity of parallel operation, condition required for parallel operation, the general procedure for synchronization of alternators, Infinite Bus, Synchronizing current, power, torque, Time period of Oscillations, effect of increasing the driving torque of one the alternator, effect of change in excitation of one of the alternator, load sharing between two alternators, operation of synchronous generator with infinite bus bar, operation of synchronous generator with other generator, various types of alternators based on prime movers, Factors affecting the size and efficiency of alternators, Transients of the alternator

5 Synchronous Motor

17% 10 Hrs

Introduction, construction, principle of operation, Equivalent circuit and phasor diagram for cylindrical and salient rotor, synchronous motor with different excitation, starting methods of synchronous motor, power flow equations for synchronous motor, examples, steady state synchronous motor operation - effect of load change on synchronous motor, effect of field current change on synchronous motor, power flow diagram, examples, Different torques of synchronous motor, stability and maximum load angle, construction of V curves and inverted V curves, O curves, Synchronous condenser, synchronous phase modifier, hunting, speed control of synchronous motor, merits, demerits and application. Comparison of three phase synchronous and induction motors

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Able to recognize the electric and magnetic circuits of AC and DC machines
CO2:	Acquire the knowledge regarding the fundamentals, working and construction of Synchronous and DC machines
CO3:	Develop skill for the practical exposure of rotating electrical machines and understand the various testing methods of Synchronous and DC machines.
CO4:	Aware with the performance characteristics of Synchronous and DC machines.
CO5:	Analyze Synchronous and DC machines performance and application various conditions.

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	1	-	-	-	-	-	-	-	-	-
CO2:	2	3	1	1	1	-	-	1	-	-	-	-	-	-
CO3:	3	2	3	2	1	-	-	-	-	-	-	2	2	1
CO4:	-	3	2	2		-	-	-	1	-	-	-	-	-
CO5:	2	3	2	3	2	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Electrical Machines, by Ashfaq Husain, Dhanpat Rai & Co.

2. A textbook of Electrical Machines by K R Siddhapura and D B Raval, Vikas Publishing House Pvt. Ltd.
3. Theory and performance of electrical machines by J.B.Gupta, S.K.Kataria and sons
4. A textbook of electrical technology VOL II (AC & DC machines) by B.L.Theraja & A.K.Theraja S. Chand Publication.

Reference Book:

1. Electrical Machines III by K Krishna Reddy, Scitech Publication (INDIA) Pvt. Ltd.
2. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
3. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill
4. The performance and design of alternating current machines by M.G.Say, CBS Publishers & Distributors
5. Electrical Machinery by Dr. P.S.Bimbhra, Khanna Publisher
6. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill
7. Fundamentals of Electrical Machines by B.R. Gupta, New Age International Publishers
8. Charles .I. Hubert, Electrical Machines (Theory, operation, application, adjustment and control) Pearson Publication, 2nd edition.
9. Theory of Alternating of Alternating Current Machinery by A.S Langsdorf.

Web Material:

- <http://nptel.ac.in/courses/108106072/>
(video lectures Prof.D. Kastha, IIT Kharagpur)
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_Machines_I
- www.nptel.ac.in/courses/IIT-MADRAS/Electrical_Machines_II/pdf/2_1.pdf
- <http://www.ece.ualberta.ca/~knight/ee332/ee332.htm>

I. List of experiments:

1. To study operating principle, working and cut model of DC machine
2. To perform the magnetization (open circuit) characteristics and load test on a DC shunt generator to obtain the internal and external load characteristics.

3. To perform the load test on a DC series generator to obtain the internal and external load characteristics.
4. To perform the speed control of a DC shunt motor.
5. To perform speed control of a DC series motor.
6. To study operating principle, working and cut model of 3 phase alternators.
7. Load test of 3 phase alternators
8. To find the voltage regulation of 3 Phase alternator using a synchronous impedance method.
9. To find the voltage regulation of 3 phase alternator using a ampere turn method.
10. To find the voltage regulation of 3 phase alternator using a Potier Triangle Method.
11. To perform a slip test on synchronous generator.
12. To perform parallel operation alternators and study of synchronization methods.
13. To obtain V curves and inverted V curves
14. To obtain O curve of synchronous motor.
15. To study AC machine windings.
16. To study DC machine windings.

EE343: POWER ELECTRONICS AND DRIVES - I
5th Semester and 3rd Year (Level III)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction	02
2	Power Semiconductor Devices	17
3	Triggering, Commutation and Driver Circuit	10
4	DC-DC Converters (Choppers)	10
5	AC-DC Converters (Controlled Rectifier)	10
6	DC Motor Drives	11

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

D. Detailed Syllabus:

1	Introduction	03.33%	02 Hrs
	Introduction, Scope and applications, Classification of power electronic converters.		
2	Power Semiconductor Devices	28.33%	17 Hrs

	Diode: Classification and its application SCR: V- I characteristics, Switching characteristics, Gate characteristics, Ratings, Effect of dv/dt and di/dt , Snubber circuit and its design Thyristor Family: Triac, DIAC, LASCR, SUS, SCS Gate commutated devices: GTO, MOSFET and IGBT		
3	Triggering, Commutation and Driver Circuit	16.67%	10 Hrs
	Triggering circuits: Resistance triggering circuit, Resistance capacitance trigger circuit, DIAC trigger circuit, UJT based trigger circuit; Micro-controller based Trigger circuit Commutation circuits: Line commutation, Load commutation, Forced commutation, External pulse commutation Driver Circuit: Optocoupler and pulse transformer based SCR driver circuit, Gate drive circuit for power MOSFET, Driver circuit for IGBT and BJT		
4	DC-DC Converters (Choppers)	16.67%	10 Hrs
	Introduction, Principle of chopper operation, Control strategies, Step down (buck) converter, Step up(boost) converter, Step up down converter, Cuk DC-DC converter, Chopper configuration		
5	AC-DC Converters (Controlled Rectifier)	16.67%	10 Hrs
	Introduction, Principle of phase control, Single phase half wave converter with different loads, Single phase full wave converter with different loads, Single phase semi converter, Three phase half wave converter with different loads, Three phase full converter, Three phase semi converter with different loads, Dual converter, Multi-pulse Rectifier		
6	DC Motor Drives	18.33%	11 Hrs
	Concept of Electric Drive, Basic DC Machine Equations and its characteristics, Control of DC separately excited motor from single phase and three phase controlled rectifier, Rectifier control of DC series motor, Chopper control of separately excited DC motor, Chopper control of DC series motor, Multi quadrant operation of, DC separately excited motor		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Enhance their skills in basics of Power Electronic devices and converters
CO2:	Know the performance and classification of Power Electronic converters
CO3:	Understand applications of converters used in industrial drives
CO4:	Design and troubleshoot hardware circuits like choppers, Inverters, Controlled rectifiers etc.
CO5:	Hardware implementation of Power Electronic Converters and its control strategies.

Mapping of course outcomes with program outcomes Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	2	3	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	3	3	-	2	-	-	-	-	-	-	-	2	3
CO5	-	3	-	-	3	-	-	-	2	-	-	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Power Electronics by P.S.Bimbhra, Khanna Publishers
2. Power Electronics Circuits, Devices and Applications by Muhammad H. Rahid, Pearson Education
3. Power Electronics by M.D. Singh and K.D.Khanchandani, Tata McGraw-Hill Education

Reference Book:

1. Fundamentals of Electrical Drives by Gopal K. Dubey, Alpha Science.
2. Power Electronics Converters, Applications and Design by Ned Mohan, Tore M Undeland and William P. Robbins, Wiley India.
3. Power Electronics and Motor Drives by Bimal K. Bose, Academic Press.
4. Power Electronics: Devices, Drivers, Applications, and Passive Components by B. W. Williams, Tata McGraw-Hill

Web Material:

- <http://www.nptel.ac.in/courses/108105066/>
(Notes from IIT Kharagpur)
- <http://www.nptel.ac.in/courses/108101038/1>
(Videos and Notes by Prof. B.G. Fernandes, IIT Bombay)
- <http://www.nptel.ac.in/courses/108108035/1>
- <http://www.nptel.ac.in/courses/108108036/>
(Videos and Notes by Prof. L. Umanand, IISC Bangalore)
- <http://www.nptel.ac.in/courses/108102046/>
- <http://www.nptel.ac.in/courses/108108077/>
(Videos and Notes by Prof. K. Gopakumar, IISC, Bangalore)

I. List of experiments:

1. (A) To perform the DC Gate control characteristics of a SCR.
(B) To perform and plot the Anode Current Characteristics of a SCR.
(C) To measure the holding current of a SCR.
(D) To perform the phase firing characteristics of a SCR.
2. (A) To perform the V-I characteristics of a DIAC.
(B) To perform application of DIAC as Sawtooth Wave Generator and Pulse Train Generator.
3. To perform the operation of UJT as Relaxation Oscillator.
4. To Study the single-phase Fully Controlled Bridge Rectifier.

5. (A) To study the gate characteristics of a TRIAC.
(B) To study the Terminal characteristics of a TRIAC.
(C) To measure the Holding current of a Triac.
6. (A) To study the application of a Triac as a Static Switch with DC control.
(B) To study the application of a Triac in control of AC with AC signal.
7. To Study the step down voltage commutated thyristorised chopper.

EE344: MINOR PROJECT I
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	04	4	2
Marks	00	100	100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		50	50	100

C. Instructional Method and Pedagogy:

- Minor project-I has 2 credit subject and 4 hours per week.
- Department/Institute will help students to find appropriate minor projects.
- The student must be find the appropriate guide for their project and make the project
- proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the minor project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which you gain during project making. The guide must verify report.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the project.

D. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done during project, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily

accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

E. Course Outcomes (COs):

After completion of the course students, will able:

CO1:	To apply knowledge and skills learned from project to real-world problems
CO2:	To solve engineering problems by the team work and work with other discipline
CO3:	Use experience related to professional and ethical issues in the work environment
CO4:	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur.
CO5:	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies

Mapping of course outcomes with program outcomes

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

EE345: ELECTRICAL PRODUCT SURVEY
5th Semester and 3rd Year (Level III)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	02	02	2
Marks	00	100	100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		50	50	100

C. Instructional Method and Pedagogy:

- Electrical product survey has 2 credit subject and 2 hours per week.
- Students can find his/her electrical product.
- The student may take the guidance from department faculty members.
- Product must be from Electrical bag round.
- The student shall submit product presentations to the Coordinator for the evaluation of the product survey.
- After the completion of product survey, a soft copy of “presentation” must be submitted to the Coordinator by the end of the semester.
- The students must make a group of 3 or 4 and submit their proposal to departmental coordinator.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision. If the revised presentation is still unsatisfactory, the student shall be requested to repeat the project.

D. Course outcomes (COs):

After completion of the course students will able:

COI:	To apply knowledge and skills learned from project to real-world engineering problems
------	---

CO2:	To function in a team work with teammates from other disciplines
CO3:	To use experience related to professional and ethical issues in the work environment.
CO4:	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
CO5:	To demonstrates knowledge of contemporary issues related with engineering in general with the use new tools and technologies

Mapping of course outcomes with program outcomes
Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2											2	
CO2	1		1	1	2				3					1
CO3	1					1		2						1
CO4	1						2		1		2		1	1
CO5	3	1		1								1		2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

EE350: SUMMER INTERNSHIP I
5th Semester and 3rd Year (Level III)
B. Tech (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week (In Summer Break)	00	30	90	3
Marks	00	150	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		75	75	150

C. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report

- Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

D. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company

- Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- **INTRODUCTION:** In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
 - **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
 - **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
 - **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
 - **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
 - **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

E. Course Outcomes (COs):

After completion of the course students will able:

CO1:	To solve engineering problems by working in a team with across other disciplines
CO2:	To apply knowledge and skills learned in company/industry/organization to real-world problems

CO3:	To use experience related to professional and ethical issues in the work environment
CO4:	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
CO5:	To demonstrates knowledge of contemporary issues related with engineering in general by adopting new tools and technologies

Mapping of course outcomes with program outcomes

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HSI24.01A: PROFESSIONAL COMMUNICATION
5th Semester and 3rd Year

I. Credits and Schemes

Credits	Teaching Scheme	Evaluation Scheme				
	Contact Hours/Week	Theory		Practical		Total
		Internal	External	Internal	External	
02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction Professional Communication • <i>Concept & Applications of Professional Communication</i> • <i>Principles of Professional Communication</i> • <i>Communication networks: personal sanctum, professional sanctum, inner circle, and outer circle; managing the networks</i> • <i>Communication strategies: communicator, audience, message, channel choice, culture</i>	04
2	Professional Communication and Rhetorics • <i>Concepts of “Communication” and “Professional Communication” and “Rhetorics”</i> • <i>Orientation towards the Concepts of Professional Communication and Rhetorics (Speaking)</i> • <i>Importance of Ethos, Logos, and Pathos in Professional Communication</i> • <i>Principles of Professional Communication (visual, oral and non-verbal)</i>	04
3	Cross-cultural Communication and Globalization • <i>Basic Concepts: Culture, Globalization and Cross-cultural Communication</i> • <i>Communicating with People of Different Cultures</i> • <i>Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them</i>	04
4	Written Professional Communication • <i>Importance of Written Professional Communication</i> • <i>Letter Writing, E-mail Writing, Report Writing</i> • <i>Resume Building</i>	06

5	Academic Writing • Importance of Academic Writing • Research Paper Writing • Article/ Review Writing • Reference and Citation	06
6	Effective Presentation Strategies • Why and How in Presentation • Audience Analysis and Supporting Material • Presentation Mechanics and Presentation Process • Managing Yourself during Q and A Session • Fundamentals of Persuasions	06
Total		30

III. Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

IV. Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	02	05	10
2	Project	01	15	15
3	Attendance			05
Total				30

External Evaluation

University Practical Examination will be for 70 marks to be conducted at the end of the semester. Details are:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical / Viva	01	70	70
Total				70

V. Course Outcome (COs):

After completion of the course, the student would :

CO1	Gain basic conceptual understanding of communication skills in Professional settings
-----	--

CO2	Develop awareness and competence in communicating across cultures in professional settings
CO3	Develop confidence and competence in speaking in formal interviews and to make formal presentations
CO4	Develop necessary soft skills to work collaboratively in a group and participate in a group Discussion for employment and in the workplace.
CO5	Develop team and group dynamics to work well in multi-disciplinary and cross-cultural work environment
CO6	Develop writing skills to prepare CV's, Resumes, Statement of Purpose and preparation of other formal documents.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	2	-	-
CO5	-	-	-	-	-	-	-	-	2	3	-	-	2	-
CO6	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VI. Reference Books

- Eckhouse, B. E. (1999). *Competitive Communication: A Rhetoric for Modern Business*. Oxford University Press. (for Module - II)
- Koneru, A. (2008). *Professional Communication*. New Delhi: Tata Mcgraw-Hill. (for Module I & III)
- Meenakshi Raman, P. S. (2006). *Business Communication*. Meenakshi Raman, Prakash Singh. (for module I and IV)
- Parul Papat, K. K. (2015). *Communication Skills*. New Delhi: Pearson. (for Module V & VI)
- Sanjay Kumar, P. L. (2015). *Communication Skills*. Oxford University Press India. (for Module I, III and V)

EE371: ADVANCED MICROCONTROLLERS

Programme Elective II

5th Semester and 3rd Year (Level III)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1.	Embedded C Language Programming for Microcontroller	12
2.	Introduction to c8051f120 Microcontroller	08
3.	Timers in c8051f120 Microcontroller	12
4.	Analog Peripherals of c8051f120 Microcontroller	12
5.	PCA & Application	10
6.	Communication Protocol	06

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

D. Detailed Syllabus:

1.	Embedded C Language Programming for Microcontroller	20%	12 Hrs
	Comparison of Assembly & High Level Programming, Concept of Super loop/Infinite loop used in Embedded C Programs, Concept of Round Robin Execution in Real Time Operating System based Programming Environment.		

	Introduction to Keil Microvision IDE & Related debugging Techniques, Basic Working of various Control Loops, Variables: Global & Local, Various types of Variables & it's Memory Scope, Mathematical & Logical Operators, Array, Structure & Union, Pointers and its use in programming. Various C library and its use in programming.		
2.	Introduction to c8051f120 Microcontroller	13%	08 Hrs
	Comparison of 8051 & CIP-51 core Microcontrollers, Peripherals available in c8051f120 Microcontroller, Watchdog Timer & it's Reset codes, System Clock Configuration with & without Internal PLL, SFR & SFR Paging, Concept of Crossbar & Crossbar Decoder, Internal Structure & Configuration of Port in Open drain & Push pull mode with & without Weak pull up disabled.		
3.	Timers in c8051f120 Microcontroller	20%	12 Hrs
	Block diagram & Operation of Timer0 & 1 with Associated SFR, Calculation of input clock frequency for various SYSCLKOUT, Various Modes of Timer & Associated Programs With & Without Interrupt enabled. Block diagram & Operation of Timer2,3 & 4 with Associated SFR. Various modes of Operations & Associated Programs.		
4.	Analog Peripherals of c8051f120 Microcontroller	20%	12 Hrs
	Basic Terminology for A/D Conversions, Specifications of c8051f120 A/D Converters: ADC0 & ADC2, Block diagram & Working of 12 bit ADC0 & 8 bit ADC2 A/D Converters, Associated SFR, Start of Conversion using Various modes, ADC0 in Window Comparator Mode, Temperature sensing using On-chip Temperature Sensor, Programs Related to ADC Operation, Digital to Analog Converter in c8051f120 with various mode of conversion, Generation of waveforms using DAC, On-chip Comparator.		
5.	PCA and Application	17%	10 Hrs
	Block diagram of PCA module, PCA Timer in Capture Compare, High Speed output, Frequency Output & 8/16 Bit Pulse Width Modulation Mode.: Associated Register & Working, Associated Programs.		
6.	Communication Protocol	10%	06 Hrs
	SPI, SMBUS and I2C, UART Communication.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	20	25	20	15

F. Course Outcomes (COs):

At the end of this course, the student should:

CO1:	To understand C programming fundamentals in detail, and be able to differentiate between super loop based and RTOS execution, and to understand fundamentals of programming in both approaches
CO2:	Have a good understanding of the architecture and programming model of the c8051f120 microcontroller and to write C program for using various internal modules and driving external hardware
CO3:	Be aware of software/code debugging and related techniques and also to rectify it and also to think about faults in hardware and hardware design and to solve it
CO4:	To be able to access peripherals in 32 bit microcontroller using pointer, structure, union and write pseudo code with registers for modules available in unknown (Which is not directly covered in course) microcontroller

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	3	-	2	3	1	-	2	-	-	-	-	2	-
CO2:	3	3	3	-	3	1	-	-	2	2	2		2	-
CO3:	3	3	-	-	3	2	-	-	-	-	-	-	2	-
CO4:	3	3	-	-	3	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials/ Assignments based on course content will be conducted/ given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build and appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

- [1] “Embedded System Design using C8051” by Han-Way Huang, Cengage Engineering, 2009. ISBN-13: 9788131512241.
- [2] “Introduction to Embedded Systems” by Shibu K V, Tata McGraw Hill.

Reference Book:

- [1] “Microprocessors & Interfacing”, Douglas Hall, Tata McGraw Hill
- [2] The 8051 Microcontroller and Embedded Systems using Assembly and C by Muhammad Ali Mazidi, Pearson
- [3] Exploring C for Microcontrollers: A Hands on Approach, Jivan S. Parab, Vinod G. Shelake, Springer

Web Material:

- [1] http://www.keil.com/dd/docs/datashts/cast/cast_c8051.pdf (Data sheet of C8051)

I. List of experiments:

1. To understand operation of CrossBar, SFR paging using C-programming and debugging on keil microvision platform.
2. To configure the internal oscillator and PLL for various SYSCLK.
3. To develop C program for timer T0 and T1 in various modes of operations.
4. To develop C program for timer T2, T3 and T4 in various modes of operation.
5. To develop C program for ADC0 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).
6. To develop C program for ADC2 in various mode of operation (Cover Single ended & Differential mode with various start of conversion method).

7. To develop C program for using window comparator mode of ADC0.
8. To develop C program for on chip analog comparator.
9. To develop C program for DAC and generating signals.
10. To develop C program for various PCA mode of operations.

EE372: VLSI TECHNOLOGY AND DESIGN
5th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction and fabrication of MOSFET	06
2	MOS Transistor	10
3	MOS Inverters: static characteristic and switching characteristic	16
4	MOS combinational circuits	08
5	MOS sequential logic circuits	08
6	MOS dynamic logic circuits	08
7	Chip input and output	04

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

D. Detailed Syllabus:

1	Introduction and fabrication of MOSFET	10%	06 Hrs
	VLSI Design Flow, Design hierarchy, Design Methodology, nMOS, pMOS, CMOS fabrication process.		

2	MOS Transistor	17%	10 Hrs
	Metal Oxide Semiconductor (MOS) structure, MOS System under external bias, Structure & Operation of MOS transistor, MOSFET Current-Voltage characteristics, MOSFET scaling & small-geometry effects, MOSFET capacitances.		
3	MOS Inverters: static characteristic and switching characteristic	27%	16 Hrs
	Introduction, Resistive load Inverter, Inverter with n-type MOSFET load (Enhancement & Depletion type MOSFET load), CMOS Inverter, Delay-time definitions, Calculation of Delay times, Inverter design with delay constraints, Estimation of Interconnect Parasitic, Calculation of interconnect delay, Switching Power Dissipation of CMOS Inverters.		
4	MOS combinational circuits	13%	08 Hrs
	Introduction, MOS logic circuits with Depletion nMOS Loads, CMOS logic circuits, Complex logic circuits, CMOS Transmission Gates (TGs).		
5	MOS sequential logic circuits	13%	08 Hrs
	Introduction, Behaviour of Bistable elements, The SR latch circuit, Clocked latch & Flip-flop circuit, CMOS D latch & Edge-triggered flip-flop.		
6	MOS dynamic logic circuits	13%	08 Hrs
	Principles of pass transistor circuits, Voltage Bootstrapping, Synchronous Dynamic Circuit Techniques, CMOS Dynamic Circuit Techniques , High performance Dynamic CMOS circuits.		
7	Chip input and output	7%	04 Hrs
	On chip Clock Generation and Distribution, Latch –Up and its Prevention		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	20	20	10	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Identify the recent trends in VLSI Technology and Design methodologies
CO2:	Understand physics of MOSFET and hence apply this in analysis of basic building block

	of digital integrated circuits
CO3:	Design combinational and sequential circuits using various topologies that realizes myriad digital function of given specification.
CO4:	Understand dynamic logic concept and implement this to design high density and high performance digital circuits
CO5:	Learn latest CAD/EDA/Verilog & Xilinx ISE suite tools used in industry for design of digital integrated circuits

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	2	2	-	-	-	-	-	-	-	-	-	-	-	-
CO3:	1	2	3	1	-	-	-	-	-	-	-	-	1	-
CO4:	2	1	2	1	-	-	-	-	-	-	-	-	1	-
CO5:	-	-	-	1	3	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Sung-Mo-Kang, UsufLeblebici ,CMOS digital integrated circuits: Analysis and Design, Tata McGraw-Hill Education.

Reference Book:

1. Douglas Pucknell, Basic VLSI Design, PHI, 1999.
2. Wayne Wolf , Modern VLSI Design., Person Education, 2001.
3. John Uyemura ,Introduction to VLSI circuits and systems, Wiley, 2002.

Web Material:

- <http://www.eng.auburn.edu/~strouce/elec4200.html>

I. List of experiments:

1. Introduction of Verilog & Xilinx ISE suite.
2. To Design and implement all logic gates using behavioral model, Dataflow modelling style.
3. To Design and implement eight different logic gates with the help of 3-bit selection line.
4. To Design and implement half adder with data flow, structural and behavioral modelling style.
5. To Design and implement half subtractor with data flow, structural and behavioral modelling style.
6. To Design and implement full-adder with data flow, structural and behavioral modelling style.
7. To Design and implement full-subtractor with data flow, structural and behavioral method.
8. To Design and implement 4-bit adder using structural modelling style.
9. To Design and implement 8:1 multiplexer using dataflow & behavioral modelling style.
10. To Design and implement 32:1 multiplexer using structural modelling style.

11. To Design and implement 2:4 decoder using dataflow & behavioral modelling style.
12. To Design and implement 4:16 decoder using structural modelling style.
13. To Design and implement BCD to Seven segment decoder.
14. To Design and implement 4-bit comparator.
15. To Design and implement D Latch and D,T,S-R,J-K Flip-flop.

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VI)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE346: DIGITAL SIGNAL PROCESSING
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Signal Processing	10
2	Z Transform and Realization of Discrete System	10
3	Fourier Analysis	10
4	Filter Design	15

Total hours (Theory) :45Hrs
 Total hours (Lab) :30Hrs
 Total hours : 75Hrs

D. Detailed Syllabus:

- 1 Introduction to Signal Processing** 22% 10 Hrs
 Basic block diagram of Digital processing system, Sampling and Sampling Theorem for Reconstruction of signal in Digital domain, Aliasing and Solution, Re-construction Filter, Errors and noise, Mathematical and graphical representation of various signals and basic operations over it, Various Discrete systems and their characterization, Convolution and Correlation. Introduction to DSP Processor.
- 2 Z Transform and Realization of Discrete System** 22% 10 Hrs

Definition and ROC for Z Transform, S plane to Z plane mapping using various techniques like impulse invariant and Bilinear Transformation, Inverse Z Transform, Realization of IIR& FIR Systems (Direct, Cascade and Parallel form Realization), Discrete Integrator and Differentiator

3 Fourier Analysis 22% 10 Hrs

Continuous time and Discrete time Fourier Series (CFS and DFS) and their properties, Discrete Time Fourier Transform(DTFT), Discrete Fourier Transform(DFT), Computational Advantage of Fast Fourier Transform(FFT) Over DFT, Bit Reversed ordering, Decimation in time (DIT) & Decimation in frequency (DIF) Algorithm for implementation of FFT.

4 Filter Design 34% 15 Hrs

Introduction to Digital Filters and their classification, Applications, Design of IIR and FIR Filters using various methods, their comparison. Introduction to adaptive Filtering concepts. Multi-rate Signal Processing: Concept and Applications, Decimation, Interpolation, Sampling rate conversion by rational factor, Multi-rate filter banks.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
10	15	20	15	20	20

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	To understand theory of various signal processing algorithms and systems like Fourier transform, wavelet Transform, Convolution and Correlation, Filters
CO2:	To understand fundamentals and mathematical representation of signal and system, mathematics related to signal processing and their computation using math and also with Computer programming and simulation
CO3:	To develop skill to identify the DSP algorithm and systems of blocks used in a real system, and in complex engineering problem and solution required for benefits to society
CO4:	To find application of signal processing algorithm and systems in current research and to present overall research in form report and block diagram

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	2	3	-	-	2	-	-	-	-	-	-	-	2	-
CO3:	-	-	3	-	-	3	2	1	-	-	-	-	2	-
CO4:	-	-	-	3	-	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session students will solve the numerical by hand computations and will also verify the results through simulations in MATLAB.

H. Recommended Study Material:

Text Book:

1. Anand Kumar “Digital Signal Processing” 2nd Edition PHI Learning Private Ltd.
2. Udayashankara "Real Time Digital Signal Processing: Fundamentals, Algorithms and Implementation using TMS Processor", Prentice-Hall Of India Pvt Limited, 2010
3. Ramesh Babu, “Digital Signal Processing”, SCITECH Publications, 4th Edition, 2009

Reference Book:

1. Alan V Oppenheim, Alan S. Willsky with Hamid Nawab, "Signals & Systems", Pearson Education, 2nd Edition (2000).
2. John Proakis, Dimitris G Manolakis, "Digital Signal Processing Principles", Algorithms and Application", PHI, 3rd Edition (2000).
3. S Salivahanan , A Vallavaraj , C Gnanapriya "Digital Signal Processing" 2nd Edition, McGraw Hill

Web Material:

1. <http://nptel.iitm.ac.in/video.php?subjectid=117102060> nptel video course by prof.s.c. duttaroy, iitdelhi.
2. DSP Video Lectures by Prof. E. Ambikairajah:
http://onlinevideolecture.com/?course_id=373
3. http://www.analog.com/en/processorsdsp/processors/beginners_guide_to_dsp/fca.htm
l a beginner's guide to digital signal processing

I. List of experiments:

1. To learn various useful MATLAB commands for Signal Processing.
2. To Understand and Simulate Concept of Aliasing in MATLAB.
3. To simulate various Discrete Signals & System (FIR and IIR) in MATLAB.
4. To develop Matlab code for Convolution & Correlation Operations on to solve the related numerical using hand calculations.
5. To do Practice for Numerical on Z Transform.
6. To Verify the Properties of Discrete Fourier Transform. Also Develop code for finding the result of Discrete Fourier Transform.
7. To develop Matlab code for design of various IIR Filters.
8. To develop Matlab code for design of various FIR Filters.
9. To simulate various IIR Filters using MATLAB model file and Analyse the frequency Response.
10. To simulate various FIR Filters using MATLAB model file and Analyse the frequency Response.
11. To implement Multi Rate Filter bank in MATLAB.

**EE347: PROGRAMMABLE LOGIC CONTROLLERS AND
INDUSTRIAL AUTOMATION
6thSemester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)**

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	50	50	200

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Control	02
2	Introduction to Programmable Logic Control (PLC)	02
3	Input and Output devices	03
4	PLC Input and Output Module	04
5	PLC Operation	03
6	PLC Programming	10
7	Designing Automation System	06

Total hours (Theory) :30Hrs

Total hours (Lab) :60Hrs

Total hours : 90Hrs

D. Detailed Syllabus:

1	Basics of Control	06.67 %	02Hrs
	Introduction to Sequential/ Logic Control, control strategy, control philosophy and control algorithm, Difference between Relay logic control and PLC, Difference		

	between Analog/ Automatic and Sequential/ Logic, Control, Evolution of Control System		
2	Introduction to Programmable Logic Control (PLC)	06.67%	02Hrs
	Definition of PLC, introduction to standard (IEC61131) used for PLC and basics of process automation (along with information about standard control signals used), History and Evolution of PLC, Block Diagram of PLC, Advantages and Disadvantages of PLC		
3	Input and Output devices	10.00%	03Hrs
	A brief overview of sensors and its types, Overview of Transmitter, difference between 2 wire Transmitter and 4 wire transmitter, Brief overview of types of output devices connected to PLC, Control Valves and its classification		
4	PLC Input and Output Module	13.33 %	04Hrs
	Power supply circuit/Wiring Diagram, Selection of power supply, Function of input and output modules, Classification and block diagram of Input and output Modules, AC input module (circuit/wiring diagram), Discrete/Digital input module (circuit/wiring diagram) Special input module cards TTL Logic, Relay and Triac output module (circuit/wiring diagram), Selection of cards/modules		
5	PLC Operation	10.00%	03Hrs
	PLC memory types and its mapping, PLC Register basics , Addressing: Internal and External, PLC Scan Cycle and response time		
6	PLC Programming	33.33%	10Hrs
	Types of programming languages, Conversion of Gate Logic and process statement into ladder and vice versa, Various PLC Instruction: NO, NC and output contacts and coils, Set, Reset, Timer (along with types), Counters (its types), PID, Logical, Arithmetic, Data handling, skip, MCR, jump, Bit operations and move instructions, Programming related to above instructions		
7	Designing Automation System	20.00%	06Hrs
	Selection of PLC, Documentation: System architecture, Piping and Instrumentation Diagram (P&ID) (Flow sheet symbols as per standard ISA S5.11984(R1992)), General Arrangement (GA) Drawing, Wiring Diagram, I/O Listing and program flowchart, Debugging (Simulation), Commissioning, troubleshooting and maintenance of PLC		

	system
--	--------

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Enhance their skills in basics of PLC
CO2:	Know the performance and classification of Input and Output Devices.
CO3:	Understand different Programming language used in PLC
CO4:	Design any automated control system for any machinery
CO5:	To program PLC and they can apply their knowledge in the field of automation and control

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	3	-	-	-	-	-	-	-	-	-	-	-	1	-
CO3:	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4:	-	3	3	3	2	-	-	-	2	-	-	-	3	-
CO5:	-	3	3	3	2	-	-	-	2	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Programmable Logic Controllers by John W. Webb and Ronald A. Reis, PHI Publication.
2. Programmable Logic Controllers by W. Bolton, Newnes

Reference Book:

1. Programmable Logic Controllers & their Engineering Application by A.J. Crispin, McGraw Hill.
2. Programmable Logic Controllers by Thomas A. Hughes, ISA.
3. Programmable Logic Controllers by John R. Hackworth, Pearson Education.
4. Exploring Programmable Logic Controllers with Applications by Pradeep Kumar Srivastava, BPB Publications
5. Programmable Logic Control: Principles and Applications, NIIT, PHI

Web Material:

- <http://coep.vlab.co.in/?sub=33&brch=97>
- <http://www.plcdev.com/book/export/html/9>
- <http://www.plcmanual.com/>
- <http://nptel.ac.in/courses/108105062/>
- <http://nptel.ac.in/courses/108105088/1>
- <http://nptel.ac.in/courses/108105063/>
- www.siemensplcweb.com
- <http://www.ab.com/programmablecontrol/plc/>

I. List of experiments:

1. To study basics of PLC & SCADA.
2. To implement logic gates in Ladder language and implement on PLC (AND, OR, NOT, EX-OR, EX-NOR, NAND, NOR)
3. To implement motor latch and unlatch Ladder on PLC (with and without using instruction)
4. To implement Liquid level control of tank using PLC
5. To implement flashing of Lamp on PLC.
6. To realize traffic light control sequence using ladder logic on PLC.
7. To implement flashing of Lamp 10 times on PLC.
8. To study interfacing of RTD with PLC.
9. To study interfacing of SCADA screens with PLC programs.

EE348: POWER ELECTRONICS AND DRIVES - II
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Inverters	15
2	Power Supplies	5
3	AC Voltage controllers	6
4	Cycloconverters	7
5	AC Drives	12

Total hours (Theory) :45Hrs

Total hours (Lab) :30Hrs

Total hours :75Hrs

D. Detailed Syllabus:

1	Inverters	33.33 %	15 Hrs
	Introduction and revision: Overview of switches, Introduction to inverters Voltage Source Inverter : Single-phase half bridge and full bridge inverter with different loads, Three-phase 180 and 120 degree mode inverter, Fourier analysis of Single-phase and Three-phase inverter, Output Voltage control in inverters, Harmonic reduction techniques in inverter output voltage Current Source Inverter: Concept of CSI, CSI with different loads		

	Load and Self-commutated inverter: Single-phase Series Inverter and Parallel Inverter		
2	Power Supplies	11.11 %	5Hrs
	DC Power Supplies: Switched-Mode DC Power Supplies, Half bridge converter, Full bridge converter, Fly back converter, Push-pull converter AC Power Supplies: Uninterruptible Power Supplies (UPS) , Batteries used in UPS Static switches and Solid State relays		
3	AC Voltage controllers	13.33 %	6Hrs
	Voltage control techniques: Principle of voltage controllers, Phase control, Integral Cycle control, Single-phase Voltage controllers with R and RL load Sequence control of AC voltage controllers: Two-stage and Multistage sequence control of voltage controllers		
4	Cycloconverters	15.56 %	7Hrs
	Introduction: Principle and Applications of Cycloconverters, Types of Cycloconverters, Single-phase to single-phase Cycloconverters: Bridge-type Cycloconverters, Mid-point Cycloconverters, Three-phase Half-wave Cycloconverters: Three-phase to Single-phase, Three-phase to Three-phase, Output voltage Equation of a Cycloconverter.		
5	AC Drives	26.67 %	12Hrs
	Basic machine equations: Analysis and Performance of AC motors, Speed Control of three-phase Squirrel-cage Induction Motors (SCIMs): Stator Voltage Control, Stator Frequency Control, Stator Voltage and Frequency Control, Stator Current Control, Speed Control of three-phase Wound Rotor Induction Motors (WRIMs): Static Rotor Resistance Control, Slip-energy recovery Control.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1	Describe the role of Power Electronics as an enabling technology in various applications such as flexible production systems, energy conservation, renewable energy, transportation etc. (1,3,5,10)
-----	---

CO2	Learn the concepts of inverters with different loads and its applications. (1,2,3,4,5,PSO1)
CO3	Understand, simulate and design single-phase and three-phase thyristor converters. (1,2,3,4,5,PSO1)
CO4	Identify a switching power-pole as the basic building block and to use Pulse Width Modulation to synthesize the desired output. (1,2,3,4,5,7,PSO1)
CO5	Learn basic magnetic concepts, analyze transformer-isolated switch-mode power supplies and design high-frequency inductors and transformers. (1,2)
CO6	Learn the requirements imposed by electric drives (dc and ac) on converters and synthesize these converters using the building block approach. (1,2,7)

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3		1	-	2	-	-	-	-	1	-	-	-	-
CO2:	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3:	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4:	3	3	3	1	2	-	3	-	-	-	-	-	2	-
CO5:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO6:	3	1	-	-	-	-	3	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Power Electronics by Dr. P. S. Bimbhra, Khanna publishers
2. Power electronics by M D Singh and K B K Khanchandani, Tata Mcgraw-Hill Publication

Reference Book:

1. Power electronics handbook, by M.H. Rashid, Academic press
2. High power converters and AC drives by Bin Wou, IEEE press
3. Power Electronics Converters, Applications and design by Mohan, Undel and, Robbins, Wiley publication.

Web Material:

1. <http://nptel.ac.in/courses/108101038/>
2. http://cusp.umn.edu/power_electronics.php
3. <http://ecee.colorado.edu/~ecen5797/notes.html>

I. List of experiments:

1. To simulate and study the performance of a single phase voltage source inverter and a single phase current source inverter.
2. To study and Compare 180 degree and 120 degree mode of conduction for 3-phase Voltage source Inverter
3. To simulate a single phase SPWM Inverter.
4. To study the operation of series inverter.
5. To simulate and compare the operation of a 6 pulse and 12 pulse converter.
6. To simulate and study operation of half bridge and full bridge topology of SMPS.
7. To compare the operation of step up Cycloconverter with a step down Cycloconverter.
8. To study and simulate single phase full wave AC voltage controller.
9. To study the operation of uninterruptible power supply (UPS).

EE349: FAULT ANALYSIS AND SWITCHGEAR
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Review of representation of power system components:	04
2	Symmetrical fault analysis	08
3	Symmetrical components	13
4	Unsymmetrical Fault Analysis	10
5	Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker. (CB) Ratings	10
6	Arc Extinction Process in Circuit Breaker	05
7	Types of Circuit Breakers	10

Total hours (Theory) : 60 Hrs

Total hours (Lab) : 30 Hrs

Total hours : 90 Hrs

D. Detailed Syllabus:

- Review of representation of power system components:** 07% 04 Hrs
Introduction, Single phase solution to three phase system, one line diagram, impedance

- and reactance diagram
- Per unit system and Examples
- 2 **Symmetrical fault analysis** 13% 08 Hrs
- Introduction, sudden short circuit on R-L series circuit, 3 – phase short circuit current calculation, examples.
- Sub-transient, transient and steady state model of synchronous machine
- 3 phase short circuit analysis of power system – Examples
- 3 **Symmetrical components** 17% 10 Hrs
- Introduction, Symmetrical Component transformation.
- Example on symmetrical components
- Sequence impedance and sequence network of transformer , Phase shift in star delta Transformers.
- Sequence impedance and sequence network of synchronous machine and transmission Line.
- Examples on sequence networks.
- 4 **Unsymmetrical Fault Analysis** 17% 10 Hrs
- Types of unsymmetrical faults and its analysis.
- Examples on unsymmetrical fault analysis
- 5 **Fundamentals of Fault Clearing, Switching Phenomena and Circuit Breaker (CB) Ratings.** 17% 10 Hrs
- Current interruption in AC circuit breaker.
- Transient recovery voltage, rate of rise of restriking voltage and Examples.
- Effect of natural frequency, power factor on TRV- effect of reactance drop on power frequency recovery voltage- effect of armature reaction on recovery voltage- effect of first pole to clear factor- single, double frequency transient.
- Interruption of low magnetizing current (current chopping), use of resistance switching for damping TRV, use of opening resistor, switching of capacitor bank, unloaded transmission lines and unloaded cables, interruption of terminal fault and short line fault (Kilometric Fault), Phase opposition switching.
- Rating of AC circuit breaker.
- 6 **Arc Extinction Process in Circuit Breaker** 08% 05 Hrs.
- Introduction, The matter and plasma- Ionization of gases – Deionization.
- Formation of electric arc and methods of arc extinction.
- Arc interruption theories.
- 7 **Types of Circuit Breakers** 22% 13 Hrs.

Air break circuit breaker, construction of air blast circuit breaker, (ABCB), arc quenching in ABCB, resistance switching in ABCB and methods of reducing overvoltage in ABCB. Physical and dielectric properties of Sulphur Hexafluoride (SF₆), arc extinction process in SF₆ CB, different puffer type circuit SF₆ CB, relative merits and demerits of SF₆ CB. Electric break down in high vacuum, arc extinction in vacuum CB, construction of Vacuum CB, contact material and switching phenomena of vacuum CB. Introduction to Direct Current Circuit Breaker (DCCB), different types of DCCB.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	35	20	5	0

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Able to calculate symmetrical fault current and voltage at any location of power system by applying allied theories
CO2:	Able to calculate unsymmetrical fault current and voltage at any location of power system using symmetrical components by applying allied theories.
CO3:	Acquire the knowledge of the phenomena of current interruption in circuit breaker.
CO4:	Acquire the knowledge of arc formation and arc interruption in circuit breaker.
CO5:	Impart the knowledge of circuit breaker construction and working. Also, gain ability to select proper rating and type of circuit breaker

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	3	1	2	2	-	-	-	-	-	-	-	-	-
CO2:	3	3	2	2	2	-	-	-	-	-	-	-	-	-
CO3:	3	1	-	1	-	-	-	-	-	-	-	-	-	-
CO4:	3	1	-	1	-	-	-	-	-	-	-	-	-	-
CO5:	3	1	2	2	-	2	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT HT control room and nearby substation will clear the concept of circuit breaker and student will realise the actual size of Circuit breaker.
- In lab session students shall solve the numerical in computer with the software “Power World Simulator”

H. Recommended Study Material:

Text Book:

1. Power System Analysis by Hadi Saadat, Tata McGraw Hill.
2. Principles of Power system by V. K. Mehta, Publisher: S. Chand & Company Ltd.
3. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath,, Tata McGraw Hill, New Delhi.
4. A Course in electrical power by Soni & Bhattnagar, Dhanpat Rai & Sons.
5. Switchgear and Protection, by S.S. Rao, Dhanpat Rai & Sons.

Reference Book:

1. Power System Analysis by Grainger & Stevenson, Tata McGraw Hill
2. Power System Analysis and Design by B R Gupta. S. Chand & Company Ltd.
3. Analysis of Faulted Power System by P.M. Anderson, IEEE Press
4. Power System Protection & Switchgear by B. Ravindranath & M. Chander, New Age International.

5. Switchgear and Protection by J B Gupta. S.K Kataria & Sons.
6. Computer Techniques In Power System Analysis By M A Pai, TMH Education Pvt. Ltd.
7. JNP Switchgear By R T Lithal.

Web Material:

- <http://courses.engr.illinois.edu/ece476/notes/?year=2010>
- <http://nptel.iitm.ac.in/video.php?courseId=1060>
- <http://freevidelectures.com/Course/2353/Power-Systems-Analysis>

I. List of experiments:

1. To study the per unit system for the power system analysis.
2. To study the symmetrical fault analysis.
3. To study the symmetrical components.
4. To study the fault clearing process and the problems associated with fault clearing process
5. To study the unsymmetrical fault analysis.
6. To study the arc extinction process in circuit breaker
7. To study air blast and SF6 circuit breaker
8. To study the vacuum and direct current circuit breaker.

EE360: MINOR PROJECT II
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	04	4	2
Marks	00	100	100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		50	50	100

C. Instructional Method and Pedagogy:

- Minor project-II has 2 credit subject and 4 hours per week.
- Department/Institute will help students to find appropriate minor projects.
- The student must be find the appropriate guide for their project and make the project
- proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the minor project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which you gain during project making. The guide must verify report.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the project.

D. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done during project, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily

accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

E. Course Outcomes (COs):

After completion of the course students will able:

CO1:	To apply knowledge and skills learned from project to real-world problems
CO2:	To solve engineering problems by the team work and work with other discipline
CO3:	Use experience related to professional and ethical issues in the work environment
CO4:	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur
CO5:	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies

Mapping of course outcomes with program outcomes

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS134 A | B | C | D | E | F | G | H :
CONTRIBUTOR PERSONALITY DEVELOPMENT
6th Semester and 3rd Year

I. Credits and Schemes:

Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				Total
		Theory		Practical		
		Internal	External	Internal	External	
02	02	--	--	30	70	100

II. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	Concept of Personality: • Meaning of Personality • Types of Personality • Factors contributing to Personality • Personality Traits	06
2	Soft Skills and Personality Development: • Critical, Creative and Positive Thinking • Leadership, Assertiveness and Negotiation Skills • Self-Management • People's Skills • Building Relationship Skills • Being a Team Player	08
3	Developing Contributor Personality – Part I • Concept of Contributor • Characteristics of a Contributor • The Contributor's Vision of Success & Career • The Scope of Contribution • Embarking on the Journey to Contributor ship	06

4	Developing Contributor Personality – Part II • Focus on Values • Engage Deeply • Think in Enlightened Self-Interest • Practice Imaginative Sympathy • Demonstrate Trust Behavior • Developing a sense of duty and morality	06
5	Contemporary Issues in CPD • Contemporary Practices & Trends in Contributor Personality Development • Case Study & Presentations	04
	Total	30

IV. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations etc.

V. Evaluation:

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment / Project Work / Term Work / Quiz	2	25	25
2	Attendance and Class Participation			05
Total				30

External Evaluation

The University Practical examination will be of 70 marks and will test the contributory personality aspects and their applications by carrying out practical assessment. The examination will avoid, as far as possible, grammatical errors and

will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
Total				70

VI. Course Outcome (COs):

After completion of the course, the student would:

CO1	Develop conceptual understanding of the term Personality and understand one's own personality Traits
CO2	Develop soft skills for holistic personality development for career and in personal life
CO3	Develop assertiveness, self management and negotiation skills to navigate the personal and professional environment successfully
CO4	Develop conceptual clarity of the term 'Contributor Personality' and develop understanding of the traits of a contributor
CO5	Develop the qualities of a contributor such as trust, strong work ethic, responsibility and accountability at workplace
CO6	Develop relevant skills required to be a contributor at workplace and achieve success in life

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	2	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	-	-	-	-	3	-	2

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

VII. Reference Books / Reading

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program Activ Guide, Illumine Knowledge Pvt. Ltd.

EE376: SPECIAL ELECTRICAL MACHINES AND APPLICATIONS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Special Electrical Machines	13
2	Special Synchronous Machines	05
3	Permanent Magnet Machines	07
4	Wind Turbine Generators	06
5	Modelling & Applications	09
6	Case Study	05

Total hours (Theory) :45Hrs

Total hours (Lab) :30 Hrs

Total hours :75Hrs

D. Detailed Syllabus:

1 Introduction of Special Electrical Machines 28.9% 13Hrs

Special DC Machines: Rosenberg Generator, Dynamotors, Printed Circuit Board Motor. Special Induction Machines: Synchronous Induction Motor, Induction Voltage Regulator, Linear Induction Motor. Special Transformers: Induction Heating Transformer, Welding Transformers, Pulse Transformer. Schrage Motor, Repulsion Motor, Hysteresis Motor, Universal Motor, DC-AC Servomotor. Applications, Examples.

- 2 **Special Synchronous Machines** 11.1% 05Hrs
Stepper Motor, Switched Reluctance Motor, Linear Synchronous Motor, Applications, Examples.
- 3 **Permanent Magnet Machines** 15.6% 07Hrs
Basics of permanent magnet property, Unique features of Permanent Magnet (PM) Machines, Permanent Magnet Material & Characteristics, Permanent magnet DC motor (PMDC), Permanent magnet brushless DC motor(PMBLDC) : construction, operation, control, advantages, dis-advantages, applications, permanent magnet synchronous motor(PMSM), Applications.
- 4 **Wind Turbine Generators** 13.3% 06Hrs
Introduction, Types of induction generator (line excited and self-excited), Introduction of fixed speed induction generator, Introduction of doubly fed induction generator. Working of Induction generator: Principle of Operation, Reactive power requirement, voltage build up process, Load and Power factor control, Effect of capacitor, Characteristics of induction generator, Advantages, Applications, Examples.
- 5 **Modelling & Application** 20.0% 09Hrs
Synchronous Reference Frames: dq Axis Theory, Conversions: abc to $\alpha\beta$, $\alpha\beta$ to dq, dq to $\alpha\beta$, $\alpha\beta$ to abc. Space Vector Pulse Width Modulation: discrete and analog implementation, Applications.
- 6 **Case Study** 11.1% 05Hrs
Case Study: Grid Connected Renewable Energy Source

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Recognize & understand the performance, operation and application of various AC & DC special machines
CO2:	Define various permanent magnet material used for manufacturing of special machine
CO3:	Acquire the knowledge and provide operational instructions for permanent magnet motors/generators, switched reluctance motor, linear motors, hysteresis motors and stepper motors as well as make effective presentations on the working, construction, advantages & drawbacks of these machines along with the needs of improvisation in the machine.
CO4:	Understand the types and operation of wind turbine generators and evaluate case studies based on renewable energy sources
CO5:	To understand the modeling of machines and reference frames for machine control.

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	2	-	-	-	-	1	2	1	3	-	-
CO2:	3	1	2	2	2	-	1	-	-	1	-	-	2	-
CO3:	3	3	3	1	2	2	1	-	-	3	2	1	2	-
CO4:	3	3	3	1	2	2	3	-	2	2	2	3	3	3
CO5:	3	-	2	3	3	-	-	2	2	3	1	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Special Electrical Machines by K Venkataratnam, University Presses.
2. Special Electrical Machines by E.G. Janardanan, PHI Learning Pvt. Ltd.
3. A Text Book of Electrical Machines by K.R.Siddhapura, D.B.Raval. Vikas Publishing House Pvt. Ltd.
4. Electric Machines and Power Systems by Syed A. Nasar, Tata McGraw Hill
5. Electrical Machines III by K Krishna Reddy, Scitech Publication (INDIA) Pvt. Ltd.

Reference Book:

1. Theory and performance of electrical machines by J.B.Gupta, S.K.Kataria and sons

Web Material:

- http://www.academia.edu/9885014/special_electrical_machines_nptel_notes
- <http://www.notesnation.com/electrical-engineering/special-electrical-machines-notes/>
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_Machines_I
- www.nptel.ac.in/courses/IIT-MADRAS/Electrical_Machines_II/pdf/2_1.pdf

I. List of experiments:

1. To perform speed control and power factor control by commutator motors.
2. Demonstration the operation of induction generator
3. To understand the behavior of induction generators considering the impact of load.
4. Working of doubly fed induction generator.
5. Study of stepper motor.
6. Study of PMSM
7. Study of SRM

8. Demonstration of induction voltage regulator to obtain variable output voltage.
9. Direct load test on synchronous induction motor.

EE377: EMBEDDED SYSTEMS
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Embedded Systems	04
2	Processors	13
3	Real Time Operating System	12
4	Wireless Devices and Communication Protocols	09
5	Embedded system design	07

Total hours (Theory) :45Hrs
Total hours (Lab) :30Hrs
Total hours : 75Hrs

D. Detailed Syllabus:

1	Introduction to Embedded Systems	9%	04 Hrs
	Embedded systems and its characteristics with examples, Embedded System's design metrics & challenges, Processor Technology , IC Technology, Design Technology, Importance of Design Metrics, Example of Embedded systems Working Models.		
2	ARM Processors	29%	13Hrs
	Introduction of Single purpose processors ,General purpose processors, Application specific instruction set processors(ASIPs), DSP processors and SHARC processors, CISC vs. RISC,		

	ARM architecture Inheritance, ARM programmer's model, Memory organization, ARM development tools, ARM data flow model, ARM processor modes, ARM instruction set, Thumb instruction set, 3-pipeline architecture, 5-stage pipeline architecture, Architectural support for high-level languages, I/O interfacing & Programming in c.		
3	Real Time Operating System	27%	12Hrs
	Multiple processes in an Applications and inter process Communication, Shared Data problem, use and types of semaphores OS system services(Linux), IO sub-system, network OS, RTOS in embedded systems, Interrupt routine in RTOS environment, RTOS, RTOS Task scheduling Techniques, Interrupt Latency and Response Times of tasks, Performance, Metric in scheduling Models, Embedded Linux Internals, OS security issues, Mobile OS.		
4	Wireless Devices and Communication Protocols	20%	09Hrs
	Serial bus communication protocols: RS232C, RS485, UART, I2C bus, CAN bus, Modbus, USB bus, Parallel bus device protocols: ISA bus, PCI and PCI/x bus, AMBA bus, Wireless devices: Irda, Bluetooth, GSM, RFID, Zigbee.		
5	Embedded system design	15%	07Hrs
	System design techniques, design methodologies, requirement analysis, specifications, system analysis and architecture design, Quality assurance, design example.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	20	20	10	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to get knowledge about the foundation education in Embedded C and Assembly language programming. Through lecture and out-of-class assignments, students are provided learning experiences that enable them to:

CO1:	To provide knowledge of architecture of CISC, RISC, ARM, DSP architecture
CO2:	To write RTOS programs for given embedded system and Hardware
CO3:	To gain knowledge of various communication Protocols and developing device driver and its VHDL implementation

CO4:	To be able to use expert system like AI and Machine learning on Microcontroller platforms.
CO5:	To understand market available embedded products, boards and systems and to recognize role of expert system in it.

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	1	-	-	-	-
CO2:	3	1	-	-	3	-	-	-	2	3	2	-	3	-
CO3:	-	-	-	-	3	-	-	-	2	3	2	-	3	-
CO4:	-	3	2	3	3	-	-	-	-	3	2	-	3	3
CO5:	-	3	2	-	-	-	-	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi -media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Raj Kamal, Embedded Systems Architecture, programming and design (Tata – MGHill)
2. Steve furber, ARM System-on-chip Architecture, Pearson

Reference Book:

1. David E. Simon, An Embedded Software Primer, Addison-Wesely.
2. Frank vahid & Tony D. Givargis, System Design: A unified Hardware/Software introduction, Addison Wesley 2002
3. Wayne Wolf, Computers as components Principles of embedded computing
4. Tammy Noergaard, Embedded Systems Architecture, Elsevier
5. Anderw N. SLOSS, Dominic SYMES, Chirs WRIGHT, ARM System Developer's Guide, Elsvier.
6. Daniel W. Lewis, Fundamentals of Embedded Software: Where C and Assembly Meet, Prentice Hall

I. List of experiments:

1. Introduction to Linux, DIR structure and basic UNIX command execution on Linux terminal.
2. Write an Application level C program and Compilation using GCC in Linux.
3. Write an Application level C program based on Pointer and Compilation using GCC in linux.
4. Write a multiple C files and execute those files using makefile.
5. Introduction to LPC2148 development board and ide (mdk-arm) software.
6. Write program to interface LEDs with ARM processor based on LPC2148 kit.
7. Write a program to glow led with its priority with the help of switchusing lcp2148 development board.
 - a. Interface 4-bit lcd with lpc 2148 and print on lcdusing lcp2148 development board
 - b. Interface 8-bit lcd with lpc 2148 board and write c code to print on lcd using lcp2148 development board.
- 8
 - c. Display counter of 00 to 99 on lcd using 8 -bit mode using lcp2148 development board.
 - d. Write a program to display the custom character on lcd using lcp2148 development board.
- 9
 - a. Interfacing of lpc2148 with 7 segment & counting 0 to 9 using lcp2148

development board

b. Write a program to display 00 to 99 on seven segment using lpc2148 development board.

10 Write a program to detect which key is pressed of keypad and display it on lcd using lpc2148 development board.

11 Write a program to rotate dc motor clock wise and anti-clock wise using lpc2148 development board.

12 Write a program for stepper motor using LCP2148 DEVELOPMENT BOARD.

Annexure – I

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN ELECTRICAL ENGINEERING											
Sem	Course Code	Course title	Teaching Scheme				Examination Scheme				Total
			Contact Hrs.			Credits	Theory		Practical		
			Theory	Practical	Total		Internal	External	Internal	External	
VII	EE441	Power System Operations and Control	4	2	6	5	30	70	25	25	150
	EE442	Power System Protection	4	2	6	5	30	70	25	25	150
	EE443	Electrical Machine Design	4	2	6	5	30	70	25	25	150
	EE444	Simulation Lab		4	4	2			50	50	100
	EE450	Summer Internship II		3	3	3			75	75	150
		Elective - I	4	2	6	5	30	70	25	25	150
	Assignment Practice/ Student Counselling	0	0	4	0	0	0	0	0	0	0
			16	15	35	25	120	280	225	225	850
VIII	EE446	Commissioning and Testing of Electrical Equipments	4	2	6	5	30	70	25	25	150
	EE447	Modeling and Control of Renewable Energy Sources	4	2	6	5	30	70	25	25	150
		Elective - II	4	2	6	5	30	70	25	25	150
	EE448	Major Project	0	18	18	9	0	0	250	250	500
	Assignment Practice/ Student Counselling	0	0	4	0	0	0	0	0	0	0
			12	24	40	24	90	210	325	325	950
	Code	Elective - I		Code	Elective - II						
	EE471	Power Electronics Applications in Power System		EE476	Advances in Power System						
	EE472	Communication Engineering		EE477	Internet of Things						
	EE473	High Voltage Engineering		EE478	Energy Conservation and Management						

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

EE441: POWER SYSTEM OPERATIONS AND CONTROL
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Load Flow Analysis	10
2	Optimal Dispatch of Generation	08
3	Automatic Generation Control (AGC)	08
4	Transients in Power System	04
5	Short Circuit Studies using Bus Impedance Matrix	06
6	Power System Stability	20
7	Power System Security	04

Total hours (Theory) : 60 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 90 Hrs

D. Detailed Syllabus:

1	Load Flow Analysis	16.66%	10 Hrs
	Introduction, Formulation of Bus Admittance Matrix and Examples using Singular transformation method, Gauss – Siedel Load Flow Method and examples, Newton –		

	Raphson Load Flow Method, Fast Decoupled Load Flow Method, Computer aided simulation of load flow analysis, Examples.		
2	Optimal Dispatch of Generation	13.34%	08 Hrs
	Introduction, non-linear function optimization, unconstrained and constrained parameter optimization, equality and inequality constraints, Operating cost of thermal power plant, economic dispatch neglecting losses and no generation limits and with consideration of generation limits, Economic dispatch including losses and examples, Derivation of loss formula, Unit commitment.		
3	Automatic Generation Control (AGC)	13.34%	08 Hrs
	Introduction, basic generation control loops, Load frequency control, generator model, prime mover model, load model, governor model, AGC in a single area system, AGC in multi-area system, tie-line bias control, Reactive power and voltage control, amplifier model, exciter model, generator model, sensor model, excitation system stabilizer.		
4	Transients in Power System	6.66%	04 Hrs
	Traveling waves on transmission lines, wave equations, specifications of traveling waves, surge impedance and wave velocity, Reflection and refraction of traveling waves, typical cases of line terminations, Successive reflections, Bewley lattice diagram, Lightning phenomena, protection of power system against lightning surges, insulation coordination.		
5	Short Circuit Studies using Bus Impedance Matrix	10.00%	06 Hrs
	Bus admittance and incidence matrices, Thevenin's theorem and Z_{BUS} , Modification of existing Z_{BUS} after addition or removal of branches, Direct determination of Z_{BUS} , Unbalanced fault analysis using Z_{BUS} , Examples.		
6	Power System Stability	33.34%	20 Hrs
	Derivation of swing equation, M and H constants, equivalent H constant and examples, Classification of power system stability, Steady state stability, derive the expression for natural frequency of oscillations and damped frequency of oscillations, example, Derivation of synchronizing power coefficient, examples. Transient stability and factors affecting it, Equal area criteria, Cases of sudden application of mechanical power input, application of three phase fault at the middle of the line or at the end of the line, Example based on equal area criteria, Point by point method for the solution of swing equation and examples, Transient stability of		

	multi machine system. Voltage stability, voltage instability and voltage collapse, voltage stability phenomena illustration by radial feeder, active and reactive power transmission by elementary models, difficulties with reactive power transmission.		
7	Power System Security	6.66%	04 Hrs
	Operating states of power system, Factors affecting power system security, Contingency analysis, Overview of security analysis, Linear sensitivity factors, AC power flow methods, Contingency relaxation.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	15	20	20	25

F. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to acquire the knowledge regarding

CO1:	The knowledge regarding to planning and designing of the new interconnected power system through power flow analysis with practical simulation
CO2:	Carry out the short circuit studies of an inter connected power system and design the circuit breaker ratings
CO3:	Understand different types of stability phenomena being observed in the power system networks including the analysis methods and the design of control measures required for the improvement of system stability with practical simulation
CO4:	Understand the concept of travelling waves on transmission lines and design the components to protect the system against over-voltages
CO5:	Techniques to get the economic operation of generating units and modeling of the components for load frequency control with practical simulation

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	2	2	2	2	-	-	-	-	-	-	-	2	-
CO2:	2	1	3	1	2	-	-	-	-	-	-	-	2	-
CO3:	1	3	2	1	2	-	-	-	-	-	-	-	2	-

CO4:	1	1	2	-	-	-	-	-	-	-	-	-	2	-
CO5:	1	-	1	-	1	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Power System Analysis and Design by B.R. Gupta, S. Chand
2. Power System Analysis by Grainger & Stevenson, Tata Mcgraw Hill.
3. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
4. Power System Analysis by Hadi Saadat, Tata Mcgraw Hill.

Reference Book:

1. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, John Wiley & Sons, New York, USA.
2. C.W. Taylor, Power System Voltage Stability, McGraw Hill.

Web Material:

1. <http://nptel.ac.in/downloads/108101040/>

2. <http://nptel.ac.in/courses/108104052/2>

I. List of Experiments:

1. To calculate Ybus matrix using singularity transformation method.
2. To calculate economic operating schedule of generator.
3. To calculate Zbus matrix for short circuit study.
4. To study transients in power system.
5. To carry out power flow using MATLAB and Power World Simulator.
6. Including Transformers in Power Flow using Power World and Confirmation by MATLAB.
7. To simulate transient stability of power system in a 3-bus system.
8. To study the dynamic interaction between two control area using Simulink.
9. To study economic dispatch using Power World Simulator.
10. To study contingency analysis in Power World Simulator.

EE442: POWER SYSTEM PROTECTION
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Philosophy of Protective Relaying System	04
2	Instrument Transformer	05
3	Overcurrent Protection	08
4	Transformer Protection	07
5	Protection of Transmission Line	16
6	Protection of Generator	05
7	Induction Motor Protection	02
8	Busbar Protection	02
9	Testing, commissioning and Maintenance of Relays	03
10	Numerical Relaying	08

Total hours (Theory) : 60 Hrs
 Total hours (Lab) : 30 Hrs
 Total hours : 90 Hrs

D. Detailed Syllabus:

1 Philosophy of Protective Relaying System 08% 04 Hrs

Introduction, faults and abnormalities in power system, need for protection, functions of protective relaying.

Basic tripping circuit, required characteristic for protective system, zones of protection, primary and back up protection, unit and non-unit protection.

Phasor diagram of voltages and currents during various faults.

2 Instrument Transformer 08% 05 Hrs

Current transformer (CT), Saturation characteristic of current transformer.

Classification of CT, General rule applicable for selecting current transformer in electrical networks, difference between CT cores for measurement and protection, problems encountered in CT.

Voltage transformer (VT), General rule applicable for using voltage transformer in electrical networks, Coupling capacitor voltage transformer (CCVT) and its transient response

3 Overcurrent Protection 13% 08 Hrs

Protection by fuse, fuse characteristic, types of fuse, physics of fuse interruption, limitation of fuse.

Operating principle of induction relay, Types of overcurrent relay, choice of different inverse characteristic, operating characteristic of other electromechanical relays.

Plug setting and time setting for overcurrent relays, rules for setting IDMT relays for phase and ground relays.

Requirement of directional relaying, operating principle of directional overcurrent relay, 30° , 60° and 90° connection

4 Transformer Protection 12% 07 Hrs

Faults in transformer, differential protection, difficulties in differential protection of transformer.

Percentage biased differential protection, harmonic restraint relay.

Inherent phase shift in star-delta transformer, differential protection of 3 – phase transformer.

Restricted earth fault protection, miscellaneous protections for transformer.

Examples.

5 Protection of Transmission Line. 27% 16 Hrs

Different discrimination schemes for transmission line and their comparison.

Protection of radial feeder, two overcurrent and one earth fault scheme for radial feeder, three overcurrent and one earth fault scheme for transformer feeder. Examples.

Drawback of overcurrent protection, Distance protection of transmission line.

Impedance relay, reactance relay, ohm relay, Mho relay, performance of distance relays for the faults involving arc resistances.

Three stepped characteristic of distance relays for single and double infeed, Examples based on setting of distance relays.

Complete distance protection of three-phase line for phase and ground faults.

Performance of distance relays in the event of power swing, out-of-step blocking and tripping scheme.

Other problems in distance relaying, examples.

Need for carrier aided protection, various option for carrier, components for carrier aided scheme, directional and phase comparison scheme.

6 Protection of Generator 08% 05 Hrs

Faults and abnormal conditions in generator, generator differential protection, stator inter-turn and earth fault protection.

Rotor earth fault protection, negative phase sequence protection.

Protection against loss of prime mover and loss of excitation, class A, B and C protection schemes for generator, Examples.

7 Induction Motor Protection 03% 02 Hrs

Faults and abnormalities in induction motor, protection against overloading, stalling, single phasing, and miscellaneous protection for induction motor.

8 Bus bar Protection 03% 02 Hrs

Differential protection of bus bar, selection of CT ratio for bus bar protection, high impedance bus bar differential protection, breaker back up protection

9 Testing, commissioning and Maintenance of Relays 05% 03 Hrs

Acceptance tests, type test, special tests, commissioning tests, laboratory set up for testing of different types of relays, dynamic testing of relays

10 Numerical Relaying 13% 08 Hrs

Advantages of numerical relaying.

Numerical relay hardware, facilities available in numerical relaying.

Digital signal processing, data acquisition system, sample and hold circuit, sampling theorem.

Anti-aliasing filter, sampling rate criteria, estimation of phasors, implementation of 2 – sample window in ideal condition and with real life conditions, three samples per data

window, least square estimation.

Full cycle Fourier algorithm, half cycle Fourier algorithm.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
15	15	15	20	20	15

F. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Interpret the fundamentals of protective relaying principles and can select correct instrument transformer for protection
CO2:	Select the proper relay for over current protection. Can able to do relay coordination for transmission line protection by over current protection, by distance protection and by carrier current protection
CO3:	Create/Design protection scheme for three phase power transformer, induction motor and bus bar protection with CT and relay connection and settings
CO4:	Develop skill for different protection scheme for generator. Design the protection scheme for generator and suggest relay settings with connections diagram of CTs for the relays applied for generator protection
CO5:	Follow various types of tests of relays, commissioning and maintenance of relays. Also, recognize the construction and working of numerical relaying. Think about relays in advanced level

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	-	-	-	-	-	1	-	-	-	-
CO2:	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3:	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4:	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO5:	3	1	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realise the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

H. Recommended Study Material:

Text Book:

1. Power System Protection and Switchgear by B.A. Oza, N.C. Nair, R.P. Mehta and V.H. Makwana, Mcgraw Hill, 2010
2. Protection and Switchgear by Bhavesh Bhalja, R. P. Maheshwari and N. G. Chothani, Oxford University Press, 1st Edition, 2011
3. Fundamentals of Power System Protection by Y.G. Paithankar, S.R. Bhide, Prentice Hall, India, 2003

Reference Book:

1. Power System Protection and Switchgear by Badri Ram and D N Vishwakarma, Tata Mcgraw Hill, 2001
2. Power System Protection Static Relays by T.S. Madhavrao, Tata Mcgraw Hill, 1989

Web Material:

- http://www.cdeep.iitb.ac.in/nptel/Electrical%20Engineering/Power%20System%20Protection/TOC_M1.html

NPTEL

I. List of Experiments:

1. To study the fundamentals of the power system protection
2. To study different protective scheme in power system
3. To perform over current protection of radial feeder using electromechanical relay
4. To perform over current protection of radial feeder using numerical relay
5. To carry out relay over current relay coordination using software.
6. To perform under voltage and over voltage protection in power system.
7. To perform over current relay co-ordination
8. To perform 3 phase Induction motor protection.
9. To perform parallel feeder protection.
10. To perform transformer protection.
11. To study different protective scheme for Generator.

EE443: ELECTRICAL MACHINE DESIGN
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	100

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basic Considerations in Electrical Machine Design	04
2	Design of Transformer	19
3	Induction Machines Design	19
4	Synchronous Machines Design	18

Total hours (Theory):60Hrs

Total hours (Lab) :60Hrs

Total hours : 120Hrs

D. Detailed Syllabus:

- | | | | |
|---|---|--------|--------|
| 1 | Basic Considerations in Electrical Machines Design | 16.66% | 04 Hrs |
| | Design factors, Limitations in design, Modern trends in design of electric machines, Temperature rise, Expression for temperature rise, heating & cooling time constants, examples. | | |
| 2 | Design of Transformer | 31.67% | 19 Hrs |
| | Specification, Output equation of transformer, Output equation-Volt per turn, | | |

Stacking factor, Ratio of iron loss to copper loss, Relation between core area and weight of iron and copper, Optimum designs, variation of output and losses in transformer with linear dimensions ,examples, Design of core, Choice of flux density and current density, Choice of window space factor, window dimensions, Design of yoke, Overall dimensions, examples, Design of high voltage and low voltage winding, examples, Estimation of operating characteristics: Primary & Secondary resistance, Leakage reactance of windings, Regulation, examples, Mechanical forces, No load current calculation, Change of parameters with change of frequency, Temperature rise of transformer, Design of tank, examples.

3 Induction Machines Design

31.67% 19 Hrs

Choice of specific electrical loadings, Choice of specific magnetic loadings, Output equation, Separation of D & L, Examples, Stator Design: Turns per phase, Stator conductors, Shape of stator slots, Number of stator slots, Area of stator slots, Length of mean turn, Stator teeth, Stator core, examples , Length of air gap, Relation for calculation of length of air gap, Examples, Squirrel Cage Rotor Design: Number of rotor slots, Rules for selecting rotor slots, Reduction of harmonic torques, Design of rotor bars & slots, Design of end rings, examples, Wound Rotor Design: Number of rotor slots, number of rotor turns, area of rotor conductors, Rotor windings, Rotor teeth, design of rotor core, examples, Estimation of operating characteristics- No load current calculation, short circuit current calculation, Stator and rotor resistance and leakage reactance calculation, examples, Dispersion coefficient – Effect on maximum power factor and overload capacity, Effect of change of air gap length, number of poles and frequency, Relation between D&L for best power factor, examples.

4 Synchronous Machines Design

30% 18 Hrs

Choice of specific electrical loadings, Choice of specific magnetic loadings, Design of Salient pole machines: Output equations, Main dimension, Short Circuit Ratio, Effect of SCR on machine performance, Length of air gap and shape of pole face, Examples, Armature design: Number of armature slots, Coil span, Turns per phase, Conductor section, Slots dimension, Length of mean turn, Stator Core, Elimination of harmonics, Estimation of air gap length, examples, Design of rotor, Height of pole, Design of damper winding, Height of pole shoe, Pole profile

drawing, examples, Design of magnetic circuit, Determination of full load field MMF, examples, Design of field winding, examples, Determination of Direct and Quadrature axis synchronous reactance.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	15	20	20	25

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Interpret the fundamentals of electrical machine design, modern trends, its operational effects on the environment like heat and gases liberated and many more.
CO2:	Design the overall transformer which includes the design of core, windings and tank and also estimate the resistance, reactance, regulations and the mechanical forces
CO3:	Design Induction machine which includes designing of stator, air-gap, squirrel cage rotor and wound rotor
CO4:	Design of salient pole machine, armature, rotor and field winding as well as determination of direct and quadrature axis synchronous reactance
CO5:	Acquire entrepreneurial outlook to design different machines for the small and medium scale manufacturer

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	3	2	2	-	2	-	1	1	2	-	2	-
CO2:	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO3:	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO4:	3	2	2	2	1	-	2	-	2	-	1	1	2	-
CO5:	3	3	-	-	-	2	-	1	3	3	3	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. A course in Electrical machine design by A. K. Sawhney & A. Chakrabarti, Dhanpat Rai & Co.

Reference Book:

1. The performance and design of alternating current machines by M. G. Say, CBS Publishers & Distributors
2. Design of rotating electrical machines by Juha Pyrhonen, Tapani Jokinen, Valeria
3. Design of electrical machines by K. G. Upadhyay, New age international publishers.
4. Design of electrical machines by V. N. Mittal & A. Mittal, Standard Publishers distributors.
5. Electric Machinery 6th Edition by A. E. Fitzgerald, Charles Kingsley, Stephen. D. Umans Tata Mcgraw Hill.
6. Elements of electrical design 2nd edition by J. G. Jamnani, Mahajan publishing house.

Web Material:

- NPTEL

I. List of Experiments:

1. Design of Core, Window & Yoke (Transformer Design)

2. Design of Overall Dimensions of Transformer.
3. Design of HV and LV winding (Transformer Design)
4. Overall Design of Transformer
5. Main Dimension, Armature Design, Design of Field System (Synchronous Machine Design).
6. Magnetic Circuit and Performance Evaluation (Synchronous Machine Design).
7. Complete Design of Synchronous Machine.
8. Main Dimensions, Stator Design, Rotor Design (Induction Machine Design).
9. Performance Evaluation (Induction Machine Design)
10. Complete Design of Three Phase Induction Motor.

EE444: SIMULATION LAB
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	4	4	2
Marks	00	50	50	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
00	00	50	50	100

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Software introduction and mathematical equations solution	10
2	Electrical Power System simulation	20
3	Electrical Machines Simulation	10
4	Power Electronics Simulation	20

Total hours (Lab) :60 Hrs
Total hours : 60Hrs

D. Detailed Syllabus:

Sr. Suggested List of Practical
No.

- 1 To study different types of solvers used in MATLAB software.
- 2 To study symbolic Toolbox in MATLAB software.
- 3 To perform Linear equations and matrix solution using MATLAB software.
- 4 Formation of bus Admittance and Impedance matrix Using MATLAB software.

- 5 Computation of transmission line parameter using MATLAB software.
- 6 To simulate single phase half wave and full wave bridge uncontrolled rectifier with R, RL & RLE load.
- 7 To simulate single phase half wave and full wave bridge controlled rectifier with R, RL & RLE load.
- 8 To simulate speed control of separately excited DC Machines.
- 9 To simulate the three phase full wave circuit using MATLAB software.
- 10 To perform Open Circuit, Short Circuit & Load tests on Single Phase transformer using MATLAB.
- 11 To perform fault analysis using Power world simulator and using MATLAB software
- 12 Calculation of leakage inductance of transformer using FEM software.
- 13 Load flow analysis using Mi Power software.
- 14 Perform relay Co- ordination for radial system using Mi Power software.
- 15 Long term load forecasting using Mi Power software.
- 16 Harmonic Analysis using Mi Power software.
- 17 Introduction of PSIM simulation software.
- 18 To study and simulate Boost convertor with PI controller.
- 19 To study and simulate Multilevel inverter.
- 20 To study and simulate Flyback convertor

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	15	20	20	25

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Acquire the knowledge regarding the simulation of different electrical machines and electrical power system network using different software
------	--

CO2:	This will be very helpful to them in dealing with real system in industries and to carry out consultancy
CO3:	Use of different software makes their software skills which help them to develop their future goals
CO4:	Symbolic toolbox study help student to develop their mathematical skill using MATLAB software
CO5:	Power Electronics model of Inverter, convertor helps to understand basic of electronic circuit design in real system

Mapping of course outcomes with program outcomes
Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	-	3	3
CO2	2	2	3	3	-	-	1	-	-	-	-	-	-	3
CO3	3	-	-	2	3	-	-	2	-	-	1	2	-	3
CO4	3	2	3	1	-	-	-	-	-	-	-	-	3	2
CO5	3	3	2	2	-	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

1. At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
2. Laboratories will be conducted with the aid of multi-media projector.
3. A student has to prepare a laboratory term work as per instruction given by lab instructor.
4. Attendance is compulsory in laboratory, which carries five marks of the overall evaluation.
5. Two viva voce will be conducted during the semester and average of two will be considered as a part of overall evaluation.

H. Recommended Study Material:

Manual of software provided by lab instructor.

Reference Book:

1. Stephen. J. Chapman, Electric Machinery Fundamentals , Mcgraw Hill

2. Hadi Saadat, Power System Analysis , Tata Mcgraw Hill

Web Material:

1. <http://www.prdcinfotech.com/business/software-engineering-group/software-products/mipower/>
2. <https://www.mathworks.com/products/matlab.html>
3. <https://www.powerworld.com/>
4. <http://www.femm.info>
5. NPTEL

EE450: SUMMER INTERNSHIP II
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week (In Summer Break)	00	30	90	3
Marks	00	150	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		75	75	150

C. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report

- Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

D. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company

- Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
 - PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
 - SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
 - CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
 - REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
 - APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

E. Course Outcomes (COs):

After completion of the course students will able:

CO1:	To solve engineering problems by working in a team with across other disciplines
CO2:	To apply knowledge and skills learned in company/industry/organization to real-world problems

CO3:	To use experience related to professional and ethical issues in the work environment
CO4:	To explain the impact of engineering solutions, developed in a project, in a global, economic, environmental, and societal context
CO5:	To demonstrates knowledge of contemporary issues related with engineering in general by adopting new tools and technologies

Mapping of course outcomes with program outcomes

Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

**EE471: POWER ELECTRONICS APPLICATIONS IN POWER SYSTEM
(PE3)**

**7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)**

A. Credits and Hours:

Teaching scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Unit	Min. no of hours
1	FACTS Controllers	30
2	HVDC Transmission	20
3	Power Quality problems & Standards	10

Total Hours (Theory) :60

Total Hours (Lab) :30

Total Hours: 90

D. Detailed Syllabus

1 FACTS Controllers 30Hours 50 %

Need of FACTS devices, classification of FACTS devices, operating principle of SVC, V-I characteristic of SVC, advantage of slope in V-I characteristic, SVC applications for transient and voltage stability improvement, mitigation of SSR, advantages of TCSC, different mode of operation of TCSC, different modeling concepts of TCSC, Operating principle of STATCOM, TCSC, SVC and their applications for power system performance improvement, Power flow solution with SVC, TCSC.

2 HVDC Transmission

20 Hours 33.33 %

Introduction, equipment required for HVDC systems, Comparison of AC and DC Transmission, Limitations of HVDC transmission lines, reliability of HVDC systems, comparison of HVDC link with EHVAC link, HVDC system configuration and components, fundamental equations in HVDC system, HVDC links, converter theory and performance equation, valve characteristic, converter circuits, converter transformer testing, multi bridge converters, abnormal operation of HVDC system, control of HVDC system, harmonics and filters. Influence of AC system strength on AC/DC system interaction, response to AC and DC system faults, Concept of reactive power compensation- reactive Power balance in HVDC substations-Effect of angle of advance and extinction angle on reactive power requirement of converters.

3 Power Quality problems & Standards

10 Hours 16.67%

Introduction: Definition & Reasons of Occurrence of following Voltage Dip, Brief voltage increases, Brief voltage interruption, Transients, Voltage Notches, Flickers, Distortion, Un-balance. Power Quality Indices, Limits of Harmonic Distortion according to IEEE, IEC, EN and NORSOK limits. Brief Introduction of Power quality Standards: IEC 61000-2-5, IEC 61000-2-1, IEC 1159 (Categories of Power quality variation according to IEEE 1159 standard with their relevant Spectral content, Duration of occurrence & Magnitude)

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	15	20	20	25

F. Course Outcomes (COs):

At the end of the course the student will be able to:

COI	Compare the HVDC Transmission and EHVAC transmission
-----	--

CO2	Identify, analyze and understand the converter configurations used in HVDC system and its components to enhance the skill development.
CO3	Understand controllers for controlling the power flow through a dc link
CO4	Able to understand the different types of Power Quality problems & Standards.
CO5	Understand the role of impedance control, phase angle control and voltage control in controlling real and reactive power in transmission systems.
CO6	Identify analyze and understand the different FACTS controller required for the power system performance improvement.

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	2	-	-	-	-	1	-	-	-	-
CO2:	3	2	2	1	2	-	-	-	-	-	-	-	2	-
CO3:	3	3	3	1	2	-	-	-	-	-	-	-	2	-
CO4:	3	3	3	1	2	-	3	-	-	-	-	-	2	-
CO5:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO6:	3	1	-	-	-	-	3	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Books and References:

1. E. W. Kimbark, Direct Current Transmission, Vol 1, Wiley-Interscience, 1971
2. P. Kundur, Power System Stability and Control, McGraw Hill, 1994.
3. S. Kamakshiah and V. Kamaraju, HVDC Transmission, McGraw Hill, 2011
4. K.R. Padiyar, HVDC Power Transmission Systems: Technology and System Interactions, New Age International
5. E. Acha, FACTS: Modeling and Simulation in Power Networks, Wiley 2004
6. T.J.E Miller, Reactive Power Control in Electric Systems, Wiley India Edition, 1982
7. R. Mohan Mathur and R. K. Varma, Thyristor-based FACTS controllers for Electrical Transmission systems, IEEE Press, 2002

Web Material:

- <http://nptel.ac.in/courses/108108077/>
- <http://nptel.ac.in/courses/108102046/>
- <http://nptel.ac.in/courses/108104011/>

I. List of Experiments:

1. Application of shunt devices for voltage control: synchronous condensers, shunt capacitors, STATCOM.
2. Application of series devices for power flow control: phase angle regulators and series voltage sources.
3. To study and simulate TCSC.
4. To study and simulate SVC.
5. Power flows with HVDC links: LCC and VSC converter based links.
6. Power System Stability Improvement Using FACTS Devices.

EE472: Communication Engineering (PE III)
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Amplitude Modulation	12
2	Angle Modulation	12
3	Receivers	08
4	Probability Theory & Random Processes	12
5	Information Theory	08
6	Principle of digital data Transmission	08

Total hours (Theory) :60Hrs
Total hours (Lab) :30Hrs
Total hours : 90Hrs

D. Detailed Syllabus:

1 Amplitude Modulation 20% 12Hrs

Amplitude Modulation, Amplitude Modulation Index, Modulation Index for Sinusoidal AM, Frequency Spectrum for Sinusoidal AM, Average Power for Sinusoidal AM, Effective Voltage and Current for Sinusoidal AM, Non sinusoidal Modulation, Double-Sideband Suppressed Carrier (DSBSC) Modulation.

2 Angle Modulation 20% 12Hrs

Frequency Modulation, Sinusoidal FM, Frequency Spectrum for Sinusoidal FM, Average Power in Sinusoidal FM, Non-Sinusoidal Modulation: Deviation Ratio, Measurement of Modulation Index for Sinusoidal FM, Phase Modulation,

Equivalence Between PM and FM, Sinusoidal Phase Modulation, Digital Phase Modulation, Angle Modulator Circuits, FM Transmitter, Angle Modulation Detectors, Automatic Frequency Control, Amplitude Limiters, Pre-Emphasis and De-Emphasis, FM Broadcast Receivers.

3 Receivers 12% 07Hrs

Homodyne, Heterodyne and Super Heterodyne Receivers, Tuning Range, Tracking, Sensitivity and Gain, Image Rejection, Spurious Responses, Adjacent Channel Selectivity, AGC, Double Conversion, Electronically Tuned Receivers (ETRS), Integrated-Circuit Receivers.

4 Probability Theory & Random Processes 20% 12Hrs

Introduction to digital communication, Probability, Conditional Probability Of Independent Events, Relation between probability and probability density ,CDF, PDF, Gaussian PDF, Threshold Detection, Random Variable, Variance of a random variable, co relation between random variables, Mean and variance of sum of random variables, Statistical Average, Chebyshev In Equality, Raleigh Probability Density, The Central Limit Theorem, correlation, random processes, PSD of random processes(along with examples), Multiple random processes, band pass random processes.

5 Information Theory 15% 09Hrs

Concept & Measure Of Information, entropy, Error Free Communication Over A Noisy Channel, The Channel Capacity Of A Discrete Memory Less Channel, Channel Capacity Of A Continuous Channel, Practical Communication Systems In Light Of Shannon's Equation.

6 Principle of digital data Transmission 13% 08Hrs

Line Coding, PSD of On/Off Signal, Bipolar Signal, Duo Binary Signal, Pulse Shaping, Nyquist First and Second Criterion for Zero ISI, Regenerative Repeaters, Detection Error Probability, M-ary System, Scrambling, Digital Carrier System, Digital Multiplexing.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	20	20	10	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Understand different techniques for amplitude modulation and demodulation
CO2:	Able to get Depth Analysis in Noise Performance of Various Receivers

CO3:	Understand and Analyze Channel characteristics
CO4:	Design a digital communication system
CO5:	Simulate communication systems and analyze their performance on MATLAB software

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	2	1	-	1	-	-	-	-	-	-	-	-	1	-
CO3:	2	1	-	1	-	-	-	-	-	-	-	-	1	-
CO4:	-	-	3	1	-	-	-	-	-	-	-	-	1	-
CO5:	-	-	-	1	3	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval. It carries a weightage of 5 Marks as a part of internal theory evaluation.
- Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Book:

1. Electronic Communications by Dennis Roddy& John Coolen IV Edition PHI.

2. Modern Digital and Analog Communication System III Edition By B.P.Lathi Pub
3. B.P.Lathi ,Digital and Analog communication system , LCBS Publication
4. B. Carlson ,Communication system , TMH Publication
5. Simon Haykin, Digital and analog communication system , Willey Publication

Reference Book:

1. Electronic Communications by Dennis Roddy & John Coolen IV Edition PHI.
2. Modern Digital and Analog Communication System III Edition By B.P.Lathi Pub
3. Taub & Schilling, Principle of communication system, TMH Publication
4. Wayne Wolf, Computers as components Principles of embedded computing
5. Tammy Noergaard, Embedded Systems Architecture, Elsevier
6. Anderw N. SLOSS, Dominic SYMES, Chirs WRIGHT, ARM System Developer's Guide, Elsvier.
7. Daniel W.Lewis, Fundamentals of Embedded Software: Where C and Assembly Meet, Prentice Hall

Reading Materials, Web Materials with Full Citations:

- 1.<http://adp.mmu.edu.my/enotes/asma/dtc5038/notes/lecture%20notes%203%20chapter%203.pdf>
2. <http://web.cecs.pdx.edu/~ece2xx/ece223/slides/communicationsx4.pdf>
- 3.<http://stattrek.com/Lesson2/Binomial.aspx?Tutorial=sat>
4. <http://www.youtube.com/watch?v=IYdiKeQ9xEI>
5. <http://www.complextoreal.com/tutorial.htm>

I. List of experiments:

1. To study and perform Double Sideband AM Generator.
2. To calculate modulation index of DSB wave by trapezoidal pattern.
3. To Study and perform Double Sideband AM Reception.
4. To Study and perform Diode Detector.
5. To perform Frequency Modulation using Varactor modulator.
6. To Perform Operation of Detuned Resonant Circuit.
7. To perform Operation of Quadrature Detector
8. To Perform Operation of Phase-Locked Loop Detector
9. To plot selectivity curve for radio receiver
10. To study Basic Block Diagram of Digital Communication System.
11. To perform Pulse Amplitude Modulation and observe its waveform.
12. To perform Pulse Position Modulation and observe its waveform.

13. To perform Pulse Width Modulation and observe its waveform.
14. To understand the concept of Pulse Code Modulation and to observe the performance of PCM system.
15. To find the channel capacity of Binary Symmetric Channel.
16. To study and perform BPSK Modulation using MATLAB.
17. To study and observe the performance of digital carrier systems ASK using MATLAB.
18. To study and observe the performance of digital carrier system FSK using MATLAB.
19. To study and perform Huffman Code using MATLAB.

EE473: High Voltage Engineering (Program Elective III)
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Breakdown mechanisms in gaseous dielectrics	08
2	Breakdown Mechanisms in Solid and Liquid dielectrics	08
3	Corona Discharge	04
4	Generation of High voltages and Currents	12
5	Measurement of High voltages and Currents	12
6	Overvoltage Phenomenon and Insulation Coordination in Electric Power Systems	03
7	High Voltage testing of equipments	13

Total hours (Theory) :60Hrs

Total hours (Lab) :30 Hrs

Total hours : 90Hrs

D. Detailed Syllabus:

1	Breakdown mechanisms in gaseous dielectrics	13.33%	08Hrs
	Introduction, ionization processes, Townsend's mechanism, Primary and secondary ionization co-efficient, breakdown in electronegative gases, Streamer theory, comparison of Townsend and Slepain's theory, Paschen's law, breakdown under		

	uniform and non-uniform fields, Post breakdown current-voltage characteristics, de-ionization, breakdown under impulse voltage, SF6 and vacuum as di-electric, vacuum BID mechanisms.		
2	Breakdown Mechanisms in Solid and Liquid dielectrics	13.33%	08Hrs
	Breakdown Mechanism in Solid Dielectrics: Introduction, solid di-electrics, intrinsic breakdown, electromechanical breakdown, thermal Breakdown, Electrochemical breakdown, treeing and tracking phenomenon of partial discharge, solid dielectrics used in practice Breakdown Mechanism in Liquid Dielectrics: Introduction, classification of liquids, liquid breakdown test cell, transformer oil purification, Testing of di-electric oil as per IS 355 and IS 6792, Breakdown in pure liquids , breakdown in commercial liquids		
3	Corona Discharge	6.67%	04Hrs
	Phenomenon, Disruptive and visual critical voltage, Corona loss, Factors and conditions affecting corona loss, radio interference, Methods to reduce corona discharges		
4	Generation of High Voltages and Currents	20%	12Hrs
	Introduction, Generation of high direct voltage, rectifier circuits, voltage doubler, cascaded circuits, Van de Graff generators, electrostatic generators, generation of high alternating voltages, cascade transformer, resonant transformer, Generation of high frequency alternating voltages, generation of impulse voltages, standard impulse voltage wave, Impulse generator, Marx circuit, constructional features of impulse generator, Trigatron gap, generation of impulse currents, related examples		
5	Measurement of High voltages and Currents	20%	12Hrs
	Measurement of high direct voltages, potential dividers, generating voltmeters, measurement of high alternating voltages, series voltmeters, Capacitance potential dividers and capacitance voltage transformers, electrostatic voltmeter, measurement with sphere gaps (IS 1876), Sphere gap construction and assembly, factors influencing the spark over voltage, Measurement of impulse voltages, measurement of high d.c. and a.c. currents, measurement of high frequency and impulse currents.		
6	Overvoltage Phenomenon and Insulation Coordination in Electric Power Systems	5%	03Hrs

	Natural Causes for Overvoltages – Lightning phenomenon, Overvoltage due to switching surges, System faults and other abnormal conditions, Insulation Coordination, BIL		
7	High Voltage testing of equipments	21.67%	13Hrs
	Introduction, related Indian standards, measurement of capacitance and loss tangent, Partial discharge detection, partial discharge measurement (IS 6209), High voltage tests on line insulators, bushings, transformers, cables, circuit breakers, lightning arrestors, Synthetic testing, non-destructive high voltage tests, Resistivity Test, Dissolved Gas Analysis, Sweep Frequency Response Analysis, Endurance test, Hot Spot measurement methods.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Understand and remember the different types of insulations, its applications and their breakdown mechanisms when subjected to high voltage stresses to get the opportunity to work as a team leader/member in multi-disciplinary branches
CO2:	Differentiate, design, simulate and analyze the performance of various types of high voltage generating and measuring techniques as per IEC, IEEE and other Indian and international standards
CO3:	Demonstrate, simulate, analyze and evaluate the performance characteristics of transmission line insulators and other high voltage assets such as transformers, circuit breakers etc. by conducting the high voltage tests under different climate conditions as per IEC, IEEE, CIGRE and other Indian and international standards to get the opportunity to work as a team member/leader in multi-disciplinary branches
CO4:	Apply the knowledge to predict the remaining life of high voltage assets based on condition assessment tools i.e. dissolved gas analysis, sweep frequency response analysis etc.
CO5:	Describe the principles behind the insulation coordination and basic impulse insulation level

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	3	-	-	-	-	-	-	2	-	-	3	3	-
CO2:	2	1	3	-	3	-	-	-	-	-	-	-	2	-
CO3:	3	2	2	3	2	-	3	3	3	3	-	2	3	-
CO4:	2	2	-	-	1	-	-	-	2	-	-	-	1	-
CO5:	3	-	-	-	-	-	-	-	-	-	-	2	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Books:

1. High Voltage Engineering by M.S. Naidu & V. Kamaraju, TMH Limited
2. High Voltage Engineering by C.L. Wadhwa, New Age International Publisher.
3. Fundamentals of High Voltage Engineering by S. K. Singh, Dhanpat Rai & Co.

Reference Book:

1. An Introduction to High Voltage Engineering by Subir Ray.
2. High Voltage Engineering by E. Kuffel, J. Kuffel and W.S. Zaengl.
3. High Voltage Engineering by D.V. Razevig, translated by M.P. Chourasia, Khanna Publishers.

4. High Voltage Engineering & Testing (2nd Edition) by Hugh M. Rayn, The Institute of Electrical Engineers, London.
5. High Voltage Engineering Practice & Theory (2nd Edition) by Mazen Abdel-Salam, Hussein Anis, Ahdab El-Morshedy, RoshdyRadwan.
6. High Voltage Engineering Practice & Theory by Dr. J. P. Holtzhausen&Dr WL Vosloo.

Web Material:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPU/HighVoltageEngg/ui/TOC.htm>
- <http://www.mv.helsinki.fi/tpaulin/Text.hveng.htm>
- <http://ocw.mit.edu/courses/physics/8-02-electricity-and-magnetism-spring-2002/video-lectures/lecture-6-high-voltage-breakdown-and-lightning.htm>
- http://www.sayedsaad.com/High_voltage/index_solids.htm

I. List of experiments:

1. An introduction to high voltage laboratory.
2. To understand the working of a lightning arrestor using horn gap arrestor.
3. Determination of dielectric strength of transformer oil based on IS: 6792.
4. Determination of dielectric strength of solid dielectrics using 30kV AC/40kV DC high voltage tester.
5. Measurement of high voltages using Sphere Gap assembly.
6. Breakdown studies of uniform field and non-uniform field gaps for various types of electrode configurations.
7. Power frequency flashover voltage test on Insulators in dry and wet conditions.
8. To study the generation of impulse voltages using multistage impulse voltage generator.
9. Tracing of Equi-potential lines and Electric Field lines between two electrodes of different geometries using Electrolytic Tank assembly.
10. Measurement of dielectric loss factor and dc resistivity of transformer oil.

B. Tech. (Electrical Engineering) Programme

SYLLABI (Semester – VIII)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

**EE446: COMMISSIONING AND TESTING OF ELECTRICAL
EQUIPMENT**

8th Semester and 4th Year (Level IV)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of commissioning, maintenance and testing	04
2	Commissioning & Testing of Transformer	15
3	Commissioning & Testing Induction Machines	13
4	Commissioning & Testing of Alternator	13
6	Commissioning & Testing of Sub Station Equipment	15

Total hours (Theory) :60Hrs

Total hours (Lab) :30Hrs

Total hours : 90Hrs

D. Detailed Syllabus:

1	Extracts from Indian Electricity Rules – 1956 and Principles and Planning of Maintenance	6.67%	04Hrs
	Extracts from Indian Electricity Rules Introduction, Routine Maintenance, Periodical Maintenance & Maintenance on Fault		
2	Commissioning & Testing of Transformer	25%	15Hrs

	Commissioning of Transformer: Measurement of impedance, Power frequency voltage withstand test, Over voltage inter-turn test, Insulation Resistance of windings, Determination of Polarization Index for transformer, High voltage dielectric tests of winding inter turn and coil insulation, Condition of oil, Oil testing, Purification & Filtration procedure of transformer oil, Drying out transformers, Temperature measurement, Partial discharge test, Radio interference, Separate source voltage testing, Induced voltage testing, Short circuit withstand test, Tan Delta test, Core insulation voltage test, Short time current rating, Parallel operation of transformer: Load sharing problem, Phase shifting/ Phase group, Correct pairing of transformer, Transformer Troubles: Switching & short circuit surges, harmonics, unbalanced loading, voltage regulation, open circuits in phase windings and lines, protective gear troubles, oil deterioration, Commissioning steps for transformer, Troubleshooting & Maintenance of transformer.		
3	Commissioning & Testing of Induction Machines	21.66%	13Hrs
	Commissioning of Motors: Insulation test and drying out of windings, Temperature rise test, Air gaps, Bearings, Preliminary run, starting torque and speed control, induction motor stator and rotor interaction, balance and vibration, ventilation and cooling, contactor starter Hammer test, Testing against variation of voltage/current/frequency, Testing of auxiliaries, Degree of protection(IP Grade), AC Motor Troubles: Insulation failure due to transient voltages, low starting torque, pull out torque and stalling, low power factor, excessive slip, crawling, single phasing, reversal of one phase winding, starting transients in squirrel cage induction motors, overheating, pull over, vibration having electrical origins, slip ring wear, shaft currents, ball and roller bearing trouble. Commissioning steps for Induction motor, Commissioning of Induction Generator.		
4	Commissioning & Testing of Alternator	21.67%	13Hrs
	Commissioning of alternator: Preparation and drying out of alternator windings before commissioning, insulation resistance measurements, high voltage tests, measurements of temperature of windings, extra high voltage alternators, alternators protective gear tests, trip circuit supplies and tripling tests, starting time and rate of picking up load, control of auxiliaries Alternator stator and rotor interaction, leakage reactance of an alternator, transient leakage reactance of an alternator, effect of excitation on stator current of an alternator in parallel with a large system Alternator		

	troubles: Instability of exciters, complete loss of field, failure of exciter voltage to build up, motor driven exciters, alternator instability, neutral inversion with grounded voltage transformer and un-grounded power system, heating of copper conductors, shaft currents and bearing troubles, slip ring wear on alternator rotor, heating of brushes and slip rings, alternator rotor faults, Unbalanced stator currents, transient torques, heating of stator core end punching and clamps, steam turbine governors, Instrument and protective transformer polarities		
5	Commissioning & Testing of Sub Station Equipments	25%	15Hrs
	Bus bar: Temperature rise test, Rated short time current test, HV test, Power frequency voltage withstand test, Impulse / surge testing, Vibration. Isolator Testing: Temperature resistance test, Short circuit test, Charging current, Making & Breaking test, Inductive current making & breaking test. Circuit Breaker& Relay testing: No load Mechanical Operation, Mechanical endurance test, Temperature rise test, Impulse & surge testing , short time current test, Short circuit making & breaking test, Line charging, current making & breaking test, Fire precautions, oil and compound fillings, insulation resistance, mechanical operation and adjustment, electrical auxiliaries, protective relay tests, relay setting, Troubleshooting & maintenance of circuit breakers C.T. & P.T. Testing, Coupling capacitors, Station Batteries for D.C. Supply, Fire Shifting equipment. Testing & Commissioning of Lightning Arrestor, Substation Commissioning by Thermography.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Memorize, recognize, create and interpret the Single Line Diagram of Sub-Stations and Indian Electricity Rules
CO2:	Understand and discriminate the designed specifications of transformer and induction

	motor
CO3:	Demonstrate, simulate, analyze and evaluate the performance characteristics of transformers, induction machines, alternators and switchgear equipments by conducting the routine/commissioning tests as per IEC, IEEE, CIGRE and other Indian and international standards to get the opportunity to work as a team member/leader in multi-disciplinary branches
CO4:	Prepare the test, maintenance and troubleshooting report for static and dynamic machines
CO5:	Acquire the knowledge regarding the installation and commissioning of electrical equipments

Mapping of course outcomes with program outcomes
Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	2	-	2	-	-	-	-	-	-	3	-	1	1	-
CO2:	3	-	2	-	-	-	2	-	2	-	-	3	2	-
CO3:	3	2	2	3	1		2	2	2	3	-	2	3	2
CO4:	3	3	-	1	-	-	-	2	1	3	-	2	2	-
CO5:	2	1	-	-	-	2	-	-	-	-	-	2	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Books:

1. The Commission of electrical plant by RCH Richardson, Chapman & Hall.

2. Substation Design and Equipment by P.V. Satnam and P. V. Gupta
3. Commissioning of electrical equipments, by Prof. M. A. and S. M. Chaudhari, Tech Max Publications.

Reference Books:

1. Testing, Commissioning and maintenance of electrical equipments by S. S. Rao, Khanna Publisher.
2. Switchgear and Protection by J. B. Gupta by S. K. Kataria and Sons.
3. Electric Machinery Fundamentals by Stephen. J. Chapman, McGraw Hill
4. Electric Machinery 6th Edition by A.E. Fitzgerald, Charles Kingsley, Stephen. D. Umans, Tata McGraw Hill
5. The performance and design of alternating current machines by M. G. Say, CBS Publishers & Distributors
6. Electrical Machinery by Dr. P. S. Bimbhra, Khanna Publisher
7. Electrical Machines by D.P. Kothari & I. J. Nagrath, Tata McGraw Hill

Web Material:

- http://nptel.iitm.ac.in/courses/IITMADRAS/Electrical_Machines_II/Testing/index.php
- http://www.ece.ualberta.ca/~knight/ee332/synchronous/s_main.html
- http://www.ece.ualberta.ca/~knight/ee332/induction/i_main.html
- http://nptel.iitm.ac.in/courses/IIT-MADRAS/Electrical_MachinesI/Testing/index.php

I. List of experiments:

1. To find out number of poles for three phase induction motor without energizing the motor.
2. To perform the polarity test on current transformer.
3. To perform the polarity test of 1-phase transformer.
4. To check the reflection of load on secondary of three phase transformer.
5. To perform the polarity test of 3- ϕ transformer.
6. To study the types of faults, causes, remedies and troubleshooting chart for transformer.
7. To study the types of faults, causes, remedies and troubleshooting chart for induction machines.

8. To study the types of faults, causes, remedies and troubleshooting chart for dc machines.
9. To study the types of faults, causes, remedies and troubleshooting chart for synchronous machines.
10. To study the types of faults, causes, remedies and troubleshooting chart for circuit breakers.
11. To prepare Maintenance Schedule for Substation Equipment.

**EE447: MODELING AND CONTROL OF RENEWABLE ENERGY
SOURCES**

8th Semester and 4th Year (Level IV)

B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction of Renewable Energy Sources	05
2	Solar Photovoltaic System	20
3	Wind Energy System	15
4	Fuel Cell	05
5	Renewable Energy Sources in Smart Grid & Electric Vehicle	15

Total hours (Theory) :60 Hrs

Total hours (Lab) :30 Hrs

Total hours :90 Hrs

D. Detailed Syllabus:

- 1 Introduction of Renewable Energy Sources 8.3% 05 Hrs
Global and Indian renewable energy scenario, various renewable energy sources, Energy units and conversions, Energy Storage, introduction to IEEE 1547 standard.

- 2 Solar Photovoltaic System** **33.4% 20 Hrs**
 History of Solar Energy Usage, Equivalent circuit of photovoltaic cell, From cell to module and from module to array, I-V curve under standard test condition, Fill Factor, impact of temperature and insolation on I-V curve, Shading impacts on I-V curve, I-V curve for various types of loads (resistor, DC motor, battery), MPPT Concept, DC-DC converters (boost, buck, buck-boost) for MPPT, MPPT methods: reference cell methods, Perturb & Observation method, power slope method, standalone PV system design, Grid connected PV system principle, PV-Grid connection topologies: unfold circuit, double stage, single stage, classification of grid connected inverter: centralized inverter, string inverter & micro inverter, PV-Grid connected system architecture, PV-Grid connected system design, Economics of PV System, PV system in micro-grid, Emerging PV cell technologies.
- 3 Wind Energy System** **25.0% 15Hrs**
 Introduction, Wind Characteristics, Wind turbines: fixed speed & variable speed wind turbine, components of wind energy conversion system, Types of generators: Induction generator, synchronous generator, Permanent Magnet Synchronous Generator, Doubly Fed Induction Generator, Pitch angle control, Grid Integration of wind energy system, Economics of wind energy conversion system.
- 4 Fuel Cell** **8.3% 05Hrs**
 Construction & Operation of fuel cell, types of fuel cells, comparison of different component and characteristic of fuel cell, fuel cell equivalent circuit, need for dynamic modeling of fuel cell, Polymer Electrolyte Membrane Fuel Cell (PEMFC) model structure, Solid Oxide Fuel Cell (SOFC) model structure.
- 5 Renewable Energy Sources in Smart Grid & Electric Vehicle** **25.0% 15Hrs**
 Introduction of Smart Grid, History of Electric Vehicle, Architecture of PV based electric vehicle charging: Conventional method, Centralized architecture, Distributed architecture, single stage architecture, Control Strategy for EV charging, Impact of EV charging on Electrical Distribution System, V2G Concept, PV powered water pumping, Smart Meters.

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Recognize the electric circuits connected to various Renewable Energy Sources
CO2:	Acquire the knowledge regarding the fundamentals of Smart Grid and Electric Vehicle
CO3:	Understand the construction, working and control of solar and wind energy system
CO4:	Aware with the performance characteristics of Renewable Energy Sources.
CO5:	Design & Analyze the performance of PV system

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	1	-	-	-	1	1	-	1	2	-
CO2:	3	2	2	1	1	-	2	-	1	-	1	-	2	-
CO3:	3	2	2	1	1	1	2	-	1	-	1	1	2	1
CO4:	3	1	-	1	2	-	2	1	-	-	2	-	2	1
CO5:	3	1	3	3	3	1	3	-	1	-	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.

- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Books:

1. Renewable & Efficient Electric Power System by Gilbert M. Masters, Wiley.
2. Design of Smart Power Grid Renewable Energy Systems by Ali Keyhani, Wiley.
3. Modeling and Control of Fuel Cells by M. Hashem Nehrir, Caisheng Wang, Wiley.
4. Solar PV and Wind Energy Conversion Systems: an introduction to theory modeling with MATLAB/Simulink and role of soft computing techniques by S. Sumathi, L. Ashok Kumar, P. Surekha, Springer.
5. Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh, Springer.
6. Smart Energy Grid Engineering by Hossam A Gabbar, ELSEVIER.

Reference Books:

1. Grid Connected PV systems Design and Installation by Global Sustainable Energy Solutions.
2. Solar Energy principles of thermal collection and storage by S.P. Sukhatme, J.K. Nayak, Tata McGraw Hill.
3. Solar Photovoltaics fundamentals, technologies and applications by Chetan Singh Solanki, PHI Learning Pvt Ltd.

Web Material:

- https://onlinecourses.nptel.ac.in/noc18_ee35/course (Design of Photovoltaic System)

I. List of experiments:

1. Modeling of PV cell to determine I-V characteristic.
2. Modeling of PV module to analyze the effect of change in insolation & temperature.
3. Determination of I-V curve of PV panel with resistive load.
4. Simulation of standalone PV system with cost analysis.

5. Simulation of grid connected PV system.
6. Modeling of Nernst equation to determine fuel cell output.
7. To study model of DFIG for wind energy conversion system.

EE448: MAJOR PROJECT
8th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	00	18	18	9
Marks	00	500	500	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
		250	250	500

C. Course Outcomes (COs):

On successful completion of the course students would be able to:

CO1:	To apply knowledge and skills learned from project to real-world problems
CO2:	To solve engineering problems by the teamwork and work with other discipline
CO3:	Use experience related to professional and ethical issues in the work environment
CO4:	To explain the impact of engineering solutions developed in a project, in a global, economic, environmental, and societal context in order to provide employability and making entrepreneur
CO5:	To implement the skills related engineering on contemporary issues related with engineering in general with advanced tools and technologies

Mapping of course outcomes with program outcomes
Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	1	-	-	1	-	1	-	1	1	1	-	1
CO2	3	3	2	2	1	2	-	-	3	2	1	2	2	-
CO3	3	2	1	-	-	-	-	3	1	-	2	-	-	1
CO4	3	-	-	3	3	3	-	-	2	1	3	2	3	3
CO5	3	1	3	2	3	3	2	-	-	2	2	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

D. Instructional Method and Pedagogy:

- Major project has 9 credit subject and 18 hours per week.
- Department/Institute will help students to find appropriate major projects.
- The student must be find the appropriate guide for their project and make the project
- Proposal and submit the project proposal to departmental co-coordinator.
- Project must be from the electrical bag round other project must be rejected.
- Student must meet at least once in week to his/her guide.
- The student shall submit two documents to the Coordinator for the evaluation of the major project:
 - Project report
 - Project presentation
- Upon the completion of project, a hard copy of “project Report” must be submitted to the Coordinator by the end of the semester.
- The report must outline with practical knowledge which student gain during project making. The guide must verify report.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory, the student shall be requested to repeat the project.

E. Format of Project Report:

Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12 pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page

- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done during project, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- INTRODUCTION: In this section, give the purpose of the project, reasons for choosing the project, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the project. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the project. Present your observations, contributions and intellectual benefits.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

EE476: ADVANCES IN POWER SYSTEM (Programme Elective-IV)
8th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Smart Grid	20
2	Restructuring of Power System	10
3	State Estimation in Power System	08
4	Substation Automation	12
5	Load Forecasting	10

Total hours (Theory) :60 Hrs
Total hours (Lab) :30 Hrs
Total hours : 90 Hrs

D. Detailed Syllabus:

1	Introduction to Smart Grid	33.33%	20 Hrs
	Introduction to Smart Grid, Architecture of Smart Grid System, Standards for Smart Grid System, Elements and Technologies of Smart Grid System, Distributed Generation Resources, Wide Area Monitoring Systems, Phasor Estimation, Digital relays for Smart Grid Protection, Islanding Detection Techniques, Smart Grid		

	Protection, Modelling of Storage Devices, Modelling of DC Smart Grid components, Operation and control of AC Microgrid, Simulation and Case Study of AC-DC Hybrid Microgrid, Demand side management. of Smart Grid, Energy Management, Design of Smart grid and Practical Smart Grid case study, System Analysis of AC/DC Smart Grid.		
2	Restructuring of Power System	16.66%	10 Hrs
	Concepts of regulation and deregulation, Characteristic of regulated power system, Need to restructured the power system, Overview of a deregulated industry, Structure of deregulated power system, Different entities in deregulated power system, Responsibilities of independent system operator, Trading arrangements: Pool, Bilateral, Multilateral, Power Exchange, Energy auction and market clearing prices, Available transfer capability, Congestion management, Ancillary services		
3	State Estimation in Power System	13.34%	08 Hrs
	Power system state estimation: Past, Present and Future, Weighted least square estimation, Hessian matrix formulation, State estimation of AC network, Detection and identification of bad data measurement, estimation of quantities not being measured, Network observability and Pseudo measurement, Application of state estimation		
4	Substation Automation	20%	12 Hrs
	Introduction to distribution automation, Layout of substations and feeders, Optimum siting and sizing of substations Distribution system load flow, configuration of distribution system, optimum capacitor placement. Optimum feeder switching for loss minimization and load control. Distribution system restoration. Distribution system monitoring and control: SCADA, Concept of modern distribution systems.		
5	Load Forecasting	16.66%	10 Hrs
	Introduction, Estimation of Average and trend terms, Estimation of periodic components, Estimation of Stochastic components: Time series approach, Auto-Regressive Model, Auto-Regressive Moving, Average Models, Kalman Filtering Approach, On-line techniques for non-stationary load prediction.		

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
05	15	15	20	20	25

F. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to acquire the knowledge regarding:

CO1:	Past, present and future of Power System. Also, Restructuring of an Indian Power System
CO2:	Requirement of power system state estimation and methodologies used for state estimation with practical simulation
CO3:	Substation Automation and Distribution system monitoring, control and design
CO4:	To learn data analysis and how to predict future energy demand for smart grid with practical simulation
CO5:	The fundamentals and detail knowledge regarding smart grid technologies with operation and control

Mapping of course outcomes with program outcomes Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	3	1	-	-	-	-	-	-	-	1	-	2	1
CO2:	3	2	1	-	2	-	-	-	-	-	-	-	2	1
CO3:	3	2	3	1	2	-	-	-	-	-	-	-	2	1
CO4:	3	3	2	2	3	-	-	-	-	-	-	-	2	2
CO5:	3	2	3	1	2	-	1	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.

- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

Text Books:

1. J. Wood and B.F. Wollenberg, Power Generation, Operation and Control, JohnWiley & Sons, New York, USA, 1996.
2. Power System Analysis by Hadi Saadat, Tata Mcgraw Hill.
3. Modern Power System Analysis by D.P. Kothari & I. J. Nagrath, Tata McgrawHill.
4. Restructured Power System by S.A. Khaparde and A.R. Abhyankar, Narosa Publication.
5. Electric Power Distribution Systems by A.S. Pabla, 6th Edition, Tata McgrawHill.

Web Material:

1. https://onlinecourses.nptel.ac.in/noc18_ee42/
2. A. P. Sakis Meliopoulos, George J. Cokkinides, Xing *Yong*Chao “A NEW PROBABILISTIC POWER PLOW ANALYSIS METHOD”, IEEE Transactions on Power Systems, Vol. 5, No. 1, February 1990.

I. List of Experiments:

1. To study smart grid architecture
2. To studyrestructuring of power system.
3. To study layout of substations.
4. To study power system estimation.
5. To study load forecasting techniques.
6. To simulate probabilities load flow analysis in MATLAB..

EE477: INTERNET OF THINGS
8th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction To Embedded Concepts	09
2	Iot Architecture And Protocols	15
3	Cloud Architecture And Computing	12
4	Computer Networks And Internet	08
5	Internet Of Things: Sensors and Actuators	08
6	Preparing Iot Projects	08

Total hours (Theory) :60Hrs
Total hours (Lab) :30Hrs
Total hours : 90Hrs

D. Detailed Syllabus:

1	Introduction To Embedded Concepts	15.00%	9Hrs
	Introduction to embedded systems, Application Areas, Categories of embedded systems, Overview of embedded system architecture, Specialties of embedded systems, recent trends in embedded systems, Architecture of embedded systems,		

	Hardware architecture, Software architecture, Application Software, Communication Software.		
2	IoT Architecture And Protocols	25.00%	15Hrs
	Overview: IoT-An Architectural Overview– Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics, Knowledge Management Reference Architecture: IoT Architecture-State of the Art – Introduction, State of the art, Reference Model and architecture, IoT reference Model - IoT Reference Architecture- Introduction, Functional View, Information View, Deployment and Operational View,		
3	Cloud Architecture And Computing	20.00%	12Hrs
	Cloud Architecture Basics: The Cloud -Hype cycle-metaphorical interpretation-cloud architecture standards and interoperability- Cloud types; IaaS, PaaS, SaaS. Benefits and challenges of cloud computing, public, private clouds community cloud, role of virtualization in enabling the cloud. End to End Design : Requirement analysis: strategic alignment and architecture development cycle-strategic impact-Risk impact-financial impact-Business criteria-technical criteria-cloud opportunities –evaluation criteria and weight-End to end design-content delivery networks-capacity planning-security architecture and design Cloud Application Architectures: Development environments for service development; Amazon, Azure, Google App-cloud platform in industry How To Move Application Into The Cloud: Web Application Design- Machine Image Design-privacy design –Database management		
4	Computer Networks And Internet	13.33%	8Hrs
	Internet - The network edge- The network core- Delay , Loss and Throughput in packet switched networks- protocol layers and their service models- TCP/IP protocol architecture- Frame relay networks- ATM networks- protocol architecture- ATM logical connections- ATM cell. Congestion Control In Data Networks And Internet: Effects of congestion – Congestion control - Traffic management - Congestion control in packet switching networks – Frame relay- Congestion control -TCP flow control and TCP congestion control.		
5	Internet Of Things: Sensors and Actuators	13.33%	8Hrs
	Introduction: Internet of Things Promises–Definition– Scope–Sensors for IoT Applications–Structure of IoT– IoT Map Device Sensors: Linear and angular displacement sensors: resistance sensor – induction displacement sensor – digital optical displacement sensor – pneumatic sensors. Speed and flow rate sensors: electromagnetic sensors – fluid flow sensor – thermal flow sensor. Force sensors: piezoelectric sensors – strain gauge sensor – magnetic flux sensor – inductive pressure sensor – capacitive pressure sensor. Temperature sensors: electrical – thermal expansion – optical. Technological Analysis: Wireless Sensor Structure–Energy Storage Module–Power Management Module–RF Module–Sensing Module		
6	Preparing IoT Projects	13.33%	8Hrs
	Creating the sensor project - Preparing Raspberry Pi - Clayster libraries - Hardware- Interacting with the hardware - Interfacing the hardware- Internal representation of sensor values - Persisting data - External representation of sensor values - Exporting		

	sensor data - Creating the actuator project- Hardware - Interfacing the hardware - Creating a controller - Representing sensor values - Parsing sensor data - Calculating control states - Creating a camera - Hardware -Accessing the serial port on Raspberry Pi - Interfacing the hardware - Creating persistent default settings - Adding configurable properties - Persisting the settings - Working with the current settings -Initializing the camera
--	--

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	20	10	20	20	10

F. Course Outcomes (COs) :

Upon successful completion of this course, a student will be able to

CO1:	Enhance their skills in basics of embedded systems
CO2:	Know the protocols of IOT and its types of protocols
CO3:	Understand applications different sensors in internet of things and in electrical engineering and programming in embedded C
CO4:	Know the all communication protocols and Design and troubleshoot hardware circuits based on different sensors

Mapping of course outcomes with program outcomes Course Articulation Matrix

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	2	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	2	-	-	-	-	-	-	-	3	-
CO4	-	2	3	2	3	-	-	-	3	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

H. Recommended Study Material:

1. Dr. K.V.K. Prasad, "Embedded / Real-Time Systems: Concepts, Design and Programming Black Book", New ed (MISL-DT) Paperback – 12 Nov 2003
2. Arnold. S. Berger, "Embedded Systems Design - An introduction to Processes, Tools and Techniques", Easwer Press, 2001
3. Dr. K.V.K. Prasad, "Embedded / Real-Time Systems: Concepts, Design and Programming Black Book", New ed (MISL-DT) Paperback – 12 Nov 2003
4. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
5. Peter Waher, "Learning Internet of Things", PACKT publishing, BIRMINGHAM MUMBAI
6. Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3-642-19156-5 e-ISBN 978-3-642-19157-2, Springer
7. Daniel Minoli, "Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications", ISBN: 978-1-118-47347-4, Willy Publications

8. Vijay Madiseti and ArshdeepBahga, "Internet of Things (A Hands-on-Approach)", 1st Edition, VPT, 2014.
9. Reese, G. (2009). Cloud Application Architectures: Building Applications and Infrastructure in the Cloud. Sebastopol, CA: O'Reilly Media, Inc. (2009).
10. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi, Mastering Cloud Computing: Foundations and Applications Programming, Morgan Kaufmann,, Elsevier publication, 2013
11. Dr. Guillaume Girardin , Antoine Bonnabel, Dr. Eric Mounier, 'Technologies & Sensors for the Internet of Things Businesses & Market Trends 2014 - 2024', Yole Développement Copyrights ,2014
12. http://www.cse.wustl.edu/~jain/cse570-15/ftp/iot_prot/index.html

I. List of experiments:

Sr No	Title
1	Using GPIO and Timer: Using inbuilt timer of ESP, blink LED at an interval of one second.
2	Using UART for Serial Print: Print a statement on Serial Terminal
3	Using ADC for analog sensing: Use POT as an analog input to ESP8266 and print its value on serial terminal
4	Using I2C communication: Use I2C pins for communication with LCD and display characters on it.
5	Using PWM: Use GPIO to change the brightness of LED using pulse width modulation
6	Use POT as an analog input to ESP8266 and use this value to change the brightness of LED using pulse and print value on serial terminal also.
7	Using ESP as Access Point: Use ESP module as access point and connect other device to it using WiFi
8	Using ESP as Station: Use ESP module as station and connect to a router (or access point).
9	Plotting data on plot.ly: Take analog input from ESP and pass that data to www.thingspeak.com and prepare an online graph.
10	Using IoT protocol: Demonstrate simple publish subscribe mechanism of MQTT protocol using MQTT protocol
11	Using OneWire interface: Interface DS18B20 with ESP module using onewire interface

12	Expanding GPIO: Interface ESP with MCP23008 to expand GPIOs using I2C protocol
13	Develop offline Webserver to control GPIO: Demonstrate offline webserver using HTML webpage which can be accessed from web browser and through which LED can be toggled
14	Control GPIO from an Android App: Use an android application to control the GPIO using HTTP GET requests.
15	EEPROM Clear, Write and Read in ESP
16	TELNET (TCP) in ESP

EE478: ENERGY CONSERVATION AND MANAGEMENT (PE IV)
8th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electrical Energy Conservation	13
2	Electrical Energy Management	15
3	Energy storage system	13
4	Energy Audit	11
5	Energy Efficient Technologies In Electrical Systems	08

Total hours (Theory) :60Hrs
 Total hours (Lab) :30 Hrs
 Total hours : 90Hrs

D. Detailed Syllabus:

1 Electrical Energy Conservation 22% 13Hrs

Introduction to energy science and energy technology, various forms of energy. Law of conservation of energy. Primary and Secondary energy, Commercial and Non-

commercial energy, Renewable and Non-Renewable energy.

Indian Energy Scenario, National energy plan and energy strategy, energy conservation act-2001 and its features, energy sector reforms. Energy scenario of India, Nodal agencies like BEE, GEDA, MNES, CEA. WEB etc.

Load Management and Maximum demand control.

Electrical Distribution System.

Electrical Furnaces

Potential energy conservation opportunities in:

- (1) HVAC System
- (2) Lighting systems
- (3) Motors and Transformers

2 Electrical Energy Management 25% 15Hrs

Concept of energy management, Introduction to global energy scenario. Carbon credit.

Electrical System Optimization: Electricity rate Tariff, Key to reduction in electrical energy consumption, methods to improve plant power factor, Load Management, Conduction Loss, Switching Loss, Magnetic Loss, Harmonic Compensation.

Cogeneration system

Intelligent buildings

Energy planning, Energy staffing, Energy Organization, Energy Requirement, Energy Costing, Energy Budgeting, Energy Monitoring, Energy consciousness, Energy Management Professionals, Environment pollution due to energy use. Need of energy planning, steps for energy planning, Role of energy manager, benchmarking.

3 Energy storage system 22% 13Hrs

Design of Energy management programmes. Energy storage: Demand for energy storage - stationary and transport applications; Integrated energy systems. Energy storage systems: heat storage- hot water, hot solids, phase change materials; Chemical storage - synthetic fuels, hydrogen, electrochemical. Mechanical. Potential energy storage: spring, compressed gas. Pumped hydro: Flywheels. Rolling mills, Electrical and magnetic energy storage systems.

4 Energy Audit

18%

11Hrs

Introduction, Procedures for energy analysis and audit. Social and economic cost benefits.

Introduction, types and walkthrough energy, audit. Energy audit at unit level, Industrial Audit approaches. Procedure for energy audit and equipment's required. Comprehensive Energy audit Site testing Measurement & Analysis of Electrical System like Induction Motors. Transformers, synchronous Machines, Illumination system

5 Energy Efficient Technologies In Electrical Systems

13% 08Hrs

Maximum Demand Controllers, Automatic Power Factor Controllers, Energy Efficient Motors, Soft Starter, Variable Speed Drives, Energy Efficient Transformers Electronic Ballasts, Energy Efficient Lighting Controls

E. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
15	15	15	20	15	20

F. Course Outcomes (COs):

Upon successful completion of this course, a student will be able to

CO1:	Understand and implement of Energy conservation Technics as per efficient technology available in Market also suggest the energy conservation technique that will be most suitable for required application
CO2:	The students will be well aware with the minimization of losses, standard benchmarking, combine generation of heat and power system .Also can able to calculate the payback period of any system installed
CO3:	Student will understand about different kind of storage system and their process
CO4:	Develop skill to carry out Procedure for energy audit of any industry and Analysis of Electrical System
CO5:	Student can apply knowledge on Energy efficient technology for different appliance available in Market
CO6:	Acquire the knowledge regarding the energy management & conservation in any electric system

Mapping of course outcomes with program outcomes

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	2	3	3	2	3	-	2		3	3	2	2	3	2
CO2:	2	3	-	-	-	-	-	-	-	-	-	3	3	-
CO3:	1	-	1	-	-	2	-	-	-	-	-	-	1	-
CO4:	3	-	3	-	3	-	-	3	3	3	3	3	3	3
CO5:	2	-	-	3	-	2	-	-	-	3	-	3	3	3
CO6:	2	-	1	-	-	-	2	-	-	-	-	3	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), No correlation “-”

G. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures, which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room, chiller plant and nearby substation will clear the concept of load calculation and student will realise the actual Audit process.
- The course shall include a project work , where students have an opportunity to find out loop holes of wastage of energy and can suggest best remedies for the same.
- Industrial visit to be arranged for practical knowledge and exposure.
- In lab session, the student will understand about practical audit of system and energy management and conservation technology in industries.

H. Recommended Study Material:

Text Books:

1. Amlan Chakrabarti, Energy engineering and management, PHI Learning Private Limited.
2. K. Nagabhusan Raju, Industrial Energy Conservation Techniques, Atlantic Publishers & Distributors (P) Ltd.

Reference Books:

1. Renewable energy sources and conservation technology By- N. K. Bansal, Kleemann and Meliss
2. Non – conventional energy sources by G. D. Rai
3. Energy technology by S. Rao.
4. A guide to energy management by Barney L Capehart, William J Kennedy, Wayne C Turner.

Web Material:

1. www.bee-india.gov.in
2. www.energymanagertraining.com

I. List of Experiments:

1. Domestic and Industrial bill Analysis.
2. To study and perform the Energy Audit of CHARUSAT University.
3. To study and perform the Energy Audit of CHARUSAT Hospital.
4. Case study for Energy Conservation Technics in industries
5. To Study standard consumption rating on Appliances.